When to Train Safety In and When to Filter at Runtime: A Co-location Regime Map for ISO 15066-Compliant Humanoid Manipulation

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Abstract

Classical safety engineering is mature for systems whose behaviour is designed by hand: lane keeping, adaptive cruise control, caged industrial arms, and speed-and-separation interlocks all rely on models that can be analysed and certified. Learned humanoid manipulation is different. A 76-degree-of-freedom robot trained from demonstrations can now work in the same space a person reaches into, but its policy optimises task reward, not human distance, ISO 15066 margins, or contact force. Combining state-of-the-art robot manipulation with collaborative-robot safety is therefore a greenfield problem: the benchmark, cost signals, perception model, training method, runtime filter, and evaluation protocol all have to be built together.

This project builds that system. I introduce `safety_bigym`, a safety-aware extension of BiGym with a moving Unitree G1 (representing a human) coworker, calibrated mock-BodySLAM++ perception noise, three ISO 15066-derived metric flavours, continuous anticipatory costs, and a deployment-faithful snapshot-evaluation harness. On top of the demonstration-driven CQN-AS backbone, I implement a value-based Lagrangian constrained policy and several runtime safety mechanisms, including learned safety-value-function vetoes, ISO speed-scaling, and critic-gated speed-scaling. This is technically hard because CQN-AS is critic-only: there is no actor network to attach a standard Lagrangian to, so the constrained objective must be re-derived in value-based form, and dense safety shaping must fit inside the C51 critic's value support or training silently saturates.

The main result is a regime map, not a single winning method. Proximity measures exposure, but a controlled sweep shows that $\approx 42\,\%$ of close proximity is human-driven and remains even when the robot is frozen. The robot-controllable axis is velocity-adaptive SSM. Under *persistent* co-location (`saucepan_to_hob`), only anticipatory constrained RL reduces proximity gracefully: a fixed-multiplier Lagrangian policy cuts proximity by $22.8\,\%$ ($0.296 \rightarrow 0.228$) at 0.76 success across three seeds, while reactive filters freeze or flee. Under *intermittent* co-location (`dishwasher_close`, `drawers_open_all`), constrained RL is inert or task-fatal, but a learned critic gating an ISO speed-scaling backstop reduces SSM-velocity violations by $50\,\%$ and $22\,\%$ with moderate success cost. The boundary between the regimes is measurable, the gate is active on $61.5\,\%$ of steps on the persistent task versus 19–$27\,\%$ on the intermittent ones, and it was validated by a cross-task test under a pre-registered rule whose failure case occurred exactly where the map predicts. The Hybrid Safety Critic is therefore a two-arm architecture, and the regime map says when to use each arm: train safety into the policy where co-location is persistent; gate a runtime speed-scaling backstop where safe windows exist.

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1 Introduction

1.1 Motivation

1.1.1 Robot Learning Is Reaching Humans

In the past five years, robot learning has progressed from controlled lab environments to platforms that share workspaces with people. Imitation-learning methods such as Action Chunking Transformers [2] and Diffusion Policy [3], demo-driven reinforcement-learning agents such as Coarse-to-Fine Q-Networks with action sequences (CQN-AS) [4, 5], and benchmarks designed around human-like manipulation morphologies such as BiGym [6] and HumanoidBench [7] have driven this transition. Commercial humanoid platforms (including the Unitree H1 [8], the Unitree G1 [9], Apptronik Apollo, and 1X Neo) are being deployed for tasks where the robot is expected to operate *near* a human, not behind a safety cage. ISO 25785-1, under development as a committee draft in 2026 [10], introduces the term "industrial mobile robots with actively controlled stability" specifically to standardise this class of system, reflecting how fast the deployment frontier is moving relative to the standards that govern it.

1.1.2 Most Learned Policies Are Not Safe Near Humans

Safety for machines near people is not new. In classical settings it is comparatively mature. Industrial arms use cages and light curtains and driver-assistance systems such as adaptive cruise control and lane keeping have shipped at scale. Those successes rely on assumptions that fail for learned humanoid manipulation. A cage assumes the human and robot do not share space. Driver assistance assumes a low-dimensional road vehicle with a controller that can be designed and verified in closed form. A 76-degree-of-freedom manipulation policy learned from demonstrations satisfies neither assumption: it shares the workspace by design, operates in the volume a person reaches into, and is a neural policy rather than a hand-derived controller. Safety for this class of system is therefore largely a greenfield problem, and closing it is what this project targets.

In particular, the policies these platforms run are largely *unaware* of the standard that governs their safety. ISO 15066 [11], now integrated into ISO 10218:2025 [12, 13], defines the two normative collaboration mechanisms: Speed and Separation Monitoring (SSM), which prescribes a velocity-dependent geometric bound on how close the robot may come to a human; and Power and Force Limiting (PFL), which caps contact forces against per-body-region biomechanical limits. Standard imitation-learning and reinforcement-learning agents on BiGym [6] optimise task success against a sparse reward. They have no representation of human position, no SSM margin, and no constraint that would distinguish a task-completing trajectory from one that endangers a nearby person.

Three independent observations frame the problem:

1. *Commercial humanoid platforms are not validated for collaborative operation.* Under ISO 10218:2025, collaboration is a property of the validated application, not of the robot [13], and no commercial humanoid had been validated for such an application by mid-2026. The Unitree G1, the most widely deployed humanoid research platform in 2026, ships with no collaborative-operation validation [9], and the H1 requires a safety cage during "lively testing". Standards-landscape reviews reach the same verdict: safety standards lag humanoid deployment [14], and the current framework "is not designed for" humanoids [15]. The deployment is happening but the validation is not.[1]

[1]The nearest milestone to date is machinery-directive conformity, not collaborative-operation validation: AiMOGA announced CE certification of its Mornine humanoid via TÜV Rheinland in September 2025 (company claim).

2. *Robot learning's safety record remains thin at deployment.* Bharadhwaj [16] argued in 2021 that safety and compliance receive too little attention in robot-learning evaluation, and the audits that have since emerged bear this out: jailbreaks have elicited harmful physical actions from a deployed commercial robot [17], and every LLM-driven robot model in a recent evaluation failed basic safety criteria [18]. Brunke et al. [19] and the newer safe-RL survey of Wachi et al. [20] show an active field of constrained and safety-filtered learning, but its representative benchmarks remain low-dimensional or abstract-cost (e.g. Safety-Gymnasium [21] and safe-control-gym [22]), and the nearest human-co-located exception pairs 6–7-DoF arms with a replayed human under an always-active shield [23].
3. *Concurrent humanoid work confirms the runtime-filter need, but not this task class.* SHIELD [24], demonstrated on a Unitree G1, adds a probabilistic CBF safety layer on top of a learned locomotion controller because black-box learned controllers are difficult to certify and expensive to retrain when constraints change. SHIELD targets navigation around people and obstacles. The analogous gap for manipulation in shared workspaces, with ISO 15066-style separation and force limits, is the target of this work.

1.1.3 Perception Adds Distribution Shift

The safety question is harder still because the human's pose is not directly observable. Deployment-time human-state estimates come from on-robot perception, typically a monocular or visual-inertial human-pose estimator such as BodySLAM++ [25], with centimetres of positional error, latency, and occasional dropouts under occlusion. Training-time data, by contrast, typically comes from motion capture or simulator ground truth. That gap is where many real safety incidents will originate, and it is a deployment challenge safe-RL benchmarks rarely model: most expose ground-truth state [21, 22, 26], and the one that perturbs human-state measurements does so with uncalibrated noise [23].

1.1.4 What This Project Targets

Commercial deployment is moving faster than published safety methods. Manipulation benchmarks largely ignore safety, safety benchmarks largely ignore manipulation, and safe-RL methods usually choose either a training-time constraint or a runtime filter, not both, while assuming cleaner human state than a real robot would have. That intersection is exactly where a safe humanoid manipulator must operate, and it is largely uncharted. This project enters it by closing three concrete gaps in safe-RL evaluation for human-co-located humanoid manipulation:

- **No benchmark exposes the full ISO 15066-specific co-working setting on a humanoid manipulator.** Safety-Gymnasium builds on Safety-Gym [21, 26] but still uses abstract cost budgets. safe-control-gym [22] targets low-DOF control. BiGym [6] and HumanoidBench [7] target manipulation and locomotion but include no human coworker. Human-Robot Gym [23] comes closest, pairing 6–7-DoF arms with a motion-capture human under a provably safe shield, but it exposes no ISO 15066-derived cost signal to the learner and no humanoid embodiment. We extend BiGym with a Unitree G1 humanoid as a sustained human-coworker stand-in and surface continuous ISO 15066-derived signals.
- **No published work characterises *when* each safe-RL mechanism helps for manipulation beside a person.** The literature is split into two families that are rarely studied together: training-time constrained RL (Lagrangian CMDP [26, 27]) and runtime safety filtering (shielding [28], CBFs [24, 29], learned safety filters [30, 31]). Cross-family comparisons exist on low-DOF tasks without humans [22] and, closest to this setting, on a single 6-DoF reaching task beside a replayed human [32]. None yields a rule for which mechanism to reach for in a given human-robot collaboration setting, or a variable that predicts when a runtime filter will preserve task throughput and when it will destroy it. We study both families together on one benchmark and derive exactly such a rule: a *regime map* keyed on the human–robot co-location pattern.

- **Safe-RL manipulation benchmarks rarely model deployment perception.** The mock-BodySLAM++ observation wrapper introduced here fills that gap for this setting: it gives the safety mechanisms a noisy, lagged, occasionally stale human-state estimate rather than perfect simulator state, and lets the report test whether the architecture survives the perception conditions a real deployment would face.

The difficulty is that these gaps have to be closed *simultaneously*, and none of the pieces has an off-the-shelf solution at this scale. A demonstration-driven, critic-only agent has no actor network to which a Lagrangian multiplier can be attached, so the constrained objective must be re-derived in value-based form (Section 4.4.1). Dense safety shaping must fit inside a distributional critic's fixed value support, or training silently saturates (Proposition 4.1). A runtime safety critic must be trained on the noisy perception it will face at deployment, not on simulator ground truth (Section 4.2). And the whole system must remain task-competent while a humanoid coworker repeatedly walks into the workspace and reaches toward the robot. The contribution is the architecture that composes these parts, the benchmark that makes the composition measurable, and the *regime map*: an empirical rule linking the pattern of human–robot closeness to the safety mechanism that works.

Terminology. Three terms carry fixed meanings throughout. The *Hybrid Safety Critic* is the two-arm architecture: one arm trains the policy to avoid risk, and the other arm filters actions at runtime. *Critic-gated speed-scaling* is the runtime arm: a learned safety critic estimates whether the proposed action is risky; only when it is risky does the robot slow the action down using the ISO 15066 speed-scaling rule (Section 4.3.3). *Policy–filter composition* is the simpler act of stacking a filter on top of the constrained policy (Section 6.2.6). The empirical finding is which arm should be used in which co-location regime.

1.2 The Problem and the Thesis

Formally, this work targets the following constrained Markov decision process (CMDP). Let an agent control a 76-DOF Unitree H1 humanoid manipulating in a domestic kitchen scene, co-located with a Unitree G1 humanoid acting as a moving stand-in for a human coworker that walks into the workspace, reaches periodically toward the robot's arm or the task object, and occasionally departs and returns. Per-step task reward r_t^{task} is the standard BiGym [6] sparse-plus-shaping signal. Per-step safety cost $c_t \in [0, 1]$ is derived from ISO 15066 SSM geometry and biomechanical force limits, with $c_t > 0$ activated *anticipatorily* as the system approaches a violation rather than only after one occurs (Equations 3.2–3.4).

Two structural properties of this CMDP shape everything that follows. First, its safety is *two-axis*: a geometric *proximity* axis (how often human and robot are close), which is partly *exogenous* because the human can close distance on a stationary robot, and secondly an ISO 15066 SSM *velocity* axis (whether the robot moves slowly enough to stop before contact at the current separation), which the robot fully controls (Section 5.5). Second, the CMDP is instantiated on three tasks (`saucepan_to_hob`, `dishwasher_close`, `drawers_open_all`) that, under the same coworker distribution, induce different *co-location regimes*: on saucepan the coworker occupies the robot's working volume nearly continuously (*persistent* co-location), while on the other two the close encounters arrive in bursts separated by genuinely-safe windows (*intermittent* co-location; Section 3.12). The central empirical claim of the report is the *regime map*, sketched in Figure 1.1 and assembled from evidence in Chapter 6:

> **Thesis.** Which safety mechanism works for a learned humanoid manipulator beside a person is decided not by the sophistication of the mechanism but by the human–robot co-location regime. Persistent co-location requires safety trained into the policy. Intermittent co-location is best served by a learned critic gating a runtime speed-

scaling backstop. The boundary between the two is measurable and predictive, and it was validated here by a cross-task test under a decision rule fixed in advance.

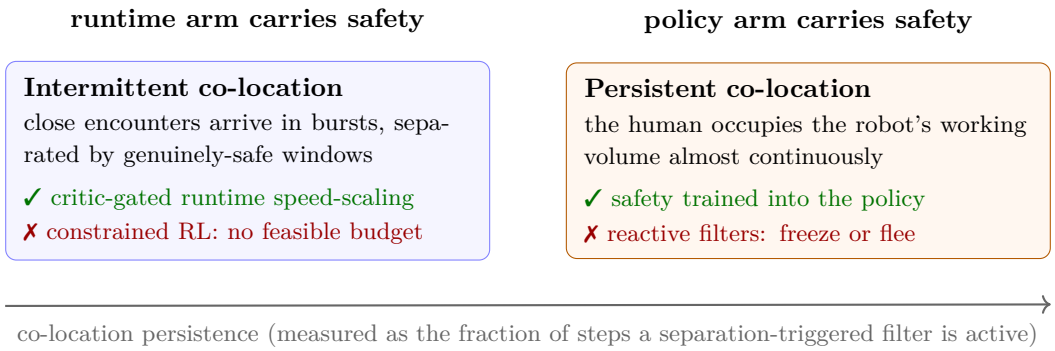


Figure 1.1: The regime map, the report's central claim, in schematic form: the pattern of human–robot co-location, not the sophistication of the safety mechanism, decides whether safety must be trained into the policy or can be enforced by a gated runtime filter. The results chapter assembles the evidence and Figure 6.9 gives the full evidence-annotated version.

The goal is a deployable system $(\pi, \mathcal{F})$ comprising a training-time policy π and an optional runtime filter $\mathcal{F}$ that:

1. achieves task success competitive with the unconstrained baseline, that is, improves safety without materially degrading task performance,
2. reports ISO 15066 SSM and proximity violations not merely in expectation but in the upper tail of the per-episode distribution (CVaR-style metrics, Section 5.5), so that whatever the system cannot control is measured rather than hidden,
3. remains stable under partial observability of the human's pose, matching the perception conditions a real deployment would face,
4. degrades gracefully when the disruption distribution is wider at deployment than at training time.

1.3 Research Questions

RQ1 (the persistent regime)
Under persistent human–robot co-location, can a training-time constrained policy reach a graceful safety–task operating point on the exposure axis, and what does it take, in cost form, budget, multiplier scheme, and checkpoint selection, to get there reproducibly, including under realistic perception noise and a held-out coworker distribution?

RQ2 (the intermittent regime)
Under intermittent co-location, which runtime mechanism reduces the robot-controllable SSM-velocity axis without destroying task throughput, and why do constrained training and binary overrides fail there?

RQ3 (the boundary)
What property of a task–coworker pair decides which safety mechanism works? Is it measurable in advance, and does it predict the outcome of transferring each regime's winning mechanism to the other regime?

RQ1 is answered by the persistent-co-location results (Section 6.2), RQ2 by the intermittent-co-location results (Section 6.3), and RQ3 by the cross-task boundary experiment (Section 6.4). The discussion revisits all three (Section 7.1), and Appendix C maps the repository's experiment codes to the sections that report them.

1.4 Contributions

This report studies the intersection of learned high-DOF manipulation, human co-working, ISO 15066-referenced safety, noisy human-state perception, and safe-RL mechanisms. Existing work covers important parts of that space, but not the whole setting at once. Five contributions, in order of importance:

1. **A validated regime map for choosing the safety mechanism.** Which safe-RL mechanism works beside a person is decided by the human–robot co-location pattern, not by mechanism sophistication. Persistent co-location requires safety trained into the policy. Intermittent co-location is best served by a learned critic gating a runtime speed-scaling backstop. The boundary is measured (a separation-triggered gate is active on 61.5 % of steps on the persistent task versus 19–27 % on the intermittent ones), it is predictive, and it was validated by a cross-task test under a pre-registered rule whose failure case occurred exactly where the map predicts (Section 6.4).

2. **A two-axis safety measurement protocol grounded in a controlled decomposition.** A frozen-robot sweep shows that ≈42 % of close proximity is caused by the human approaching the robot and is outside robot control. Proximity is therefore reported as the *exposure* axis (the policy's) and velocity-adaptive SSM as the *robot-controllable* axis (the filter's), for every configuration, with three accept/reject rules fixed before the corresponding sweeps ran (Section 5.5).

3. **`safety_bigym`, an open-source benchmark for the question.** An extension of BiGym with a moving Unitree G1 coworker stand-in, three ISO 15066-derived safety metrics, continuous anticipatory costs, calibrated mock-BodySLAM++ human-state noise, three manipulation tasks spanning the two co-location regimes, and a deployment-faithful snapshot-evaluation harness (Chapter 3).

4. **A reproducible value-based constrained-RL formulation, with a support-bound guarantee.** Because CQN-AS is critic-only, the Lagrangian must be written as action selection on $Q_r - \lambda Q_c$ rather than as an actor update. The report also proves a C51 support-bound condition stating when dense safety shaping can be added to a distributional critic without silent saturation (Section 4.4).

5. **Mechanism-level lessons with deployable operating points.** A binary veto breaks the action coherence of the chunked CQN-AS policy (max velocity 2.46 → 6.03 m/s), learned vetoes freeze, and geometric dodges flee, so the successful runtime design is: *let the critic decide when to intervene, but slow smoothly rather than veto*, trained on unfiltered rollouts (Sections 6.2.5 and 6.3.2). The deployable recipes: a fixed-multiplier constrained policy reduces proximity by 22.8 % at 0.76 success on `saucepan_to_hob`, and critic-gated speed-scaling reduces SSM-velocity violations by 50 % and 22 % at moderate success cost on `dishwasher_close` and `drawers_open_all`, with WCSAC reimplemented as an external safe-RL reference (Chapter 6).

1.5 Report Outline

The remainder of the report follows the arc of the project. Chapter 2 reviews the relevant literature in collaborative-robot safety, safe reinforcement learning, and human-aware sim-to-real, and positions this work against it. Chapter 3 presents `safety_bigym`: the benchmark environment, the Unitree G1 coworker, the ISO 15066-derived safety metrics, the three-task suite and its two co-location regimes, the calibrated perception-noise model, and the snapshot-evaluation harness. Chapter 4 presents the two arms of the Hybrid Safety Critic, the value-based constrained policy and the runtime filter family from learned veto to critic-gated speed-scaling, together with the WCSAC external baseline and the implementation work that makes the comparison trustworthy.

Chapter 5 defines how safety is measured: the two axes and their ownership, the pre-registered decision rules, the statistical methodology, and the unconstrained baselines. Chapter 6 reports the results in three movements, persistent co-location, intermittent co-location, and the cross-task boundary condition that unifies them into the regime map. Chapter 7 interrogates threats to validity and broader impact, and Chapter 8 concludes with the practitioner lessons and future work, in which the persistence-dial experiment is the named next step. A declarations section covering originality, generative-AI use, ethics, sustainability, and code availability precedes the references; hyperparameter tables, the full reproducibility checklist, and an index mapping repository experiment codes to report sections appear as appendices.

2 Background and Related Work

2.1 ISO 15066: Speed-Separation and Force Limiting

ISO 15066 [11], recently integrated into ISO 10218:2025 [12, 13], is the international standard governing contact safety in collaborative robotics. It defines two normative mechanisms relevant to autonomous manipulation.

2.1.1 Speed and Separation Monitoring (SSM)

SSM requires that the protective separation distance between any human and any robot link exceeds the velocity-dependent *required-separation distance* S_p:

$$S_p \;=\; v_h\,(T_r + T_s) \;+\; v_r\,T_r \;+\; \tfrac{v_r^2}{2a_{\max}} \;+\; C \;+\; Z_d \;+\; Z_r, \tag{2.1}$$

where v_h is human velocity, v_r is robot-link velocity, T_r and T_s are reaction and stopping latencies, $a_{\max}$ is the robot's maximum deceleration, C is the intrusion contribution, and Z_d, Z_r are uncertainty terms [11, 33, 34]. The standard mandates $v_h \le v_{h,\max} = 1.6\,\mathrm{m/s}$ as a conservative cap on assumed human motion. Under typical domestic-manipulation geometry the conservative S_p over-fires ($\sim 93\,\%$ violation rate on random rollouts in our environment), motivating the velocity-adaptive variant introduced in Section 3.8. Our implementation evaluates S_p with the human and robot-velocity terms and the braking term, plus a single lumped constant C into which the sensor-specific position-uncertainty terms Z_d and Z_r are folded, since their per-axis budgets are not defined for a simulated camera; this keeps the geometric check faithful to the standard's dominant velocity-dependent structure while avoiding fabricated uncertainty figures.

2.1.2 Power and Force Limiting (PFL)

PFL caps inadvertent contact forces against per-body-region biomechanical thresholds. Annex A of the standard tabulates quasi-static and transient force limits per body region (forehead, chest, abdomen, upper arm, forearm, hand, etc.), carried forward into ISO 10218-2:2025 as Annex M with the PFL requirement now normative [13]. [35] measures these forces experimentally on UR10e and KUKA LBR iiwa arms and finds they vary by more than 100 % within the workspace, a finding confirmed at journal scale by the same group's 2,250-collision campaign [36] and one that motivates the force-ratio formulation

$$f_t^{\text{ratio}} \;=\; F_t^{\text{contact}} / F_t^{\text{limit}} \tag{2.2}$$

adopted in Equation 3.3 of the cost signal. $f_t^{\text{ratio}} \ge 1.0$ constitutes a PFL violation; the anticipatory cost activates at $f_t^{\text{ratio}} > 0.8$, giving the policy headroom to react.

2.1.3 Why ISO 15066 Is Hard to Target with Generic Safe-RL

Generic safe-RL formulations target a single CMDP cost budget $\mathbb{E}_\pi[\sum_t c_t] \le d$ on a per-step cost *usually treated as a binary indicator* [21, 26, 27, 37, 38]. ISO 15066 compliance, by contrast, is both *geometric* (the required separation depends on instantaneous velocity) and *distributional* (the standard's intent is a worst-case bound on *every* step, not an expectation; a 1 % mean cost is consistent with a 1 % violation rate at unit severity or with a 0.1 % violation rate at 10× severity, very different worst-case implications). The architecture in Chapter 4 addresses both gaps: continuous, anticipatory cost signals (Section 3.10) replace binary indicators, and CVaR-style evaluation metrics (Section 5.5) align the reported numbers with the standard's worst-case intent.

2.2 Reinforcement Learning Fundamentals

The control problem is formalised as a Markov decision process (MDP) $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with states $s \in \mathcal{S}$, actions $a \in \mathcal{A}$, transition kernel $P(s' \mid s, a)$, reward $r(s, a)$, and discount $\gamma \in [0, 1)$. A policy $\pi(a \mid s)$ induces a state–action value $Q^\pi(s, a) = \mathbb{E}_\pi[\sum_t \gamma^t r_t \mid s_0 = s, a_0 = a]$, the unique fixed point of the Bellman operator $(\mathcal{T}^\pi Q)(s, a) = r(s, a) + \gamma\, \mathbb{E}_{s', a'}[Q(s', a')]$. Value-based agents learn Q and act greedily, $\pi(s) = \arg\max_a Q(s, a)$. The safety architecture of Chapter 4 adds a second value function over a cost signal and selects actions against a combination of the two. Safety is imposed by lifting the MDP to a *constrained* MDP (CMDP), which augments the objective with the cost constraint of Equation 2.3; Section 2.3 reviews how that constraint is solved.

Rather than estimate the scalar Q^π, *distributional* RL learns the full return distribution $Z^\pi(s, a)$ whose expectation is Q^π [39]. The C51 algorithm represents Z^π as a categorical distribution over a fixed support $[v_{\min}, v_{\max}]$ and minimises a projected Bellman error. This fixed support is central for the safety extension: a shaped reward whose discounted return exceeds the support is silently clipped, which breaks value learning unless an invariant holds (Proposition 4.1). Our backbone is built on a C51 critic, which is why distributional value learning is introduced here.

Three agent families recur as baseline or backbone. *Imitation learning* (ACT [2], Diffusion Policy [3]) is strong on BiGym but, lacking a reward channel, cannot respond to a safety cost, a limitation we confirm empirically (Section 6.2.1). *Off-policy pixel RL* (DrQ-v2 [40]) learns from reward but struggles on long-horizon sparse-reward manipulation without a demonstration prior. *Demonstration-driven, value-based RL*, the family we adopt, combines both; CQN-AS [4] is the instance used throughout, for the reasons given next.

2.2.1 Why CQN-AS (Choice of Backbone)

CQN-AS [4, 5] (the coarse-to-fine Q-network of [4] with the action-sequence extension of [5]) was chosen because BiGym demonstrations [6] are a strong inductive prior on long-horizon manipulation, and CQN-AS is a demo-driven, value-based agent that achieves high success on long-horizon BiGym tasks (we obtain around 85 % success on `saucepan_to_hob`; Section 5.8), with author-run BiGym evaluations as external validation [4, 6]. The critic-only, coarse-to-fine, C51-headed [39] structure is unusually well-suited to the safety extension introduced here: a second *cost* critic of the same shape composes cleanly via dual-Q action selection (Section 4.4.3), without requiring an actor network at all. The non-trivial consequence is that the standard actor-critic Lagrangian formulation [26, 27] must be re-derived in value-based form; this re-derivation is itself a methodological contribution. An imitation backbone such as Diffusion Policy [3] was rejected for the reason given above: without a reward channel it cannot accept a per-step cost gradient at all.

2.2.2 Curriculum Learning for Long-Horizon RL

Curriculum learning orders the training distribution from easier to harder variants of a task so that an agent can bootstrap competence it could not acquire by training on the hardest setting from scratch [41]. The technique is standard in high-dimensional robotics: OpenAI's dexterous-manipulation work [42] used an automatic difficulty curriculum to make an otherwise-intractable problem learnable, massively parallel legged-robot training rests on a game-inspired terrain curriculum [43], and current Unitree H1 pipelines adjust terrain difficulty on success [44]. In our setting the difficulty axis is the coworker's intrusiveness: direct training of the CQN-AS backbone on the full coworker disruption distribution fails to discover the manipulation task at all (the agent is dominated by avoidance and never gets the saucepan onto the hob). We therefore adopt a staged idle → easy → full curriculum, justified and specified in Section 4.6, and treat the curriculum dependence itself as a reported limitation (Section 7.2.2).

2.3 Safe Reinforcement Learning

2.3.1 Lagrangian Constrained MDPs

The standard formulation [26, 45] augments the reward objective with a cost constraint:

$$\max_{\pi} \mathbb{E}_{\pi}\left[\sum_{t} \gamma^{t} r_{t}\right] \quad \text{subject to} \quad \mathbb{E}_{\pi}\left[\sum_{t} \gamma^{t} c_{t}\right] \leq d, \tag{2.3}$$

solved by introducing a Lagrange multiplier $\lambda \geq 0$ that trades off task return against cost. Gradient ascent [26, 37], PID control [27], and primal-dual variants all appear in the literature. PID outperforms gradient ascent on convergence and oscillation when constraints are tight, and we adopt the PID variant (Section 4.4.7). García and Fernández [46] is the foundational safe-RL survey. Brunke et al. [19] connect safe learning to robotics; and the newer survey by Wachi et al. [20] summarises current constraint formulations. These surveys establish the method families, but not the co-location regime boundary studied here.

2.3.2 Distributional and Tail-Aware Safe RL

Mean-cost Lagrangian methods bound only $\mathbb{E}[Z_c]$. *Distributional* safe RL replaces the expectation with a coherent risk measure, typically the Conditional Value at Risk:

$$\mathrm{CVaR}_{\alpha}(Z_c) \;=\; \mathbb{E}[\, Z_c \,|\, Z_c \geq \mathrm{VaR}_{\alpha}(Z_c)]\,, \tag{2.4}$$

the expected cost conditional on being in the worst $1-\alpha$ fraction of outcomes. WCSAC [1, 47] parametrises the cost return as Gaussian and minimises CVaR_{α} directly; quantile critics [48] are a non-parametric alternative. This work adopts CVaR *as the evaluation metric* (Section 5.5.4), reports CVaR for the mean-cost-optimised policy as a diagnostic (Section 6.2.9), and identifies explicit CVaR-optimised training as natural future work (Section 8.1.3).

2.3.3 Recovery, Switching, and Filters

The runtime-override family has a formal-methods anchor in *shielding* [28, 49], which synthesises a reactive monitor that overrides any action violating a temporal-logic safety specification. Learned variants replace the synthesised shield with a value function: Recovery RL [31] switches between a task policy and a recovery policy based on a learned safety value, and Srinivasan et al. [30] train a safety critic offline and use it as a deployment-time filter. All are structurally related to the runtime filters in Section 4.2: each consults a safety signal to decide whether to override the task action. The distinguishing choices are *when* the switch happens (training time for Recovery RL, deployment for shields and for this work) and *what the override does*, a design axis that Section 6.3.2 shows is decisive: a hard override (veto) interacts destructively with chunked action execution, while a graded speed reduction does not.

2.4 Safety Filters and Control Barrier Functions on Humanoids

Hamilton-Jacobi reachability [50, 51] gives the theoretical foundation for safety value functions. Practical implementations span hand-engineered Control Barrier Functions (CBFs) [29], predictive filters [52], and learned safety value functions [30, 31], a spectrum unified under a single safety-filter view by Hsu et al. [53]. Recent CBF-RL hybrids [54] apply CBF-style filtering during RL training so the deployed policy needs no filter. SHIELD [24], demonstrated on a Unitree G1, applies a probabilistic CBF filter over high-level commands without modifying the learned controller.

Our offline safety filter (Section 4.2) is in this learned-value-function family but is trained offline with CQL [55] rather than via online HJ reachability iteration. The trade-off is explicitness versus data efficiency: HJ and CBF methods provide tighter formal guarantees, while CQL-trained filters scale to higher-DOF observation spaces without solving a constrained optimisation

at every step. Solver cost alone no longer rules out CBF-QPs at moderate scale. A whole-body CBF-QP with 15 collision pairs solves in ∼0.2 ms on a 24-DOF humanoid in simulation [56], and operational-space formulations sustain hundreds of constraints at kilohertz rates on a 7-DOF arm [57]. The combination posed here remains undemonstrated, however. Hardware humanoid filters to date restrict themselves to reduced-order models with a single constraint [24, 58] or to safe-set control of an upper-body subspace [59], and the most direct attempt at human-link × robot-link CBF filtering on a humanoid is simulation-only, commands an 8-DoF upper-body subspace, and solves its 72 capsule constraints at ∼33 Hz on a GPU [60], below this benchmark's control rate and before certified barrier synthesis under exogenous human motion is even addressed. A learned value function reduces the runtime decision to one forward pass at any embodiment dimension. This practical consideration, alongside the data-efficiency argument above, is why we adopt the learned-value-function path.

2.4.1 Filter Conservatism and the Freeze-versus-Flee Limit

A recurring tension runs through the runtime-filter literature: a filter strict enough to guarantee safety against an unmodelled human tends to be too conservative to let the task proceed. This is the field's own stated problem. Recent latent safety filters [61] observe that the standard *least-restrictive* construction "potentially undermines the task performance that makes modern visuomotor policies valuable", degrading task completion sharply in their own experiments. Smoother variants [62] are proposed specifically to soften the resulting freeze. A recent separation result [63] gives the theoretical counterpart: a runtime filter need *not* degrade asymptotic return, but only when it is least-restrictive (an identity on already-safe actions), and the prescription, deploy the most permissive filter available, is precisely the condition a baseline-calibrated learned veto violates once the policy it guards has itself learned to avoid, a mechanism we observe directly in Section 6.2.5.

The most pointed evidence concerns an *approaching* human whose motion the robot cannot predict. Tian et al. [64] show that a worst-case Hamilton–Jacobi filter "maintain[s] safety, but inhibit[s] the robot's ability to make progress by intervening excessively",the freeze-versus-flee mode, and that the obvious remedy, trusting a human-motion model to relax the bound, is unsafe: their model-trusting variant suffered a 25 % collision rate against an unmodelled, distracted human, while their safe, confidence-aware variant recovers low task cost *only* while the human follows the model, reverting to the conservative worst-case filter (and its cost) whenever the human does not. The implication for sustained co-working is direct: against a coworker who repeatedly walks into the workspace, a reactive filter can buy task-cheap safety only when the human is predictable, and must pay the conservatism cost exactly when they are not. The freeze half of the dilemma is old enough to have a name: the *freezing-robot problem* [65], identified in crowd navigation, where a planner that treats the human as an unconstrained obstacle concludes that every forward path is unsafe and stalls. Proactive policies are not bound by this trade in the same way: a learned navigation policy keeps its freezing rate below 1 % where an analytical velocity-obstacle solver with ground-truth state freezes and stalls in dense crowds [66].

What the literature does *not* provide, on either side, is a characterisation of *when* each mechanism helps. The constrained-RL line reports tasks where a cost budget is feasible, and the shielding and filtering line reports settings where interventions are rare enough to be cheap. The nearest cross-family evidence is a single 6-DoF reaching task beside a replayed human, on which Thumm et al. [32] found a PID-Lagrangian agent "unable to learn a suitable policy" while their always-on shield held violations at zero, an observation the intermittent-regime results here reproduce independently at 76 DOF (Section 6.3.1). Within the filter family, Krasowski et al. [67] offer guidance on choosing an intervention type, but no work selects between the families, names the variable that separates their successes from their failures, or tests both across co-location regimes on a high-DOF manipulator under ISO 15066. These results frame the central empirical question of Chapter 6: not only *whether* a reactive filter can add safety at acceptable task cost against an approaching coworker, but *under what observable conditions*. The regime map of

Section 6.4 is this report's answer.

2.5 Conservative Q-Learning

CQL [55] is a standard pessimistic offline-RL method [68]. The regulariser pushes down Q-values for out-of-distribution actions, preventing a safety filter from falsely certifying unseen actions as safe. Applying the same pessimism to a cost critic is established practice in safe offline RL [69], and the safety critic here follows that pattern. A bounded-output network (scaled sigmoid to $[0, 1/(1-\gamma)]$) provides additional protection against overestimation. See Section 4.2 for the full safety-critic treatment.

2.6 Sim-to-Real for Human-Aware Robotics

2.6.1 Dynamics vs. Perception Sim-to-Real

Most sim-to-real literature focuses on the *dynamics gap* [70, 71], that is, the mismatch between simulated and real-world physics: contact and friction modelling, actuator response, mass and inertia estimates, and control latency. The deployment story for human-aware robotics also includes a *perception* gap: human-state estimates on a real robot come from on-robot pose estimators, not from motion capture or simulator ground truth. The mock-BodySLAM++ model of Section 3.13 captures the perception side; dynamics-side sim-to-real is a documented limitation, discussed in Section 7.2.3.

2.6.2 Human Pose Estimation in Co-Located Settings

BodySLAM++ [25] is a tightly-coupled visual-inertial human-and-camera state estimator that achieves 15+ FPS on an Intel i7 CPU, improving over its predecessor BodySLAM [72] by 26 % on human-state accuracy and 12 % on camera-state accuracy. Typical positional error is on the order of ∼10 cm, with higher errors under occlusion. The mock-BodySLAM++ noise model (Section 3.13) is calibrated against these published characteristics: Ornstein-Uhlenbeck temporally correlated position noise, a 3-step latency buffer matching the 15 Hz perception clock, and stochastic dropout matching reported tracking failures. Newer world-grounded estimators such as WHAM [73] are more accurate but run offline on GPUs, so the noise model stays calibrated to the on-robot CPU estimator.

2.7 Benchmark Landscape

Table 2.1: Existing benchmarks in robot learning and safe RL, positioned against the needs of evaluating ISO 15066-compliant humanoid manipulation under sustained co-working. No prior benchmark combines all five axes. Human-Robot Gym comes closest but enforces SSM/PFL through an always-active shield on 6–7-DoF arms, rather than exposing graded cost signals to the learner.

Benchmark	High-DOF	Live human	ISO 15066 signals	Perception noise	Sustained interact.
Safety-Gym [26]	✗	static obstacles	✗	✗	✗
Safety-Gymnasium [21]	✗	static obstacles	✗	✗	✗
safe-control-gym [22]	✗	✗	✗	✗	✗
Safety-CHORES [74]	✗	✗	✗	✓(egocentric)	✗
RLBench [75]	✗	✗	✗	✗	✗
BiGym [6]	✓	✗	✗	✗	✗
HumanoidBench [7]	✓	✗	✗	✗	✗
Assistive Gym [76]	✗	✓(seated)	✗	✗	✓
Habitat 3.0 [77]	✗	✓(kinematic)	✗	✗	partial
Human-Robot Gym [23]	✗	✓(mocap)	shield	✓(optional)	✓
`safety_bigym` (this work)	✓	✓(G1)	✓	✓	✓

The landscape splits along three axes. **Safety-axis** benchmarks (Safety-Gym [26], Safety-Gymnasium [21], safe-control-gym [22], and most recently Safety-CHORES [74]) expose cost budgets without human coworkers. **Manipulation-axis** benchmarks (RLBench [75], BiGym [6],

HumanoidBench [7]) expose realistic tasks without safety constraints. **Human-axis** benchmarks place a simulated person beside the robot, from assistive care (Assistive Gym [76]) to social navigation (Habitat 3.0 [77]), and Human-Robot Gym [23] adds an ISO-grounded shield with optional measurement noise on 6–7-DoF arms. None exposes ISO 15066-derived continuous cost signals on a high-DOF humanoid, and the intersection this work targets remains uncovered.

2.8 Positioning of This Work

Table 2.2: Positioning of this work's two-arm architecture (the Hybrid Safety Critic) against representative safe-RL methods. "Tail-aware" indicates whether the method targets a coherent risk measure (CVaR, VaR) rather than expected cost. "Dual critic" separates the task-value and cost-value estimators to avoid the non-stationarity of single-critic Lagrangian. "Runtime guar." indicates whether a deployment-time mechanism (not just a training-time bound) is provided. "High-DOF" indicates evaluation on a $\geq$ 16-DOF embodiment. The contribution of this work is the *combination* of all five columns on a 76-DOF humanoid with a live G1 humanoid coworker.

Method	Noise-aware	Runtime guar.	Tail-aware	Dual critic	High-DOF
Safety-Gym Lagr. [26]	✗	✗	✗	✗	✗
WCSAC [1]	✗	✗	✓	✓	✗
Recovery RL [31]	✗	training-time	✗	✓	✗
CBF-RL [54]	✗	training-time	✗	✗	✗
SHIELD [24]	✗	✓	probabilistic	✗	✓(G1)
Learned SVF [30]	✗	✓	✗	N/A	✗
This work	✓	✓	diagnostic*	✓	✓(76 DOF)

* B-value-mean optimises the mean cost; CVaR-optimised training is future work (Section 8.1.3). CVaR is reported as an evaluation diagnostic.

WCSAC [1] introduced the distributional cost critic on Safety-Gym, much lower-DOF than our setting. We reimplement it faithfully on the 76-DOF humanoid tasks and report it as the external safe-RL reference (Section 6.3.1). CBF-RL [54] and SHIELD [24] provide runtime CBF-style filtering on top of RL but without a distributional treatment of cost. SHIELD is the closest concurrent work (G1 humanoid, runtime safety) but targets navigation around humans/obstacles rather than manipulation in a shared workspace. Recovery RL [31] provides the switching formulation but at training time, on lower-DOF arms, and without distributional safety. This work evaluates representatives of both families, constrained RL and runtime filtering, on the same benchmark, under the most challenging combination of embodiment, disruption pattern, and perception conditions in the safe-RL literature to date; its distinguishing output is not a new mechanism but the *regime map* that says when each existing mechanism class works, a characterisation neither family's own literature provides.

3 The `safety_bigym` Benchmark

3.1 Why a New Benchmark

This chapter presents `safety_bigym`: a safety-aware extension of BiGym [6] that we contribute as one of the two principal deliverables of this project. `safety_bigym` is a *benchmark*, not a method: the environment and metric scaffolding against which any present or future safe-RL approach for humanoid manipulation can be evaluated.

Table 2.1 in Section 2.7 established that no existing benchmark covers the intersection of (i) a high-DOF humanoid embodiment, (ii) a live moving human coworker, (iii) ISO 15066-derived safety signals, (iv) calibrated perception noise, and (v) sustained interaction. `safety_bigym` fills this gap. The aspects of human-robot interaction that the benchmark is designed to test are:

1. **ISO 15066 SSM compliance while completing the task**: the velocity-dependent bound exposes fast-moving-near-human behaviour, not mere task incompetence.
2. **Response to *anticipatory* safety signal**: the continuous cost activates before violations, and the benchmark emits margins, not just binary flags.
3. **Robustness to noisy human-state observation**: ground-truth and calibrated mock-BodySLAM++ modes are switchable, so the compliance delta is measurable.
4. **Task competence under sustained co-working**: a "stop until the human leaves" policy trivially achieves zero contact and trivially fails the task; the success-rate axis catches it.
5. **Generalisation beyond the training disruption band**: train/eval `ParameterSpace` separation supports an explicit OOD experiment (Section 6.2.10).
6. **Graceful tail behaviour**: raw per-step `min_separation` is exposed, so CVaR and percentile metrics are computable for any policy.

These six items are the explicit design targets of `safety_bigym`; every design choice in this chapter is motivated by one or more of them.

3.2 Why Build on BiGym

We build on BiGym [6] rather than starting from scratch for three reasons. First, BiGym provides a 76-DOF Unitree H1 humanoid kinematic chain, 40 manipulation tasks, and human-collected teleoperation demonstrations, all non-trivial assets to recreate. Second, BiGym is an established manipulation benchmark with published baselines: its authors benchmark mainstream imitation learning (ACT [2], Diffusion Policy [3]) and demo-driven RL, and CQN-AS [4], the agent we adopt, was itself evaluated on BiGym by its authors. This gives us a reference set of methods and reported scores against which to position safety-aware variants. Third, the BiGym task distribution (manipulation in domestic kitchen scenes) is exactly the deployment context where ISO 15066 compliance matters most: tasks such as `saucepan_to_hob`, which requires the robot to open a cupboard, take out a saucepan and place it on the hob, are representative of close-range domestic manipulation carried out near a person [6].

3.3 Robot Embodiment and Simulator

We adopt BiGym's 76-DOF Unitree H1 humanoid as the manipulating robot, with the **bi-manual** action mode ($\mathcal{A}_{\text{bm}} \in \mathbb{R}^{16}$, with a 4-DOF floating base). Whole-body locomotion adds a substantial learning challenge orthogonal to the safety question this project targets. Bi-manual mode isolates the manipulation-near-human problem and is the mode BiGym's authors themselves benchmark on. The proprioceptive observation is $s^{\text{bm}}_{\text{proprio}} \in \mathbb{R}^{64}$.

3.4 The Human Coworker: A Unitree G1 Stand-In

A second articulated body is added to the scene to stand in for a human coworker. **We model the human with a Unitree G1 humanoid [9]**, rather than a parametric human body. The choice is deliberate, for three reasons:

(i) The stand-in is a physically grounded, well-characterised body. We model the person with an off-the-shelf Unitree G1, whose mass distribution, geometric extent, joint limits and achievable velocities are published platform quantities [9]. The ISO 15066 limits are instantiated against concrete numbers rather than a hand-tuned parametric body, and the stand-in lives inside the same MuJoCo contact and dynamics model as the robot, so separations and contact forces are computed consistently for both bodies.

(ii) The safety-relevant properties are biomechanical, not specifically human. SMPL-H [78] and similar parametric models remain the standard for human-pose research [79], but the quantities driving an ISO 15066 SSM/PFL assessment (velocity, geometric extent, contact response) are functions of mass and articulated degrees of freedom, not of human appearance. The limits are applied to the stand-in exactly as if it were a person, under the conservative assumption that any human sharing the workspace is at most as massive and as fast as the G1.

(iii) The benchmark is body-agnostic. Planner, scenario sampler, and IK chains route through a single `HumanConfig.human_model` selector, so the G1 can be swapped for any articulated body with no architectural change. Alternative embodiments are supplementary (Section 8.1.4).

Modelling decisions.

- **Pelvis as a mocap body, not a freejoint.** The pelvis is implemented as a MuJoCo `mocap="true"` body, with position and orientation written directly by the trajectory planner each control step. The original (and natural) choice, a `<freejoint/>` pelvis, was discarded because the implicit pelvis velocity computed by MuJoCo as $\Delta_{\text{pos}}/\Delta_{\text{phys}}$ at the physics timestep ($\Delta_{\text{phys}} = 2\,\text{ms}$) reached $\sim$120 m/s during teleport-style updates, causing the SSM-required separation distance from Equation 2.1 to balloon to $\sim$18 m. The mocap pelvis exposes physically-realistic velocity to the safety wrapper at the control timestep.
- **Cross-paired collision channels.** The collision-channel scheme was redesigned so that the G1 coworker emits contact on MuJoCo channel bit 1 and accepts bit 2. The manipulating H1 and the static scene emit bit 2 and accept bit 1. This eliminates G1-self-collision between adjacent body segments (originally producing $\sim$220 kN spurious forces and crashing the simulator within $\sim$1 s) and G1–scene penetration with dishwashers and cupboards. Without this fix, training is infeasible regardless of the safety architecture.
- **PD actuator gains.** The G1's body joints are driven by position actuators with $k_p = 200, k_v = 20$, matching the per-joint mass and inertia of the G1 platform [9] and aligning the closed-loop dynamics with a representative humanoid co-worker.

3.5 The CQN-AS Adapter Layer

BiGym exposes a Gym-style `step/reset` API. CQN-AS [4] consumes a `dm_env`-style `ExtendedTimeStep` API. The adapter bridges these and carries per-key frame-stack deques, a bidirectional action normaliser for CQN-AS's discrete bin centres, and a safety-payload extractor that routes per-step `info["safety"]` into `ExtendedTimeStep.cost` and per-episode aggregates into trajectory metadata. The layer is what makes the pipeline body-agnostic: swapping the G1 coworker for SMPL-H [78] (Section 8.1.4) requires no adapter changes.

3.6 Bounded Workspace Reward Shaping

BiGym's task reward is sparse-plus-shaping: $r_t^{\text{task}} \in \{0, +1\}$ for the sparse component plus a small dense distance term where the task allows. Under sustained co-working disruption, this is insufficient to prevent an *evacuation* local optimum: the policy learns that moving the end-effector arbitrarily far from the human eliminates safety cost, and accepts the task-reward cost. The behaviour is observable as a robot that parks 2–3 m from the workspace and stays there.

`safety_bigym` adds a bounded penalty for excess distance between the end-effector and the task object:

$$r_t^{\text{ws}} \;=\; -\beta \cdot \min\Big(\max(0,\, \|p_t^{\text{ee}} - p_t^{\text{task}}\| - r_{\text{ws}}),\; c_{\text{ws}}\Big)\,, \tag{3.1}$$

with $r_{\text{ws}} = 0.4\,\text{m}$, $\beta = 0.05$, $c_{\text{ws}} = 1.0\,\text{m}$. The cap c_{ws} is critical and is derived in Proposition 4.1 as a value-support invariant of the C51 distributional critic; the bounded form is the one that *works*, and is therefore the form the benchmark ships with. The shaped reward is $r_t^{\text{task,shaped}} = r_t^{\text{task}} + r_t^{\text{ws}}$.

3.7 The Snapshot-Evaluation Harness

A practical benchmark needs a reproducible evaluation pipeline. The `safety_bigym` `benchmark_policy.py` harness takes any policy snapshot (a CQN-AS checkpoint, an ACT snapshot, or a user-supplied `Policy` subclass) and produces a CSV row per (task, disruption, observation-mode, seed) cell with the full safety-metric schema of Section 3.8: mean and percentile separation, three SSM-flavour rates, time in near-proximity bins, time-to-first-violation, and per-region PFL violation counts. The harness is what makes Tables 6.1, 6.7, and 6.4 reproducible from snapshots alone.

The harness is delivered alongside the benchmark and is documented in the project's `README.md`; see Section 5.6 for the smoke-gate command that verifies an installation.

3.8 The Three Safety-Metric Flavours

The ISO 15066 SSM formula (Equation 2.1) is velocity-dependent and, under the standard's conservative $v_h = v_{h,\max} = 1.6\,\text{m/s}$ human-velocity cap, the resulting required-separation distance S_p varies between $\sim$0.3 m and $\sim$5 m on our environment's robot speeds. The conservative S_p over-fires on the kinematic scale of a domestic-manipulation scene: random-policy rollouts in `safety_bigym` register a $\sim$93 % SSM violation rate, more a function of robot-arm capability than of actual human proximity. A direct "SSM compliance rate" number is therefore not the canonical thesis metric.

We expose *three* flavours of the relevant signal at every env-step:

`ssm_violation` (boolean) The conservative formal ISO 15066 check: S_p computed with $v_h = v_{h,\max}$, observed v_r, and the geometric closest-pair distance. Kept for ISO traceability and historical comparability against earlier experimental data.

`ssm_violation_actual` (boolean) S_p computed with observed v_h and observed v_r. Formal ISO compliance *under the realised motion*; this is the reportable ISO number when the deployment can measure v_h.

`proximity_violation` (boolean) Geometric: min-separation $< \tau_{\text{prox}}$, with $\tau_{\text{prox}} = 0.3\,\text{m}$ matching the safety-filter dataset labelling threshold. This is the *exposure* measure: how often human and robot are close, regardless of who closed the distance and how fast anyone is moving. It is the most directly comparable number to the human-aware-robotics literature.

The three flavours are not redundant: a fast robot at 1 m separation from a still human registers `ssm_violation` but not `proximity_violation`; a still robot at 0.25 m from a still

human registers `proximity_violation` but not `ssm_violation_actual`. They are also not equally *attributable to the robot*: Chapter 6 shows that ≈42 % of time-in-proximity on these scenes is the coworker approaching an already-stationary robot, a component no robot policy can remove, whereas the velocity-adaptive SSM margin is a function of the robot's own speed and is fully robot-controllable. The evaluation protocol therefore treats the two as distinct *axes* with explicit ownership, proximity as the (partly exogenous) exposure axis on which policy-level avoidance is judged, and `ssm_violation_actual` as the robot-controllable axis on which runtime filters are judged, and reports *both* for every configuration in every results table (Section 5.5).

3.9 Calibrating the Proximity Threshold

The threshold τ_{prox} is the single most consequential parameter of the proximity (exposure) axis, so we fix it by two speed-independent analyses rather than by intuition.

Contact geometry. Rendering the coworker at controlled separations (robot frozen in its reset pose, the coworker pelvis slid along the approach axis; Figure 3.1) shows the collision geometries interpenetrating at small min-separations (the 0.09 and 0.21 m columns), with a clearly visible gap open by ≈ 0.31 m. A usable threshold must therefore sit above the contact band, so that a violation *precedes* contact rather than coinciding with it; this is the geometric half of the case for $\tau_{\text{prox}} = 0.3$ m.

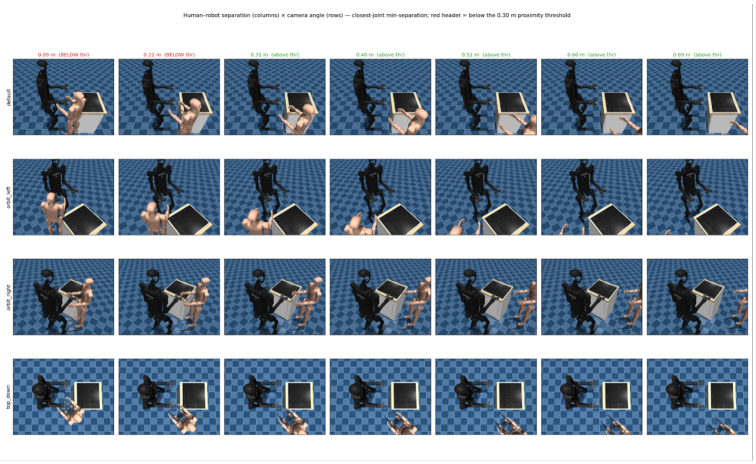


Figure 3.1: Contact-geometry calibration of τ_{prox}. The H1 robot (dark) and the G1 coworker (light) are rendered at seven controlled closest-joint separations (columns) from four camera angles (rows), with the `saucepan_to_hob` table between them. Red column headers are below the 0.30 m proximity threshold: at 0.09 and 0.21 m the body geometries visibly interpenetrate, whereas by 0.31 m (green) a clear gap has opened. This confirms that 0.3 m sits just above the physical contact band, so a proximity violation fires *before* contact, complementing the empirical-CDF knee of Table 3.1.

Empirical distribution. Because the proximity-violation rate at threshold τ is exactly the empirical CDF $P(\text{min-separation} < \tau)$, the threshold can be read off the separation distribution directly. On the G1 stage-2 baseline (3 seeds, 20 episodes, 30,000 steps) the distribution is bimodal, a near-contact mode at 0–0.2 m plus a far mid-range and patrol tail, with a knee at $\tau \approx 0.2$–0.3 m (Table 3.1); the knee is consistent across two independently trained policies, so it reflects scene contact geometry rather than a single controller.

We fix $\tau_{\text{prox}} = 0.3$ m: it sits at the knee, ∼0.2 m above the contact band, and captures the genuine near-contact population without absorbing the benign mid-range proximity that a 0.5 m (36 %) or 1.0 m (63 %) threshold would. Being purely geometric it is speed-independent, so the proximity-violation rate is comparable across policies, tasks and disruption bands, and it is held fixed across every experiment in this report. The ≈25 % near-contact rate this calibration

Table 3.1: Proximity-violation rate as a function of the threshold τ (the empirical CDF of min-separation on the G1 stage-2 baseline, 3 seeds × 20 episodes, 30,000 steps). The rate climbs steeply once τ crosses into the benign mid-range, which is why a 0.5 m or 1.0 m threshold would absorb a large benign population.

τ (m)	0.20	**0.30**	0.40	0.50	0.60	0.75	1.00
Violation rate	20.1 %	**25.3 %**	29.9 %	35.8 %	39.5 %	47.0 %	62.9 %

rollout induces is precisely the behaviour the constrained policy and the runtime filters are meant to reduce (the headline benchmark reports 0.296 for the same baseline under per-episode aggregation, Table 6.1).

3.10 Continuous Cost Signals

The wrapper computes two continuous, anticipatory cost signals from the ISO 15066 margins:

$$c_t^{SSM} = \max\left(0,\ 1 - \frac{m_t^{SSM}}{d_{\text{buffer}}}\right), \tag{3.2}$$

$$c_t^{PFL} = \max\Big(0,\ f_t^{\text{ratio}} - 0.8\Big), \tag{3.3}$$

$$c_t = \min\Big(1,\ \max(c_t^{SSM},\, c_t^{PFL})\Big), \tag{3.4}$$

where m_t^{SSM} is the instantaneous SSM-actual margin (positive when safe, negative when violating), $d_{\text{buffer}} = 0.3\,\text{m}$ is the anticipation buffer, and f_t^{ratio} is the force-ratio from Equation 2.2. The aggregator takes the worst-case across both criteria and saturates at $c_t = 1$.

Why continuous rather than binary. A binary indicator $c_t \in \{0, 1\}$ on violation supplies only *post-hoc* gradient signal. The policy learns that crossing the threshold is bad but receives no information about how close to the threshold it is approaching. The continuous formulation in Equations 3.2–3.3 gives a non-zero cost before violations occur: a SSM-actual margin of 0.15 m carries a cost of ∼0.5, and any margin beyond the 0.3 m buffer carries zero. We adopt it for this gradient richness. We note in advance, however, that the cost-signal *form* turns out *not* to be the operative variable: Experiment E3.1 (Section 6.2.2) shows that at a tight budget both the continuous and the binary forms collapse, because the binding quantity is the Lagrangian budget relative to the task's inherent cost, not the encoding of the cost. The continuous form is retained as the more principled design, not because it is shown to dominate the binary one.

The choice of $d_{\text{buffer}} = 0.3\,\text{m}$ matches the typical ISO 15066 SSM robot-stopping contribution at the H1's observed end-effector speeds; the 0.8 threshold in Equation 3.3 matches the 20 % headroom that ISO 15066 Annex A motivates for "approaching the limit".

The PFL contact-detection limitation. PFL flag computation reads MuJoCo's `data.ncon` for human–robot contact pairs. BiGym's runtime robot-attachment in MOJO currently produces `data.ncon = 0` for these pairs despite geometric overlap, so $c_t^{PFL} \equiv 0$ throughout the project. The wrapper exposes a `use_pfl` flag wired through every component; the flag can be flipped on once the contact-detection issue is resolved without architectural change. **All quantitative compliance claims in this report are qualified as SSM/proximity-only** (Section 7.2.1).

3.11 The coworker Disruption: Design and Rationale

The dominant deployment pattern for human-aware manipulation is *sustained co-working*: the human is in the workspace for the whole task, not for a one-shot intrusion, which is the situation a household robot in a kitchen actually faces. The original BiGym distribution had no human at all, and the legacy disruption types in the `safety_bigym` ecosystem all encoded a single intrusion event per episode, inadequate as a training distribution for this setting. We introduce Disruption.coworker as the primary distribution for both training and evaluation.

3.11.1 Design Goals

The COWORKER disruption was designed against three concrete targets:

1. *Realism.* The human's behaviour should resemble what a real co-worker would do in a shared kitchen: enter the workspace, approach the task area, occasionally reach toward the task object or toward the robot's own arm, and occasionally step away for a moment. Episodes should feel like minutes of collaboration, not seconds of intrusion.

2. *Sufficient disruption to expose unsafe behaviour.* If the human's reaches are rare or always far from the robot, a fast-but-careless policy is safe by luck and the benchmark cannot discriminate. The COWORKER band tightens the closest *pelvis-to-pelvis* approach (0.60–0.95 m) and the NEAR dwell (9–14 s); the reaching arm extends well inside the body-centre separation, so the closest geometry pair routinely enters the $\tau_{\text{prox}} = 0.3$ m zone. The resulting baseline `ep_proximity_violation_rate` of 0.296 (Section 5.8) is high enough to expose a 20 + % improvement as statistically significant.

3. *Compositional disruption modes.* A real co-worker exhibits at least three distinct behaviours over a session: staying put while reaching; walking in then staying; walking in, leaving briefly, and returning. The COWORKER sampler covers all three but weights them unequally, drawing the most demanding mode (patrol) the large majority of the time so that the training distribution concentrates on the hardest behaviour while still exercising the simpler ones.

3.11.2 Three Trajectory Modes

A COWORKER scenario samples one of three trajectory modes, with the patrol mode drawn ∼90 % of the time and the two simpler modes sharing the remaining ∼10 %:

stationary G1 spawns at NEAR distance and stays put. Arm reaches throughout the episode.

approach-loiter-depart G1 spawns far, walks in to NEAR, stays put. Arm reaches after arrival.

coworker-patrol G1 walks in to NEAR, then cyclically departs to an AWAY position (90°–270° offset, ∼2–3 m), loiters, returns to a *new* NEAR position (different angle, slightly different distance), and repeats. Arm reaches only while at NEAR.

The patrol mode is the most realistic operational pattern and is therefore the dominant mode at full strength; the two simpler modes are retained both as occasional variety in the headline distribution and as the bulk of the earlier curriculum stages (Section 4.6), which ramp up to the patrol-dominant mixture.

3.11.3 Arm State Machine and Reach Gate

Independently of trajectory, the G1's reaching arm runs a four-state cycle, EXTEND → HOLD → RETRACT → IDLE (fractions 0.15/0.20/0.15/0.50 of a period $T \in [1.3, 2.2]$ s in the train band), sampling per cycle whether to reach for the robot's end-effector or the task object (`target_mix_p_ee`), with the target re-resolved each step so the reach tracks a moving robot. A *reach gate* suppresses the IK during patrol's AWAY phase. Because the arm idles half of each cycle and patrol periodically vacates the NEAR zone, the task remains completable throughout: the success-rate axis rewards progress in those windows over a trivial "freeze until the human leaves" strategy.

3.11.4 Train and Eval Parameter Spaces

Five COWORKER axes are exposed as continuous `ParameterSpace` ranges, with train and eval distributions configured separately for the OOD generalisation experiment (E5.2):

Table 3.2: The COWORKER train and eval `ParameterSpace`. Train is moderately tight, deliberately set to produce a high baseline violation rate (so a 20 + % safety improvement is statistically detectable). Eval is wider, capturing the generalisation question. The 0.60 m closest-approach floor is the smallest body distance that keeps the G1 mocap-pelvis collision capsule clear of the H1 pelvis collision capsule; tighter values generate spurious contact forces and are pathological.

Axis	Train (`coworker_train`)	Eval (`coworker_eval`)
Closest-approach distance	0.60–0.95 m	0.6–1.8 m
Reach period T	1.3–2.2 s	3.0–9.0 s
P(reach EE)	0.45–0.72	0.1–0.9
Near-loiter dwell time	9.0–14.0 s	4–16 s
Walk speed	1.0–1.4 m/s	0.6–2.2 m/s

3.12 The Task Suite: Three Tasks, Two Co-location Regimes

The evaluation uses three BIGYM tasks. `saucepan_to_hob` is the primary deep-dive task; `dishwasher_close` and `drawers_open_all` complete the suite. The saucepan choice is motivated by:

1. **Long horizon for violation opportunity.** `saucepan_to_hob` is a pick-and-place task with a vertical clearance constraint; episodes last 30–50 s, several reach cycles of the COWORKER arm. Shorter reach-target tasks end before the coworker has cycled enough to expose representative unsafe behaviour.

2. **Workspace overlap with the coworker.** The saucepan pickup and the hob target are both within the H1's primary reach envelope, which is also where the G1 reaches when the reach target is the task object. Episode trajectories therefore pass through the same volume the coworker occupies, exposing real SSM violations rather than incidental closeness.

3. **BiGym-RL competence.** Of the four candidate BIGYM tasks we screened with ACT baselines, `saucepan_to_hob` combines the longest horizon with the greatest workspace overlap, and the unconstrained CQN-AS agent (Section 5.8) reaches a strong `success_rate` on it after the curriculum (Section 4.6). The unconstrained baseline is a non-trivial reference: any safety improvement we report is against a competent task policy, not a policy that was failing the task anyway. CQN-AS task-performance on BIGYM is documented in [4].

Two co-location regimes. Although all three tasks run under the *same* COWORKER disruption distribution, the task geometry and the policy's motion pattern make the resulting human–robot co-location qualitatively different, and this difference turns out to be the variable that governs which safety mechanism works (Chapter 6). On `saucepan_to_hob` the coworker's reach destination coincides with the robot's working volume for most of the episode: the co-location is *persistent*, with the closest geometry pair inside the interaction envelope nearly continuously. On `dishwasher_close` and `drawers_open_all` the close encounters arrive in *bursts*, the coworker's patrol cycle and the appliance-facing geometry leave genuinely-safe windows between approaches: co-location is *intermittent*. We do not assert this distinction from task descriptions alone; it is *measured* in Section 6.4 via the activity rate of a separation-triggered runtime filter, which is the operational definition used throughout (persistent ≈ the filter would be active most of the time; intermittent ≈ it would not). Figure 3.2 shows the three settings at a close-approach instant.

3.13 Mock BodySLAM++: Calibrated Perception Noise

Training-time runs use ground-truth human position; deployment-time runs use a noisy estimate. To characterise the perception sim-to-real gap, we expose a configurable `BodySLAMWrapper` with three modes:

`off` No human-state channel in the observation.

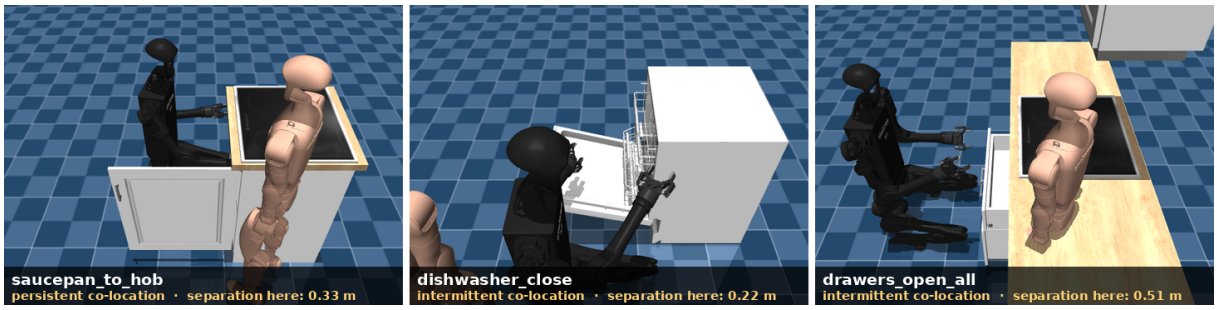


Figure 3.2: The three-task suite, photographed mid-episode from live rollouts of each task's unconstrained baseline policy under the COWORKER disruption: the Unitree H1 manipulator (black) shares its working volume with the Unitree G1 human-coworker stand-in (light) in the BIGYM kitchen. Left: `saucepan_to_hob`, whose pick-and-place trajectory and the coworker's reach destination coincide, so co-location is *persistent*. Centre and right: `dishwasher_close` and `drawers_open_all`, where close encounters arrive in bursts separated by genuinely-safe windows (*intermittent* co-location). The annotated separation is the environment's ground-truth closest-pair distance at the pictured instant; the same coworker distribution drives all three tasks, so the regime is a property of the task–coworker pair, not of the disruption.

`oracle` Ground-truth human position injected as `human_pos_estimate` (6D: position + occluded flag + staleness counter + confidence).

`noisy` The full mock-BodySLAM++ pipeline, calibrated against the published noise characteristics [25].

The noisy pipeline applies, in order:

1. Ornstein-Uhlenbeck temporally-correlated position noise ($\alpha = 0.9$, $\sigma = 0.05\,\text{m}$, stationary std $\approx 0.115\,\text{m}$), calibrated against BodySLAM++'s reported positional accuracy [25].
2. 3-step latency buffer ($\approx 60\,\text{ms}$ at $50\,\text{Hz}$ control, matching the 15+ FPS perception clock typical of monocular visual-inertial pose estimation).
3. Stochastic tracking dropout ($p = 0.02$/step) with staleness counter; OU state continues to update during dropout, so recovery is smooth rather than discontinuous.
4. Optional ray-cast occlusion from the H1's head camera (opt-in).

Demo-replay path. Cached BIGYM demonstrations [6] were recorded *without* a coworker; their `human_pos_estimate` would be missing if read live. The wrapper supports a `mode=demo_replay` path that synthesises `human_pos_estimate` from an `AMASSDemoPositionProvider` [80] playing back a real motion clip, so a behaviour-cloning pretraining phase sees realistic human state at training time without a train→eval distribution shift on the channel.

3.14 Demonstrations and the Demo-Conversion Pipeline

CQN-AS [4] is a demo-driven RL agent (num_demos > 0 assumed at construction); demo-free runs on long-horizon manipulation never discover the sparse reward and collapse to workspace evacuation. The demo path uses the raw, non-safety-wrapped BIGYM environment to match `DemoStore`'s class-name lookup, fetches 10–36 cached demonstrations, and converts each to `ExtendedTimeSteps` carrying the synthesised `human_pos_estimate` channel and a safe-side placeholder $c_t = 0$ (the cached demos contain no live human; the curriculum of Section 4.6 makes this benign in practice). The most consequential of the action-normalisation pitfalls fixed during integration (Section 4.9): CQN-AS's normalisation must use *demo-derived* action statistics so the live and demo distributions share a binning.

4 Methods: The Two Arms of the Hybrid Safety Critic

4.1 Overview: Two Arms, One Open Question

Chapter 3 introduced `safety_bigym` as the benchmark and Section 3.1 listed six concrete properties the benchmark is designed to test. The unconstrained CQN-AS [4] baseline that ships with BiGym fails almost all six: it has no representation of human position, no SSM margin, no constraint that would distinguish a task-completing trajectory from one that endangers the coworker, and Section 5.8 will show that it registers proximity violations on ≈30 % of steps (mean separation only 0.13 m, with the worst approaches within a few millimetres), far above what any plausible co-working deployment would accept.

This chapter presents **our solution to that gap**: the Hybrid Safety Critic, a two-arm safety architecture that turns the unconstrained imitation-or-RL policy into an ISO 15066-aware collaborator. The architecture has two mechanisms with deliberately different roles:

1. A **training-time mechanism** that *internalises* safety into the policy via a value-based Lagrangian formulation of the continuous cost signal. At deployment, the policy is used as-is; the safety knowledge is baked into the learned Q-functions (Section 4.4).
2. A **deployment-time mechanism**: a runtime *filter slot* that intercepts the policy's proposed actions. The slot is mechanism-agnostic, and we specify and evaluate the complete reactive taxonomy through it: an offline-trained, frozen safety value function (SVF) used as a veto, the original design (Section 4.2); two closed-form alternatives, a graded ISO 15066 speed-scaling rule and a model-based geometric dodge (Section 4.3); and the *critic-gated speed-scaling* combination, the SVF deciding *when* to intervene and the graded scaler deciding *how* (Section 4.3.3), which is the Hybrid Safety Critic's runtime arm.

Both mechanisms share a network shape and an observation interface, which keeps the two safety critics directly comparable and leaves SVF reuse available as an option, though the constrained policy's cost critic is in fact cold-started (Section 4.4.9); they differ in training objective and operational role. The full pipeline at deployment is sketched in Figure 4.1. One question is deliberately left open in this chapter: *which arm should carry the safety burden in a given deployment*. That is the empirical question the thesis answers, and the chapters that follow answer it with the regime map rather than presupposing either arm.

4.2 The Offline Safety Value Function

4.2.1 Dataset and Labelling

The SVF is trained offline on a dataset of transitions collected by rolling out CQN-AS policy snapshots from the base-policy curriculum (Section 4.6) in the COWORKER distribution under the `noisy` observation mode. Earlier iterations of the critic mixed random-policy and demonstration transitions; the current critic is collected from policy snapshots alone. The reasoning is that the regions a deployed policy actually visits are precisely the regions the filter must judge accurately, and snapshot rollouts sample exactly that on-policy distribution, including the imitation-learning artefacts a competent-but-imperfect policy exhibits under sustained co-working disruption. The filter is *noisy-native*: it is trained and evaluated on `noisy` human-state estimates, and on ground-truth (`oracle`) input its Q-estimates collapse and it intervenes almost everywhere, so all filtered evaluations are reported on `noisy` (Section 5.3). The exact transition count for the current critic

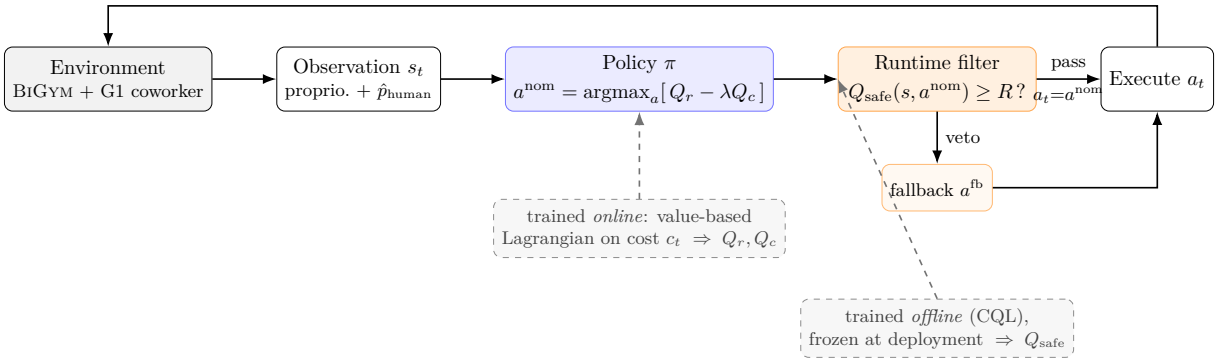


Figure 4.1: The Hybrid Safety Critic at deployment: a two-arm architecture. At each step the Lagrangian-trained policy proposes the nominal action $a^{\text{nom}} = \text{argmax}_a [Q_r(s,a) - \lambda Q_c(s,a)]$; a runtime filter slot passes it through when its safety check holds and otherwise modifies or replaces it (veto, dodge, graded speed-scaling, or critic-gated speed-scaling, Sections 4.2–4.3.3). The two arms have deliberately different jobs: the constrained policy internalises safety *proactively* during training, while the filter is a *reactive* runtime mechanism. Chapter 6 shows that which arm carries the safety burden is governed by the co-location regime: under persistent co-location only the policy reduces exposure gracefully (Table 6.1), while under intermittent co-location the critic-gated filter is the working arm (Section 6.3.4).

is 37,883, of which 16.8 % are labelled unsafe ($r^{\text{safe}} = 0$) at the $\tau = 0.3$ m proximity threshold; the full figures are in the repository's filter-training write-up.

Each transition is labelled with the binary safety reward

$$r^{\text{safe}} \;=\; 1 - \mathbb{1}[\texttt{proximity\_violation}], \tag{4.1}$$

using the geometric proximity-violation flavour of Section 3.8 at the calibrated threshold $\tau_{\text{prox}} = 0.3$ m (the calibration is derived in Section 3.9). We label on `proximity_violation` rather than `ssm_violation_actual` for two reasons. First, the conservative-ISO SSM flag is saturated: it fires on ∼93 % of random rollouts and remains high even for competent policies because it assumes the worst-case human velocity, so it carries almost no gradient for a value function to learn. Second, and by design, labelling on geometry keeps a clean separation of concerns between the two safety mechanisms. The runtime filter is made responsible for the *geometric* danger of being too close, while the velocity-dependent SSM response is left to the cost signal and the constrained policy (Section 4.4). Coupling the filter's training target to instantaneous velocity would make its label depend on the very action it is meant to veto, conflating "how dangerous is this configuration" with "how fast is the robot moving", the latter being a decision the policy, not the filter, should own.

Because the violating class is the minority, a `WeightedRandomSampler` over-samples it to a 1:1 ratio of safe to violating transitions per batch, so the critic is not dominated by the safe majority.

A second, task-agnostic critic for the intermittent-co-location tasks. The same pipeline produces the critic used on `dishwasher_close` and `drawers_open_all`: a single shared SVF trained once on 52,794 transitions collected from curriculum-policy rollouts on both tasks under COWORKER disruption, labelled by geometric *near-contact* (min-separation < 0.10 m). The observation specification (proprioception + human-position estimate + action) is identical across the two tasks, so one critic serves both, evidence that the learned safety signal is task-agnostic at matching embodiment. The tighter 0.10 m label (versus the saucepan critic's 0.30 m) makes this critic a *near-contact* detector rather than a proximity detector; an ablation with looser, earlier-warning labels (0.20 / 0.30 m) confirms the selective near-contact label is the better gate (Section 6.3.5).

4.2.2 Architecture and Training

The safety critic is a 3-layer MLP with hidden sizes $[256, 256, 256]$ and ReLU activations. The input is the concatenation of robot proprioception, the mock-BodySLAM++ 6D human-position estimate, and the proposed action. The output is bounded to $[0, q_{\max}]$ with $q_{\max} = 1/(1-\gamma)$ via a scaled sigmoid, which *mathematically* prevents Bellman overestimation: Q cannot exceed the maximum possible discounted return of a safe trajectory.

Training combines a Bellman residual with the CQL [55] regulariser:

$$\mathcal{L} \;=\; \mathcal{L}_{\text{Bellman}} \;+\; \alpha_{\text{CQL}} \cdot \left(\mathbb{E}_{a\sim\text{uniform}}[Q(s,a)] - \mathbb{E}_{a\sim\mathcal{D}}[Q(s,a)]\right), \tag{4.2}$$

with $\alpha_{\text{CQL}} = 5.0$ as the headline value, Polyak target update $\tau = 0.005$. Full training hyperparameters in Appendix A.1.

4.2.3 No-Pixel Critic: Justification

The filter does not consume RGB, for three reasons: *compute* (a pure-MLP critic costs ∼0.3 ms per forward against a ∼20 ms control budget, versus a substantial CNN-encoder fraction); *robustness* (no lighting/texture domain shift, only the upstream pose estimator's calibration [25]); and *compositionality* (a no-pixel filter wraps any policy whose observation includes proprioception, the human-position estimate, and the proposed action).

4.2.4 Runtime Wrapper

At deployment, the `SafetyFilterWrapper` intercepts the proposed action and compares the safety value against a threshold R:

$$a_t^{\text{exec}} \;=\; \begin{cases} a_t^{\text{nom}} & \text{if } Q_{\text{safe}}(s_t, a_t^{\text{nom}}) \geq R, \\ a_t^{\text{fb}} & \text{otherwise.} \end{cases} \tag{4.3}$$

R is calibrated by sweep over a held-out validation set (Experiment E2.3, Section 6.2.3); the headline operating point for the G1 critic on the COWORKER distribution is $R = 2.25$. The intended role of the veto at this operating point is an ISO 15066 SSM *velocity backstop* rather than a proximity reducer: much of the residual proximity is the human walking up to an otherwise stationary robot, which no veto of robot actions can prevent; Section 6.2.3 quantifies the resulting limits.

The veto's fallback is zero-velocity braking on the floating base and on every joint velocity actuator. This can produce abrupt motion mid-trajectory; proportional damping is the natural upgrade and is deferred to future work (Section 8.1.3).

4.3 Closed-Form Runtime Filters: Speed-Scaling and Geometric Dodge

The filter slot of Figure 4.1 is mechanism-agnostic: anything that maps the state and the proposed action (s_t, a_t^{nom}) to an executed action can occupy it. The learned SVF veto above is its learned instantiation. Two *closed-form* mechanisms, specified here and evaluated alongside the veto in Chapter 6, complete the reactive taxonomy: stop (veto), slow (speed-scaling), and move away (dodge). Both consume the same $\hat{p}^{\text{human}}$ estimate as the SVF and contain no learned component.

4.3.1 Graded ISO 15066 Speed-Scaling

ISO 15066 SSM's intent is graded: the robot should *slow* as separation shrinks, not merely stop at a threshold. The speed-scaling filter implements this directly. At each step a scale factor is computed from the closest-pair human–robot separation sep_t,

$$\sigma_t \;=\; \text{clip}\left(\frac{\text{sep}_t - d_{\text{stop}}}{d_{\text{slow}} - d_{\text{stop}}},\; 0,\; 1\right), \qquad a_t^{\text{exec}} \;=\; q_t + \sigma_t\,(a_t^{\text{nom}} - q_t), \tag{4.4}$$

where q_t is the current joint position for the absolute-mode arm dimensions (so the per-step *motion* from the current pose is scaled) and zero for the delta-mode floating base (so the commanded increment is scaled directly). Above d_{slow} the filter is the identity ($\sigma_t = 1$, the action is unchanged); at or below $d_{\text{stop}} = 0.15\,\text{m}$ the robot holds position ($\sigma_t = 0$); in between, speed falls linearly with separation. The slow radius d_{slow} is the single operating-point knob, swept over $[0.25, 0.60]\,\text{m}$ with the headline at $0.40\,\text{m}$; an intervention is any step with $\sigma_t < 1$. Unlike the veto, this filter preserves the *direction* of the policy's motion and modulates only its magnitude, which is why it can maintain the graded velocity margin the standard prescribes where a binary stop-or-go veto cannot (Section 6.2.4).

4.3.2 Model-Based Geometric Dodge

The second closed-form mechanism is a geometric control-barrier-style fallback: when the closest-pair separation falls below a target d_{target}, the filter adds a correction step directed *away* from the human, proportional to the violation depth ($d_{\text{target}} - \text{sep}_t$) and capped per step. Two variants differ in which part of the robot dodges: the *base-dodge* pushes the floating base along the escape direction and leaves the arm untouched, while the *end-effector retract* ("flinch") keeps the base planted and retracts the end-effector along the escape direction via a damped-Jacobian step. d_{target} is swept over $[0.30, 0.55]\,\text{m}$. These variants exist to test whether a *dodging* reactive response escapes the limits of the *stopping* one; Section 6.2.5 shows it does not (it trades the veto's freeze for a flee). The veto and the dodges are reported as taxonomy points (Table 6.2), with operating-point values for all mechanisms in Appendix A.3.

4.3.3 Critic-Gated Speed-Scaling: The Hybrid Safety Critic's Runtime Arm

The unconditional speed-scaler of Section 4.3.1 pays its task cost at *every* close approach, including those where the situation is genuinely safe (the human is close but stationary, or moving away, or the robot is barely moving). The learned SVF of Section 4.2 knows the difference; the binary veto is simply the wrong way to act on that knowledge. The *critic-gated speed-scaling* filter splits the decision:

$$a_t^{\text{exec}} = \begin{cases} a_t^{\text{nom}} & \text{if } Q_{\text{safe}}(s_t, a_t^{\text{nom}}) \geq R \quad \text{(critic says safe: full speed)}, \\ q_t + \sigma_t \, (a_t^{\text{nom}} - q_t) & \text{otherwise} \quad \text{(critic flags risk: graded slow-down, Eq. 4.4)}. \end{cases} \tag{4.5}$$

The critic decides *when* to intervene; the speed-scaler decides *how*. Both halves matter. Replacing the graded slow-down with a hard veto breaks the action coherence of a chunked policy (Section 6.3.2); removing the gate makes the scaler fire on every close approach and pay the full unconditional task cost. The gate threshold R is the single operating-point dial: $R \to 0$ recovers the unfiltered policy, $R \to \infty$ recovers unconditional scaling, and intermediate values trace a frontier between them (Section 6.3.4). Whether that frontier *bends* toward a high-success/low-violation corner or merely *slides* between its two endpoints is precisely the regime-dependent question answered in Section 6.4.

4.4 The Value-Based Lagrangian Constrained Policy

4.4.1 Why Value-Based

CQN-AS [4] is *critic-only*: the policy is implicit, $\pi(s) = \text{argmax}_{a \in \mathcal{A}_{\text{bin}}} Q(s, a)$, over a coarse-to-fine discretisation of the joint action space. The classical actor-critic Lagrangian formulation [26, 27] updates a policy network π_θ to maximise $\mathbb{E}[Q_r(s, a)] - \lambda \mathbb{E}[Q_c(s, a)]$, which presumes that π_θ exists as a separately-trained network. There is no π_θ in CQN-AS. The Lagrangian must be re-expressed in value-based form.

The starting point is the constrained objective

$$\max_\pi J_r(\pi) \quad \text{s.t.} \quad J_c(\pi) \leq d, \qquad J_r(\pi) = \mathbb{E}_\pi \left[\sum_t \gamma^t r_t \right], \quad J_c(\pi) = \mathbb{E}_\pi \left[\sum_t \gamma^t c_t \right]. \tag{4.6}$$

For any fixed multiplier $\lambda \geq 0$, the Lagrangian relaxation is $\mathcal{L}(\pi, \lambda) = J_r(\pi) - \lambda(J_c(\pi) - d)$. The λd term is constant with respect to action choice, so a critic-only policy can implement the same trade-off at decision time by learning separate action-values

$$Q_r(s,a) \approx \mathbb{E}\left[\sum_t \gamma^t r_t \mid s_0 = s, a_0 = a\right], \qquad Q_c(s,a) \approx \mathbb{E}\left[\sum_t \gamma^t c_t \mid s_0 = s, a_0 = a\right],$$

and selecting the action that maximises $Q_r - \lambda Q_c$. This is the key reproducibility point: *the policy improvement step is changed, not the Bellman target of the reward critic.* The reward and cost critics can therefore be trained on stationary targets, while λ acts only as a deployment-time or training-time selection parameter.

Three options were considered, in increasing principledness; the choice is itself an empirical comparison (Experiment E3.1).

4.4.2 Option A-value: Shaped-Reward Single Critic

The simplest re-formulation folds the cost into a modified reward and trains a single critic:

$$r_t^{\text{mod}} \;=\; r_t^{\text{task,shaped}} \;-\; \lambda \cdot c_t, \qquad Q(s,a) \approx \mathbb{E}\left[\sum_t \gamma^t r_t^{\text{mod}}\right], \tag{4.7}$$

with action selection $a^* = \text{argmax}_a Q(s,a)$. Its known pathology: the regression target is *non-stationary* in λ, which is itself non-stationary in the rolling cost, so the critic chases a moving target. Used here as a prototype baseline only.

4.4.3 Option B-value-mean: Dual Q-Functions on Mean Cost (Headline)

Decouple the reward and cost critics:

$$Q_r(s,a) \approx \mathbb{E}\left[\sum_t \gamma^t r_t^{\text{task,shaped}}\right], \tag{4.8}$$

$$Q_c(s,a) \approx \mathbb{E}\left[\sum_t \gamma^t c_t\right], \tag{4.9}$$

each regressing toward its own *stationary* target. The action selection combines them at decision time:

$$a^* \;=\; \underset{a \in \mathcal{A}_{\text{bin}}}{\text{argmax}} \Big[\, Q_r(s,a) \;-\; \lambda \cdot Q_c(s,a) \,\Big]. \tag{4.10}$$

λ now enters only at selection, not at regression. The cost critic shares its network shape with the offline SVF, so SVF reuse is available, though here the cost critic is cold-started (Section 4.4.9). This is the **headline architecture** for this work.

4.4.4 Option B-value-CVaR: Distributional Cost Critic (future work)

A natural extension replaces the mean-cost critic with a distributional one: Q_c is replaced by either a Gaussian parametrisation (μ_c, σ_c) [1] or a quantile head $\{\theta_k\}_{k=1}^K$ [48], with action selection

$$a^* \;=\; \underset{a \in \mathcal{A}_{\text{bin}}}{\text{argmax}} \Big[\, Q_r(s,a) \;-\; \lambda \cdot \text{CVaR}_\alpha(Z_c \mid s,a) \,\Big], \tag{4.11}$$

with $\alpha \in \{0.95, 0.99\}$. This would align training with the CVaR evaluation metric and answer the standard critique that mean-cost methods bound only $\mathbb{E}[Z_c]$; the cost head is already C51-shaped, but calibrating a rolling-CVaR target and re-verifying the support invariant (Section 4.4.5) for the cost distribution is non-trivial engineering. **We adopt B-value-mean as the headline and defer explicit-**CVaR **training to future work (Section 8.1.3)**, evaluating the mean-policy's *achieved* CVaR as a diagnostic (Section 6.2.9).

4.4.5 The C51 Value-Support Invariant

CQN-AS uses a C51 distributional reward critic with fixed value support $[v_{\min}, v_{\max}]$ [39]. The Bellman target $T_z = r + \gamma \cdot \text{support}$ is clamped to this support at each update. Adding a dense shaped reward term such as Equation 3.1 introduces a discounted return whose value can fall outside the support; once the target is clamped, the critic loses the ability to distinguish values in the saturating region, and the gradient that should pull the end-effector back toward the task is clipped away.

Proposition 4.1 (Support-bounding invariant)**.** *Let a C51 distributional critic have value support $[v_{\min}, v_{\max}]$, discount γ, and an additive shaped reward $r_t^{\text{shaped}} \in [-\beta \cdot c_{\text{ws}}, 0]$ with $\beta, c_{\text{ws}} > 0$. The shaped-reward contribution to the discounted return is bounded by $\beta \cdot c_{\text{ws}}/(1-\gamma)$ in magnitude. The Bellman target clamp is informative (i.e., the support is not saturated) only if*

$$\beta \cdot c_{\text{ws}} \,/\, (1-\gamma) \;\le\; |v_{\min}|. \tag{4.12}$$

Proof. The geometric-series sum of the per-step shaping under the worst case gives $\sum_{t=0}^{\infty} \gamma^t \beta c_{\text{ws}} = \beta c_{\text{ws}}/(1-\gamma)$, and the C51 target clamp clips below $v_{\min}$. □ □

Consequence. The original workspace shaping ($\beta = 0.2$, no cap on per-step excess distance) gave a worst-case discounted return of $-0.2/0.01 = -20$ against the default CQN-AS support of $[-2, +2]$, saturating every state outside $\sim$0.5 m of the task. Training diverged: `ep_reward` fell monotonically from -78 to -775 over 50k frames as the policy parked further from the workspace and the support flattened. The fix is two-sided: cap the per-step penalty ($c_{\text{ws}} = 1.0$ m), reduce β to 0.05, and widen the support to $[v_{\min}, v_{\max}] = [-6, +2]$. With these settings the invariant holds ($0.05 \cdot 1.0/0.01 = 5 \le 6$) and training proceeds normally; the next-attempt `ep_reward` reaches $\sim +1.8$ on stage 1 of the curriculum (Section 4.6).

The engineering lesson generalises beyond `safety_bigym`. It generalises to any distributional critic + dense shaping setting and applies as a checking principle independent of `safety_bigym`.

4.4.6 Per-Env-Step Cost Backup

CQN-AS executes *action sequences* of length K between policy queries (coarse-to-fine refinement). A natural but subtly wrong design is to backup the cost critic on the chunk-averaged cost $\bar{c} = (1/K)\sum_k c_{t+k}$. The policy can satisfy a mean-over-chunk budget while spiking violations *inside* a chunk. The cost critic must be backed up on *every* env-step's c_t. Our training pipeline preserves c_t through the replay buffer at per-env-step granularity (Section 4.9).

4.4.7 PID Lagrange Multiplier Update

The Lagrange multiplier is updated by a PID controller [27] on the running constraint violation. Let m_t be the rolling per-episode cost mean and d the cost budget; the update is

$$e_t = m_t - d, \tag{4.13}$$

$$\lambda_{t+1} = \text{clip}(\lambda_t + K_I \cdot e_t + K_P \cdot e_t + K_D \cdot (e_t - e_{t-1}),\; 0,\; \lambda_{\max})\,. \tag{4.14}$$

PID is preferred over gradient ascent ($\lambda \leftarrow \lambda + \eta \cdot e_t$) because the integral term provides faster convergence to the constraint boundary while the derivative term damps oscillation [27]. We use $K_I = 10^{-3}$, $K_P = 10^{-2}$, $K_D = 0$, and $\lambda_{\max} = 100$. The clamp prevents the policy from collapsing into a fully-frozen "safe-and-useless" attractor when the cost budget is briefly violated.

PID auto-tuning is seed-unstable at the feasibility boundary; a fixed multiplier is the fix. The task's inherent mean cost is $\approx$0.25–0.30, so any budget $d \le 0.2$ is infeasible (abandon-the-task is the only policy under it) and $d = 0.5$ is slack ($\lambda \to 0$); the single viable

budget, $d = 0.3$, sits *at* the inherent cost, where the dual variable is decided by each seed's stochastic cost trajectory rather than by design. Across three seeds at $d = 0.3$, λ converged to 0.000, 0.267, and 3.855, unconstrained, graceful, and windup-collapse respectively; the evidence is presented with the budget sweep in Section 6.2.2 (Figure 6.1). The fix is to remove the auto-tuning: freezing λ (zero PID gains, Q_c still trained and used in dual-action selection) makes all three seeds behave consistently and turns λ into a clean proximity–success knob; we sweep it and adopt $\lambda = 0.1$ as the headline ($\lambda = 0.27$ over-constrains, $\lambda = 0$ recovers the baseline; Figure 6.2). The lesson generalises: when the only feasible budget coincides with the task's inherent cost, the dual variable carries too little signal to auto-tune, and a fixed or scheduled multiplier is preferable. The PID path remains in the codebase as the documented negative.

4.4.8 Filter During Training (optional)

The runtime safety filter of Section 4.2 can also be applied at training time, interposed between the proposed action and the environment, vetoing catastrophically unsafe proposals with the fallback [24, 54]. We report results without the training-time filter as the default; the filter-during-training ablation is deferred to future work (Section 8.1.3).

4.4.9 Cost-Critic Initialisation and Policy Warm-Start

Two initialisation choices matter here, and they point in opposite directions. The actor, the reward critic and the shared visual encoder are *warm-started* from the unconstrained base policy: the stage-1 snapshot of the curriculum (Section 4.6) is loaded so that constrained training begins from a task-competent policy rather than from scratch. The cost critic Q_c, by contrast, is *cold-started* (freshly initialised).

This is deliberate. Although Q_c shares its network shape with the offline SVF, the two carry opposite sign conventions (Q_c high means dangerous, Q_{safe} high means safe) and different learning targets (offline CQL value of a binary proximity label vs. online expected value of the continuous cost under the current policy); seeding Q_c from the SVF would initialise it on a surface the dual-Q selection step would have to unlearn. A `warm_start_from_svf` path (with the required sign-flip) remains in the codebase, unused in the reported runs. One loader subtlety matters for reproduction: the unconstrained base snapshot contains no cost-critic keys, so the loader guards them explicitly, task weights load, Q_c stays freshly initialised.

4.5 External Baseline: Worst-Case SAC

To position the value-based Lagrangian against the published safe-RL state of the art rather than only against our own ablations, we faithfully reimplement WCSAC [1, 47], the canonical distributional safe actor-critic, on the same 76-DOF tasks (`agent=wcsac` on the same training stack). WCSAC maintains a Gaussian distributional cost critic and constrains CVaR_α of the discounted cost return ($\alpha = 0.9$) to a budget d via an adaptively-tuned multiplier. Two faithfulness points matter. First, WCSAC is actor-critic, so it cannot load the critic-only CQN-AS warm-start snapshots; it trains *from scratch* (num_demos = 0, 150k frames), which is the honest configuration and is reported as such, the comparison measures the published method as a practitioner could actually deploy it on these tasks, not a demo-warm-started variant that does not exist in the literature. Second, because it trains demo-free, its action de-normalisation must use identity statistics; the evaluation harness originally forced demo-derived statistics onto every snapshot, a train/deploy mismatch that produced plausible-but-wrong numbers and was fixed and cross-validated before any WCSAC result was recorded (the same class of silent confound as Section 4.10). Results in Section 6.3.1.

4.6 The Base-Policy Curriculum

Direct training on the full COWORKER disruption distribution from a randomly-initialised CQN-AS agent fails. The empirical signature is a base policy that learns to evacuate the workspace

within 5–10k frames and never attempts the task; `ep_reward` stays at the floor and `success_rate` stays at 0. The cause is a *distribution-shift cascade*: cached BiGym demonstrations contain no coworker, so behaviour-cloning teaches "go to the task, ignore everything"; at runtime, the live coworker physically blocks the task path, producing off-distribution states; the C51 critic has no reliable signal in those states and the policy drifts toward whatever physics + workspace shaping push it toward, which is away from the human and toward the workspace boundary.

The fix is a three-stage curriculum that resumes from each prior stage's snapshot [41]. The environment is *stateless* with respect to training step, so the curriculum is implemented as three sequential training runs with snapshot resume. The `BodySLAMWrapper` channel and the obs width are preserved across stages by keeping a coworker in the scene at all stages (just distant in stage 0), so the architecture and replay buffer carry over cleanly.

The three stages are: *stage 0*, `coworker_idle` (G1 present but ∼3 m off and never reaching, doubling as a "can the agent learn the task at all?" sanity check; 30k frames); *stage 1*, `coworker_easy` (moderate distance, occasional reaches; 30k); *stage 2*, the full `coworker_train` band of Table 3.2 (40k). Stages 0–1 saturate well before their budgets; the largest budget is reserved for the hardest distribution.

Why a curriculum (against alternatives). Domain randomisation [70] on the human velocity from step 0 was considered, but it addresses dynamics-estimate error, not the actual failure, never discovering the sparse reward; curriculum learning addresses the discovery problem [41, 42]. The empirical anchor: stages 0–1 complete with `ep_reward` $\sim +1.8$ (positive, near the $+2$ support ceiling, the unambiguous un-saturated-support signal; contrast the failed direct run's $-78 \to -775$ divergence), and stage 2 provides both the unconstrained baseline on the real distribution and the Lagrangian-stage warm-start.

4.7 Algorithm

4.8 Software Architecture and Metric Provenance

With the architecture and training loop specified above, the remaining design decisions are matters of engineering; the following sections document the ones that materially affected the reported results. The codebase is partitioned: `safety_bigym` (env layer + ISO 15066 wrappers + perception wrapper + scenario sampler), `safety_bigym/filters` (offline SVF: dataset, critic, runtime wrapper, sweeps), `safety_bigym/agents/cqn_as` (vendored CQN-AS [4] + adapter + Lagrangian agent), and a top-level training entrypoint `train_cqn_as.py`. Configuration is managed via Hydra; W&B is the experiment tracker; per-run JSONL metric dumps provide W&B-independent reproducibility. Hardware target: a single NVIDIA A100 40 GB for training, with CPU-only smoke gates for every script.

Per-step c_t flows from the cost-signal module through the `cost` field of the env adapter's `ExtendedTimeStep` and the replay buffer into the cost-critic Bellman target at *per-env-step* granularity (not per K-step chunk, Section 4.4.6). The snapshot harness (`benchmark_policy.py`, Section 3.7) takes any policy snapshot and produces a CSV row per evaluation cell with the full safety-metric schema; it is the canonical source for every numerical entry in this report's results tables.

4.9 CQN-AS Vendor Integration

The upstream CQN-AS [4] reference implementation was vendored at commit `8cf806e` and adapted to the `safety_bigym` setting. Eight non-trivial bugs and distribution mismatches were resolved during integration; the most informative are summarised here.

Demo distribution gap. Cached BiGym demos contain no human, so demo-derived `human_pos_estimate` would be empty. The wrapper's `demo_replay` mode injects an

Require: Task reward $r_t^{\text{task,shaped}}$ (workspace shaping per Equation 3.1), cost signal c_t per Equation 3.4, cost budget d, PID gains (K_P, K_I, K_D), optional offline filter Q_{safe} + threshold R

Warm-start the actor, reward critic Q_{r,ϕ_r} and encoder from the stage-1 curriculum snapshot; initialise the cost critic Q_{c,ϕ_c} from scratch (cold start)
Initialise replay buffer, demo replay buffer; load num_demos demonstrations
$\lambda \leftarrow \lambda_0$ ▷ headline: fixed $\lambda_0 = 0.1$ (PID gains zero); PID-tuned only in the seed-instability ablation
for each environment step t **do**
    $a_t^{\text{nom}} \leftarrow \text{argmax}_{a \in \mathcal{A}_{\text{bin}}}[Q_r(s_t, a) - \lambda \cdot Q_c(s_t, a)]$
    if using training-time filter **and** $Q_{\text{safe}}(s_t, a_t^{\text{nom}}) < R$ **then**
        $a_t \leftarrow a_t^{\text{fb}}$ ▷ fallback
    else
        $a_t \leftarrow a_t^{\text{nom}}$
    end if
    Step env; store $(s_t, a_t, r^{\text{task,shaped}}, c, s_{t+1})$ in replay
    Sample batch (demos + replay)
    Update ϕ_r: C51 Bellman regression on $r^{\text{task,shaped}}$ per-env-step
    Update ϕ_c: Bellman regression on c per-env-step
    if episode end **and** PID-tuning λ (ablation only) **then**
        $m \leftarrow$ rolling mean cost over recent episodes; $e \leftarrow m - d$
        $\lambda \leftarrow \text{clip}(\lambda + K_I e + K_P e + K_D(e - e_{\text{prev}}),\ 0,\ \lambda_{\max})$
    end if
    (headline runs keep $\lambda \equiv \lambda_0$; see Section 4.4.7)
end for

Algorithm 4.1: The value-based Lagrangian training loop (headline dual-critic, mean-cost formulation). Because the backbone is critic-only, the constraint enters through action selection on $Q_r - \lambda Q_c$ rather than through an actor update: both critics train on stationary Bellman targets, the cost critic is backed up at per-env-step granularity, and λ is fixed in the headline configuration because PID auto-tuning is seed-unstable at the feasibility boundary (Section 6.2.2).

AMASS [80] clip during demo conversion so that demo and live obs share the same channel width under `bodyslam` $\in$ {`oracle`, `noisy`}.

Demo–live action-stat sharing. CQN-AS normalises actions against `action_stats`; the live env and the demo replay must share one normalisation, or demos teach a policy that lives in a different bin than the policy executes. The adapter overrides `self._action_stats` with the demo-derived statistics at construction.

C51 projection CUDA overflow. The vendored C51 target-projection built its per-row scatter offset with an integer `torch.linspace`, which CUDA computes via float32; past 2^{24} the offsets round (here the endpoint is $\approx 3.97 \times 10^7$ at our batch size), so the boundary row addressed out of bounds and triggered an opaque, data-dependent `index_add_` device-side assert that never reproduces on CPU (exact integer arithmetic). The fix is an exact `torch.arange(B) * N_atoms` offset plus defensive index clamps; the diagnostic playbook (pin the op with `CUDA_LAUNCH_BLOCKING=1`, `isfinite`-guard the batch, log index extents) is recorded in the repository. This bug also affects the cost critic, which reuses the projection.

Other gotchas. A hidden `tensordict` version pin, a Python-3.12 `random.seed` type regression, and a custom `state_dict` round-trip for the non-`nn.Module` agent were also required; the full catalogue lives in the project's documentation.

4.10 Making the Offline Filter Deployment-Faithful

A runtime safety filter is only as good as the match between the data it was trained on and the control loop it runs inside. The offline SVF is trained on transitions collected by replaying a policy snapshot through a hand-built collection environment; if that environment drifts from the RoboBase factory `_create_env` the agent actually deploys through, the critic is calibrated against actions and states the deployed robot never produces, and it over-vetoes. The deployment-faithful harness (Section 3.7) exposed four such drifts, each in a separately-replicated piece of `_build_live_env`, and all four had to be fixed before any filter number was trustworthy:

1. **Action de-normalisation.** Collection de-normalised against `env.action_space` rather than the agent's demo-derived statistics, so the critic saw mis-scaled actions.
2. **Action-execution mode.** CQN-AS deploys with 16-step action sequences under temporal-ensemble blending; collection used receding-horizon `chunk[0]`. This mismatch alone drove a filtered-row intervention rate of 89.5 % at deployment against a 7.9 % sweep estimate.
3. **Control frequency (the root cause).** The collection env ran at 500 Hz rather than the deployed 20 Hz; the 25× mismatch shrank each action's effect so the collection policy never completed the task (0 % vs. 85 %) and the coworker barely approached within the truncated wall-clock.
4. **Coworker scenario drift.** The collection scenario came from a Python preset that had drifted looser than the YAML the deployment factory reads.

The validation that the fixes worked: the filterless ($R = 0$) sweep proximity rate, 0.286, matches the deployment benchmark baseline 0.296 to within sampling error; the earlier "31.7 % reduction at 21.6 % intervention" knee was an artefact of the broken collection. The general lesson, an offline safety component must be collected through the exact control loop it will be evaluated in, is a silent confound class invisible to within-collection sweeps; it directly underwrites every filter number in Chapter 6.

5 Measuring Safety Beside a Person: Tasks, Axes, and Protocol

5.1 Tasks

Three tasks, spanning the two co-location regimes of Section 3.12. `saucepan_to_hob` is the persistent-co-location deep-dive: a long-horizon pick-and-place with a vertical clearance constraint whose trajectory overlaps the G1 coworker's reach envelope, and on which the unconstrained CQN-AS baseline reaches a strong `success_rate` after the curriculum. `dishwasher_close` and `drawers_open_all` are the intermittent-co-location tasks: shorter-horizon appliance manipulation on which the same curriculum recipe produces competent unconstrained baselines (0.77 and 0.82 success respectively under COWORKER disruption). All headline evaluation is under the same COWORKER distribution and `noisy` perception unless stated.

5.2 Disruption Distributions

Training uses `Disruption.coworker` sampling from the `coworker_train` `ParameterSpace` (Table 3.2). Evaluation uses the same distribution for all in-distribution numbers and the wider `coworker_eval` distribution for the held-out generalisation check (Section 6.2.10).

5.3 Perception Mode (oracle vs. noisy)

The `BodySLAMWrapper` of Section 3.13 exposes three modes for the `human_pos_estimate` channel: `off` (no channel), `oracle` (ground truth), and `noisy` (the calibrated mock-BodySLAM++ pipeline). The per-experiment policy is uniform: every *policy* trains on `oracle` and is evaluated on `noisy`; every *filter* (SVF dataset, sweeps, gated runs) is collected and evaluated on `noisy` throughout; E3.6 is the deliberate exception that evaluates the same checkpoints under both modes to measure the perception gap; and WCSAC trains and evaluates on `oracle` (flagged as an upper bound where reported).

Rationale. The deployment-relevant condition is `noisy`, so every headline number in this report is a `noisy`-eval number (the one exception, WCSAC, is flagged where it appears). Training the *policy* on `oracle` isolates the treatment variable: any safety improvement is attributable to the constraint rather than to noisy training acting as an incidental regulariser, and the matched `oracle`-eval pass then upper-bounds what perfect perception would buy. The avoidance turns out perception-robust (Section 6.2.8), which is the property a sim-to-real claim needs. The *filter* is trained and evaluated on `noisy` throughout, because a runtime filter must match the perception it will face at deployment. A fully noisy-trained headline is a natural follow-up (Section 8.1.4).

5.4 Baselines and Configurations

On the persistent-co-location task, the headline comparison (E4.1, Section 6.2.4) evaluates four configurations: the *unconstrained baseline* (end-of-curriculum stage-2 snapshot); the *proactive policy* (that baseline fine-tuned with the fixed-$\lambda = 0.1$ value-based Lagrangian); the *speed-scaling filter on the baseline* ($d_{\text{slow}} = 0.40\,\text{m}$, no retraining); and the *policy–filter composition*. Two further families serve as controls and as the reactive-filter taxonomy: a workspace-shaping confound control (the baseline with the bounded shaping term ablated), and the learned SVF veto on both baseline and policy plus the model-based base-dodge and end-effector-retract filters (Section 6.2.5).

On the intermittent-co-location tasks, five method classes are compared on the same success/SSM-violation plane (Section 6.3): the unconstrained curriculum baseline; the PID-Lagrangian across a cost-budget sweep (with its best basin checkpoint reported); the SVF binary veto; unconditional speed-scaling; and critic-gated speed-scaling at a swept gate threshold R. The boundary-condition experiment (Section 6.4) then runs the critic-gated filter on `saucepan_to_hob` under a pre-registered decision rule.

External baseline. **WCSAC** [1], the canonical distributional safe actor-critic, is faithfully reimplemented (Section 4.5) and evaluated from scratch on both intermittent-co-location tasks across three CVaR budgets. It is an *external reference* rather than a like-for-like ablation: it is actor-critic and demo-free where the project's methods are value-based and demo-warm-started, and that asymmetry is part of what it measures (Section 6.3.1).

5.5 The Two Safety Axes and the Evaluation Protocol

5.5.1 The Exogenous Decomposition of Proximity

The protocol starts from a measurement, not a convention. The intuitive safety number for human–robot co-working is `ep_proximity_violation_rate`: the fraction of env-steps with min-separation $< \tau_{\text{prox}} = 0.3\,\text{m}$ (threshold calibrated in Section 3.9). But how much of that quantity does the *robot* control? A controlled sweep answers this directly: driving a runtime filter from no intervention to a *fully frozen* robot (vetoing 100 % of steps) removes only ≈58 % of time-in-proximity (Figure 5.1). The remaining ≈42 % is the coworker walking up to a robot that is already stationary: an *exogenous*, human-driven component that no robot-side mechanism, policy or filter, can remove. Three provenance points matter for how this finding is used. First, the decomposition comes from a sweep run *before* any of the runtime-filter comparisons it now organises, on the saucepan SVF threshold sweep; the axis split was derived in advance of the results it frames, not fitted to them. Second, *both* axes are reported for every configuration in every results table, so no result is hidden by the framing. Third, the framing cuts both ways: the constrained policy's headline win lives on the proximity axis, and the runtime filters' wins live on the velocity axis; this is axis *ownership*, not metric replacement.

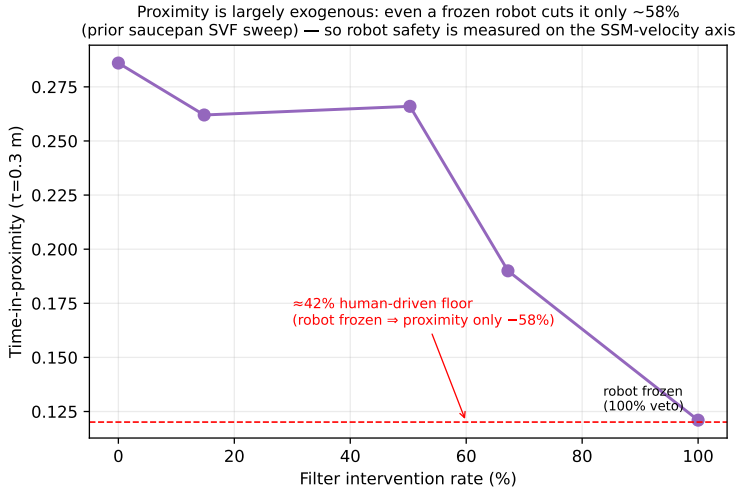


Figure 5.1: The exogenous decomposition of proximity (the protocol's organising measurement). Sweeping the runtime filter's intervention rate from 0 to 100 % (a fully frozen robot) removes only ≈58 % of time-in-proximity; the residual ≈42 % is the coworker approaching a stationary robot, a floor outside robot control. Provenance: this sweep is the *prior* saucepan SVF threshold sweep, run before the intermittent-co-location experiments existed; the two-axis protocol was fixed on its basis, not after seeing the later results.

5.5.2 Axis Ownership

Proximity measures *exposure*, but ≈42 % of it is human-driven and no robot policy can remove it. We therefore retain proximity as the exposure axis, it remains the natural axis for *policy-level* avoidance, whose anticipatory repositioning is the one mechanism that can reduce how often co-location arises at all, and use the velocity-adaptive SSM margin (`ep_ssm_violation_actual_rate`, Section 3.8) as the robot-controllable safety axis on which *runtime filters* are judged: whether the robot is moving slowly enough to stop before contact at the current separation is a function of the robot's own commanded speed. Every results table reports both axes for every configuration; the conservative worst-case `ep_ssm_violation_rate` appears only in appendix tables (it saturates at ∼93 % on random rollouts and carries little discriminative signal, Section 3.8).

Pre-registered decision rules. Three accept/reject rules were fixed before the corresponding sweeps ran, and are reported with their outcomes: (i) the E1.1 observation-ablation bar, ≥ 20 % SSM-rate reduction on at least one (method, task) pair (Section 6.2.1); (ii) the reactive-filter gracefulness bar on the proximity axis, ≥ 30 % proximity reduction at ≤ 25 % intervention (Section 6.2.5); and (iii) the boundary-condition rule for critic-gated scaling on `saucepan_to_hob`, success ≥ 0.60 *and* SSM-actual ≤ 0.08 for the gate to count as recovering throughput (Section 6.4). Rule (iii)'s failure on saucepan is itself a headline result, because it matches the predicted boundary condition.

5.5.3 Task Performance and the Safety Tax

Two complementary task metrics so the *cost of safety* on the task is explicit:

- `success_rate` (terminal task completion) is the primary task axis. **The Δ success column in the headline Table 6.1 is the explicit "did safety reduce task performance?" answer**, computed as $(s_{\text{row}} - s_{\text{row 1}})/s_{\text{row 1}}$ with s = success rate.
- `steps_to_completion` (mean env-steps to terminal, among completed episodes) captures whether the policy preserves task completion but takes longer. A safe-and-slow policy and a safe-and-fast policy both register `success_rate` = 1.0; `steps_to_completion` discriminates them. Under ISO 15066 SSM, the robot is expected to slow when the human approaches, so an increase here is expected behaviour rather than a failure mode, but it should be quantified.

`episode_reward` (sum of $r_t^{\text{task,shaped}}$ over the episode) is reported in appendix tables as a tertiary axis that combines task progress and the workspace-shaping signal.

5.5.4 Tail Risk

Tail-risk metrics:

- $\text{CVaR}_{0.95}$ of `ep_cost_integral` (the tail metric matching the cost signal the policy was trained on).
- $\text{CVaR}_{0.95}$ of `ep_min_separation` (a more human-readable tail metric).
- 99^{th} percentile of `ep_min_separation`.

Each is computed over the union of 60 evaluation episodes per configuration. Contact-force tail metrics are not reported because of the open PFL contact-detection limitation (Section 3.10).

5.5.5 Policy / Filter Interaction

Three complementarity metrics for the runtime-mechanism question (RQ2):

- *Filter intervention rate*: fraction of env-steps where the runtime filter substitutes its fallback (zero-velocity for the veto, scaled velocity for speed-scaling). Reported in the headline Table 6.1 and the taxonomy Table 6.2; passthrough rate is its complement (1− intervention).

- *Internalisation curve*: filter intervention rate evaluated on intermediate snapshots of the Lagrangian-trained policy as training progresses. A *falling* curve (rising passthrough) would be evidence the policy is learning the filter's job; E4.3 (Section 6.2.7) finds the opposite, a *flat, elevated* curve, which is itself the diagnostic for the out-of-distribution veto.
- *Passthrough–task correlation*: at deployment, episodes with higher filter passthrough should also exhibit lower `steps_to_completion`, because every intervention swaps in the zero-velocity fallback and costs the policy a step or more of task progress. The harness records both per episode for the headline cells.

5.6 Statistical Methodology

Every evaluation cell runs 20 episodes per evaluation seed across three evaluation seeds (60 episodes), sampled from the relevant `ParameterSpace`. Configurations involving a *trained* policy additionally pool three independently trained seeds, giving 180 episodes for the headline saucepan policy and composition rows and for every row of the boundary-condition sweep; each table states which applies. 95 % bootstrap confidence intervals (10,000 resamples over episodes) are reported for the headline safety metric of each table. Directional claims are assessed by bootstrap-CI overlap rather than parametric tests: episode-level safety metrics are bounded, zero-inflated and heavily non-Gaussian, so t-statistics would be poorly calibrated, whereas the CI criterion is conservative and assumption-free. Two deliberate exceptions to the three-trained-seed rule are flagged where they occur: the intermittent-co-location Lagrangian budget sweep trains a single seed per budget cell (six cells; the saucepan arm had already established the seed-instability finding that would make per-cell seed replication uninformative), and the WCSAC external reference is a single seed per (task, budget) cell at 30 evaluation episodes, reported with that caveat (Section 6.3.1).

Every script also carries a `-smoke` flag (a 5-minute CPU validation); no multi-hour training was launched without passing the relevant smoke gate first.

5.7 Compute Budget

All training ran on single NVIDIA A100 (40 GB) GPUs. The persistent-co-location arc cost ≈220 GPU-hours (log-reconstructed), dominated by eighteen 40k-frame Lagrangian fine-tunes at ≈9 h each across the cost-form, budget, PID, and fixed-λ grids, plus the three-stage curriculum, the confound control, the offline SVF pipeline, and snapshot evaluation. The intermittent-co-location arc adds ≈230–280 h, of which the six 150k-frame from-scratch WCSAC runs dominate (≈120–180 h, estimated rather than log-reconstructed, since they ran on a shared multi-GPU dispatcher). The project total is roughly 450–500 GPU-hours; the carbon accounting (≈40 kg CO_2e under the 2024 UK grid emissions factor) is derived in Appendix F.

5.8 The Unconstrained Baselines: Task-Competent, Not Compliant

The results below are organised by research question rather than by phase; this section first establishes the unconstrained baseline against which every later number is measured. The unconstrained CQN-AS policy at the end of stage 2 of the curriculum (Section 4.6) is task-competent on `saucepan_to_hob`: under the headline `noisy` perception condition it reaches `success_rate` = 0.85. The safety picture is poor (Table 5.1): `ep_proximity_violation_rate` = 0.296 (≈30 % of steps within $\tau_{\text{prox}} = 0.3$ m) averaged across 60 eval episodes, the velocity-adaptive ISO 15066 SSM-actual flag fires on 14.6 % of steps, and the mean robot–human separation is only 0.132 m. The worst-case ISO bound (conservative S_p with $v_h = v_{h,\max}$) is saturated, firing on ∼93 % of steps, which is why it is reported only for traceability. Tail behaviour is dominated by the human's closest lunge: $\text{CVaR}_{0.95}$ of `ep_min_separation` is ≈ 0.005 m, with the single worst approach reaching ≈ 0.002 m. The episode length is ∼449 control steps.

Table 5.1: Baseline ISO 15066 compliance for the unconstrained CQN-AS policy on `saucepan_to_hob` under the `coworker_train` distribution (means ± 95 % bootstrap CI, 3 seeds × 20 episodes). This is the "before" picture and the floor against which every later result is measured: the policy is task-competent yet violates proximity on a large fraction of episodes, because nothing in its objective represents the human.

Metric	Unconstrained baseline	95 % CI
`success_rate`	0.85	$[0.75, 0.93]$
`ep_proximity_violation_rate`	0.296	$[0.23, 0.37]$
`ep_ssm_violation_actual_rate`	0.146	$[0.10, 0.19]$
`ep_min_separation` (mean, m)	0.132	–
$\mathrm{CVaR}_{0.95}$ `ep_min_separation` (m)	≈ 0.005	–

Reading. The unconstrained baseline is the floor: a task-competent policy that is *not* ISO 15066 compliant under sustained co-working. The deployment posture that CQN-AS ships with [4] on BiGym would, under `safety_bigym`'s COWORKER distribution, be within 0.3 m of the coworker on ≈30 % of steps. Every reduction reported below is relative to this floor.

Intermittent-co-location baselines. The same curriculum recipe yields competent unconstrained baselines on the other two tasks (60 episodes each, `noisy`, COWORKER): `dishwasher_close` reaches 0.77 success with SSM-actual 0.176 and proximity 0.246; `drawers_open_all` reaches 0.82 success with SSM-actual 0.100 and proximity 0.211. The drawers policy is naturally slower (mean robot velocity 0.19 vs. dishwasher's 0.44 m/s), which is why its baseline SSM-violation rate is half the dishwasher's, a fact that later bounds how much a velocity-axis filter can add there (Section 6.3.4).

6 Results: Two Regimes, One Law

6.1 The Shape of the Evidence

This chapter reports the evaluation in three movements. The first (Section 6.2) takes the persistent-co-location task, `saucepan_to_hob`, from the unconstrained baseline through every safety mechanism to the division of labour that works there. The second (Section 6.3) repeats the exercise on the two intermittent-co-location tasks, where the verdicts reverse. The two movements will appear to contradict each other, and the third (Section 6.4) shows that they are one finding: the chapter closes by transplanting the intermittent-regime winner onto the persistent task under a decision rule fixed in advance, with the regime map predicting, correctly, that it must fail there.

The evaluation is deliberately ablation-led: each major component is removed, replaced, or retuned to test whether it actually contributes. Observation without a safety gradient does not produce safety (Section 6.2.1). The cost budget, not the cost form, decides whether the constraint binds, and PID auto-tuning is seed-unstable at the feasibility boundary (Section 6.2.2). The runtime override form is decisive, in that a binary veto breaks chunked-policy coherence where graded scaling preserves it (Sections 6.2.5 and 6.3.2), and the gate-design ablations show the gate threshold to be the dominant dial (Section 6.3.5). Each retained component of the final architecture survives a specific failure mode of a simpler alternative.

6.2 Persistent Co-location: Safety Must Be Trained In

This section is the persistent-co-location arm of the regime map. On `saucepan_to_hob` the coworker occupies the robot's working volume nearly continuously: a separation-triggered filter would be active on ≈44–62 % of steps (measured directly in Section 6.4). The claim this section establishes is the first half of the regime map: *where co-location is persistent, no runtime mechanism reduces exposure gracefully, and a fixed-multiplier constrained policy does, cutting proximity violations by* 23 % *at* 0.76 *success, reproducibly across seeds.* The evidence runs in order: observation alone is not a safety mechanism; what it takes to make the constraint bind; what reactive filtering can and cannot contribute on each axis; the headline comparison and the composition's ceiling; and how far the result stretches under perception noise, a held-out coworker, and the worst-case tail.

6.2.1 Observation Alone Is Not a Safety Mechanism

Before any constraint was trained, a behaviour-cloning pilot (E1.1: ACT, four tasks, observation modes `off`/`oracle`/`noisy`, no violation penalty) tested whether merely *seeing* the human improves safety, against a criterion fixed in advance: a $\geq 20\,\%$ SSM-violation reduction. No cell cleared the bar. The best case (`drawers_open_all`) improved by 13.7 %, and on three of four tasks the oracle channel made violations *worse*, most sharply on `saucepan_to_hob` ($0.135 \rightarrow 0.203$) even as it raised task success ($0.22 \rightarrow 0.58$). The policy uses human state to route around obstructions and complete the task, not to keep distance, because nothing in its objective rewards the latter. Observation without a safety gradient is at best inert and at worst counterproductive, which is what motivates training the constraint into the policy. (Behaviour cloning alone cannot distinguish "channel useless" from "cloning marginalises a channel uncorrelated with demonstrated actions". The perception sweep of Section 6.2.8 resolves the ambiguity under non-degenerate constrained training.)

6.2.2 The Cost Budget, Not the Cost Form, Is the Operative Variable

Two ablations locate what makes the constraint bind. The first (E3.1) ablated the *form* of the per-step cost c_t, holding the rest of the agent fixed: a continuous graded signal ($c_t = \min(1, \max(c^{SSM}, c^{PFL})) \in [0, 1]$, Equation 3.4) against a binary worst-case indicator, at the nominal tight budget $d = 0.01$. The comparison is degenerate: both forms collapse to 0 % task success (at deployment proximity 0.013 and 0.024 respectively). The cause is feasibility, not form. The dual variable, updated as $\lambda \leftarrow \lambda + k\,(\bar{c} - d)$, can stabilise only if some policy achieves average cost $\bar{c} \leq d$, and no policy can: ≈42 % of the baseline's 0.296 proximity is exogenous (Section 5.5.1), so the achievable cost floor is $\approx 0.42 \times 0.296 \approx 0.12$, an order of magnitude above the budget. The multiplier winds up regardless of the cost signal's shape, the $-\lambda Q_c$ term swamps the reward objective, and the policy abandons the task to minimise cost. (A fixed reward penalty with no dual variable sidesteps the windup, but its apparent training-time reduction did not survive deployment-faithful evaluation, so it is not adopted either.)

The second ablation (E3.2) sweeps the variable that *is* operative, the budget d. The sweep does not trace a smooth safety/throughput Pareto. It exposes a sharp feasibility boundary: every budget at or below the task's inherent per-step cost collapses task success, with continuous-cost runs at $d \in \{0.001, 0.05, 0.1\}$ reaching success 0.00, 0.00, 0.13 respectively, all essentially task-abandoning. The single budget that is both feasible and non-trivial, $d = 0.3$ at the task's inherent cost, does not give a stable operating point under PID auto-tuning: it sits *on* the feasibility boundary, so the dual variable is decided by each seed's stochastic cost trajectory. Figure 6.1 shows the three seeds at $d = 0.3$ converging to $\lambda = 0.000, 0.267, 3.855$, producing an unconstrained policy, a graceful avoidance basin, and a windup collapse to zero success. The graceful regime reproduces in only one of three seeds. This is why the headline policy (Section 6.2.4) pins λ at a fixed value rather than auto-tuning it: the budget loop is ill-conditioned exactly where the only feasible budget lies.

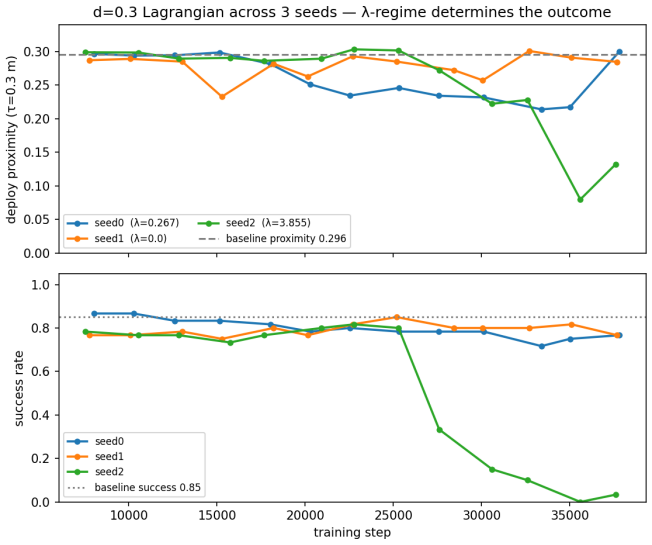


Figure 6.1: The budget-feasibility experiment (E3.2; RQ1). PID-Lagrangian at the only feasible budget $d = 0.3$, three seeds. Top: deployment proximity-violation rate ($\tau = 0.3$ m) versus training step; bottom: task success. The seeds converge to $\lambda = 0.000$ (seed 1, unconstrained, flat at the baseline ≈0.30), $\lambda = 0.267$ (seed 0, a graceful sub-0.26 avoidance basin), and $\lambda = 3.855$ (seed 2, windup collapse, success $\to$ 0). Because $d = 0.3$ coincides with the task's inherent cost, the dual variable carries too little signal to auto-tune stably, motivating the fixed-λ headline.

6.2.3 The Reactive-Filter Pareto Frontier

The offline SVF was trained at $\alpha_{\text{CQL}} = 5.0$ and swept over the veto threshold R on a held-out validation set (three seeds, 20 episodes each). With the filter disabled ($R = 0$) the swept

`ep_proximity_violation_rate` is 0.286, which matches the unconstrained benchmark baseline of 0.296 to within sampling error; this agreement is the validation check that the sweep environment reproduces the deployment environment.

The sweep does *not* expose an operating point that reduces geometric proximity cheaply: $R = 2.25$ cuts proximity by only ≈8 % at ≈15 % intervention, a one-third reduction needs ≈67 % intervention, and even vetoing *every* step caps the reduction at ≈58 %, the decomposition of Section 5.5.1 seen from the filter side. We therefore pin the operating point at the light backstop $R = 2.25$ and read the veto as an ISO 15066 SSM velocity backstop, not a proximity reducer; proactive proximity reduction is delegated to the constrained policy (Section 6.2.4).

The reactive-fallback dilemma. A natural objection is that a more aggressive fallback would let the filter reduce proximity directly. A fallback sweep on the unconstrained baseline at matched intervention rates rules this out. Zero-velocity braking holds station: it trims mean robot speed (the velocity-backstop effect) but leaves proximity unchanged (0.296 → 0.303), because the robot simply dwells while the human approaches. An aggressive retreat cuts proximity sharply (0.296 → 0.095) but only by having the robot flee: success collapses from 0.85 to 0.18, mean speed rises sixfold (0.29 → 1.70 m/s), and the SSM-actual rate *worsens* (0.147 → 0.180) because a fast-retreating robot is itself an SSM hazard. A gentle retreat does neither job. No reactive setting buys geometric separation at acceptable success and speed: against an approaching human, reducing proximity requires sustained, anticipatory robot motion, which a one-step reactive filter cannot supply. The veto remains independently useful as a deployment backstop that trims mean speed without any retraining, but the graded margin ISO 15066 prescribes needs the speed-scaling filter (Section 6.2.5). This is the empirical case for assigning the exposure axis to the policy arm, revisited in Section 7.2.4.

6.2.4 The Headline Comparison: A Division of Labour

Main comparison. The central question of the architecture is which mechanism reduces which ISO 15066 axis, and how the mechanisms compose. The answer is a *division of labour*: the proactive constrained-RL policy owns the geometric-proximity axis and a complementary-axis runtime filter owns the velocity (SSM-actual) axis. Table 6.1 reports the four configurations that establish this; the two reactive-veto configurations the architecture was originally built around are covered in Section 6.2.5, and the both-axis composition is analysed in Section 6.2.6.

The proactive policy reduces proximity, reproducibly. The fixed-λ ($\lambda = 0.1$) Lagrangian policy reduces `ep_proximity_violation_rate` from the baseline 0.296 to 0.228, a 22.8 % reduction (bootstrap 95 % CI $[0.194, 0.264]$), at 0.76 task success (CI $[0.70, 0.82]$), pooled over three seeds (180 episodes). Each seed's selected operating point is below the baseline; the velocity-adaptive SSM-actual rate also improves (0.146 → 0.112, −23 %) and mean robot velocity rises only ∼8 %. This is the graceful proximity reduction the reactive filter could not achieve (Section 6.2.3). The magnitude is bounded by the ≈42 % exogenous floor (Section 5.5.1): it captures roughly 40 % of the *reducible* (robot-driven) proximity. Modest in absolute terms, but the reactive filter captured essentially none of it at acceptable cost.

Operating-point selection must be safety-aware. A methodological point underwrites the policy row. At the feasible budget the avoidance does not live at the peak-success checkpoint: `snapshot_best` (peak success) and the final checkpoint both deploy at ≈baseline proximity (≈0.30), because selecting on task success picks *against* the cost constraint. The avoidance lives in a mid-training basin (∼20k–35k steps, several consecutive checkpoints at 0.21–0.25 with success 0.72–0.80 and SSM-actual improved). The operating point must therefore be chosen by a *safety-aware* rule (lowest deployment proximity at a success floor), not by peak success; selecting on success produced two false-null results before the basin was found. Each seed's selected basin

Table 6.1: Experiment E4.1 (**the headline**). The division of labour between the proactive policy and a complementary-axis runtime filter on `saucepan_to_hob` under `coworker_train`, `noisy` perception. Rows 1 and 3 evaluate the single stage-2 baseline policy over 60 episodes (3 evaluation seeds × 20); rows 2 and 4 pool three independently trained seeds (180 episodes). Brackets are 95 % bootstrap CIs. The policy owns the geometric-proximity axis ($\tau = 0.3$ m); the graded speed-scaling filter owns the velocity-adaptive SSM-actual axis; only their composition reduces both at once, at a multiplicative success cost. Δ columns are relative to the unconstrained baseline (negative is safer). The two SVF-veto configurations are in Table 6.2.

Configuration	`succ.`	`prox.viol`	Δ prox.	`ssm-act.`	Δ ssm.	Interv.
Unconstrained baseline	0.85	0.296 [0.23, 0.37]	–	0.146 [0.10, 0.19]	–	–
Proactive policy (λ=0.1)	0.76	**0.228** [0.19, 0.26]	**−23 %**	0.112 [0.09, 0.13]	−23 %	–
Speed-scaling filter (on baseline)	0.53	0.273 [0.21, 0.35]	−8 %	**0.048** [0.03, 0.07]	**−67 %**	45 %
Policy + speed-scaling (composition)	**0.44**	0.250 [0.21, 0.29]	−15 %	0.065 [0.05, 0.08]	−55 %	40 %

checkpoint is shown in Figure 6.2; that the basin appears consistently across all three seeds under fixed λ is what makes the reduction reproducible, in contrast to the PID instability of Figure 6.1.

The basin is constraint-driven, not selection noise. Selecting a checkpoint by a deployment metric invites the objection that the "basin" is evaluation noise harvested by the rule. Four observations rule this out: the $\lambda = 0$ seed, processed by the *same* rule, shows no basin (flat at ≈0.30 with isolated dips); the basin is a *run* of consecutive checkpoints at 0.21–0.26 proximity with success 0.72–0.80 (Figure 6.2), not a single dip; SSM-actual co-improves inside it, which uncorrelated proximity noise would not produce; and the selected point also sits below the *no-shaping* baseline at matched success (the −12 % comparison below). When λ engages, the constraint does work; when it is slack, no selection rule manufactures a basin.

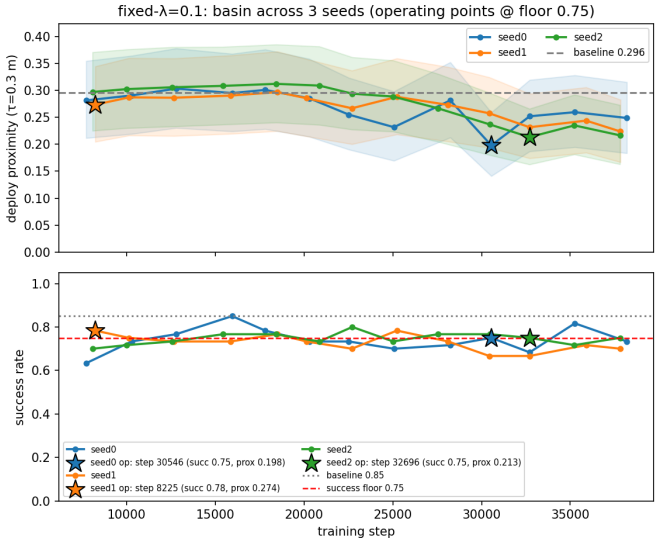


Figure 6.2: The fixed-$\lambda = 0.1$ policy across three seeds (stars mark the safety-aware operating point at a success floor of 0.75). Top: deployment proximity-violation rate; bottom: success. Unlike the PID runs (Figure 6.1), all three seeds behave consistently and each reaches a sub-baseline proximity basin in mid-training, giving the pooled 0.228 at 0.76 success. Fixing λ removes the ill-conditioned budget loop that made the PID controller seed-unstable.

Controlling for the workspace-shaping confound. The unconstrained baseline carries a mild workspace-distance reward ($\beta = 0.05$, Section 3.6) that anchors the end-effector in the task workspace. A natural objection is that this inflates the baseline's proximity and hence the policy's apparent reduction. Ablating it: the no-shaping baseline is 0.75 success / 0.258 proximity, versus the shaping baseline's 0.85 / 0.296. So shaping is a *task-success aid that incidentally*

raises proximity (it keeps the robot where the coworker reaches), not a safety term. Crucially, the Lagrangian policy (0.228 at 0.76) sits below the no-shaping baseline too, a −12 % proximity reduction at matched success, so the headline −23 % (against the matched-recipe baseline) is not an artefact of an inflated reference. We report both: −23 % (isolated constraint effect, shaping held fixed) and −12 % (confound-controlled, against the vanilla baseline).

The velocity axis is the filter's, and only a graded filter reaches it. A runtime filter's canonical ISO 15066 role is the velocity axis. The graded speed-scaling filter delivers it: on the baseline it drives SSM-actual from 0.146 to 0.048 (−67 %) at its optimum ($d_{\text{slow}} = 0.40\,\text{m}$), with proximity essentially unchanged (0.296 → 0.273). It is the only filter in the taxonomy that meaningfully reduces *any* ISO 15066 axis on its own terms; the binary veto, despite also slowing the robot, leaves SSM-actual at 0.148 ≈ baseline, a stop-or-go veto does not maintain the graded margin the standard prescribes. The success cost, however, is not an artefact of the 0.40 m setting: even the gentlest swept point ($d_{\text{slow}} = 0.25\,\text{m}$) intervenes on 37 % of steps, co-location is persistent, so some part of the robot is nearly always inside the slow radius, and still costs a third of task success at *less* velocity-axis gain. The reactive cost is paid at every operating point; only its exchange rate moves.

The composition is the only both-axis-compliant configuration, and it costs success. Composing the policy with the speed-scaling filter is the *only* configuration that reduces both ISO 15066 axes at once: geometric proximity 0.250 (retained from the policy's 0.228, CIs overlap) *and* SSM-actual 0.065 (−42 % against the policy's 0.112, the filter's contribution). Each component alone leaves the *other* axis near baseline (policy: SSM-actual 0.112; filter-alone: proximity 0.273 ≈ baseline). The cost is task success: 0.85 → 0.44. The proactive cost (0.85 → 0.76, ×0.89) and the reactive cost (×0.58) stack *multiplicatively*, and the composition beats neither specialist on that specialist's own axis (the policy still wins proximity 0.228 < 0.250; the filter still wins velocity 0.048 < 0.065). The composition is the joint-coverage compromise, not a per-axis champion. Both measured safety axes are improved in this regime only by composing the proactive policy with a complementary runtime filter, and that composition makes the safety–throughput trade explicit, with a cost dial tunable through λ and d_{slow} (Section 6.2.6).

Sim-to-real perception. The whole table is evaluated under `bodyslam=noisy`, the realistic deployment condition and the filter's native distribution (Section 5.3). The policy's reduction survives this noisy, lagged perception unchanged (E3.6, Section 6.2.8): the architecture claim is reported under deployment-relevant perception, not only under oracle state.

6.2.5 The Reactive-Filter Taxonomy: Freeze versus Flee

The headline filter is the graded speed-scaling one. The architecture was, however, originally built around a learned CQL safety value function used as a veto, and the journey from that design to the graded filter is itself a result. We evaluated every reactive modality available, learned and model-based, and none reaches the proactive policy's graceful operating point. The reason is structural: a reactive filter acts only *once the human is already close*, when the only moves are to *freeze* (and dwell in the danger zone) or to *flee* (and abandon the task). Table 6.2 summarises the taxonomy. The claim is scoped deliberately: *no reactive filter gracefully reduces proximity under persistent co-location.* It is a statement about the exposure axis in this regime, not about reactive filtering in general; Section 6.3 shows what reactive mechanisms *can* deliver, on the velocity axis, where co-location is intermittent and genuinely-safe windows exist.

The learned veto freezes. On the baseline, the SVF veto with a zero-velocity fallback ($R = 2.25$, 15 % intervention) does not reduce geometric proximity (0.296 → 0.303, within noise); it only trims mean robot velocity (−8 %), acting as an SSM *velocity backstop* (Section 6.2.3). Applied

Table 6.2: The reactive-filter taxonomy on `saucepan_to_hob` (COWORKER, `noisy`); every row is 60 evaluation episodes, and rows involving the constrained policy use the seed-0 trained policy (the pooled three-trained-seed policy numbers are in Table 6.1). Every reactive modality, learned or model-based, is bounded by the freeze-versus-flee dilemma: it either freezes (and proximity does not fall) or flees (and task success collapses). None reaches the proactive policy's frontier (success 0.75, proximity 0.198 at seed 0; 0.76/0.228 pooled). The SVF veto applied to the *policy* (the architecture's original composition design) is worse than the policy alone.

Filter	Mechanism	Proximity	Success	Interv.
Learned veto (SVF), on baseline	velocity backstop (freeze)	0.303	0.78	15 %
Learned veto (SVF), on policy	freeze → dwell	0.265	0.62	48 %
Model-based base-dodge (CBF)	flee (retreat base)	0.150	0.57	1.3 %
Model-based EE-retract (CBF)	flee (retract arm)	0.155	0.50	5.8 %
Proactive policy (λ=0.1)	**anticipatory avoid**	**0.198**	**0.75**	–

on top of the already-avoiding policy, it is *counterproductive*: proximity worsens (0.198 → 0.265) and success falls (0.75 → 0.62), at a 48 % intervention rate (versus 15 % on the baseline). The mechanism is distribution shift: the SVF critic was collected on *baseline* rollouts, so it is out of distribution on the avoiding policy and over-vetoes, and each veto zeroes robot velocity, re-introducing the freeze/dwell failure. A threshold sweep confirms this is not mis-thresholding, $R \in [1.0, 2.25]$ gives 39–48 % intervention and proximity 0.27–0.28 at every setting, and re-collecting the critic on the policy's own rollouts (on-policy) still yields no graceful operating point: at ≤ 25 % intervention every threshold *increases* proximity, and proximity falls only at 65–100 % intervention where the robot is essentially frozen. Neither critic meets the pre-registered gracefulness bar (≥ 30 % proximity reduction at ≤ 25 % intervention, Section 5.5). Both the baseline-trained and on-policy critics fail at every threshold, so the limit is the reactive paradigm, not the critic's calibration.

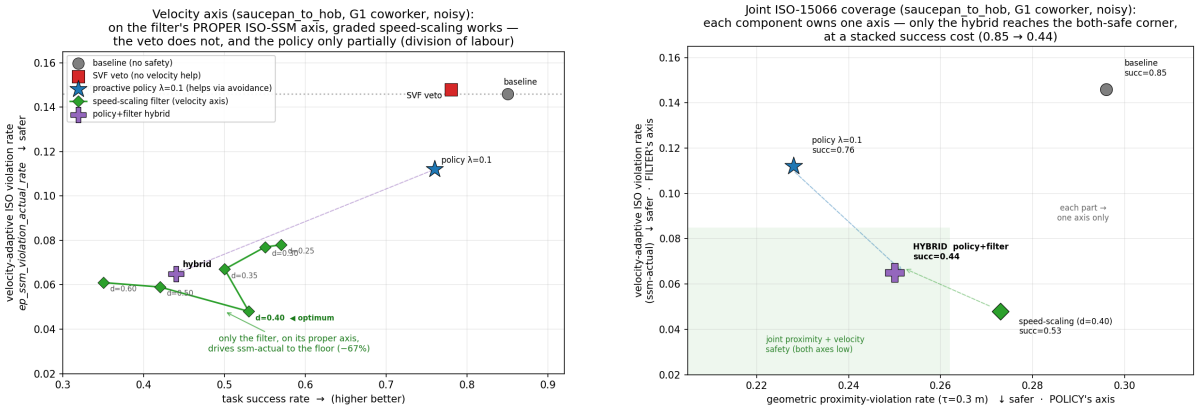
The model-based filter flees. A geometric control-barrier-style "directional-dodge" filter (no learned critic) *does* reduce proximity where the veto cannot, 0.198 → 0.150 at $d_{\text{target}} = 0.35$ m and only 1.3 % intervention, but by *fleeing*: the base retreats, episodes stretch from ∼450 to 620–770 steps, and success falls to 0.57/0.43/0.35 as $d_{\text{target}} = 0.35/0.45/0.55$, with proximity flooring at ≈0.14 regardless (the exogenous floor). There is no harmless-backstop setting: even at the minimal $d_{\text{target}} = 0.30$ m (1.1 % intervention) success falls 0.75 → 0.58, because even rare dodges land at task-critical moments. The same flee appears whichever part of the robot dodges: retracting the *end-effector* (an arm "flinch" via a damped-Jacobian step) is worse in the useful regime (0.50 success at $d = 0.35$; the EE is the closest pair, so it intervenes far more often, and retracting the arm *is* pausing the task). The flee is structural: the task and the hazard are co-located, so a separation barrier is satisfiable only by radial retreat, which is retreat from the workspace. Eliminating the flee requires anticipation, which is exactly what the constrained policy provides; "fixing" the flee inside a reactive filter amounts to reinventing the policy.

6.2.6 Joint ISO Coverage: The Composition and Its Ceiling

The two preceding subsections separate cleanly onto the two ISO 15066 axes (Figure 6.3). Figure 6.3a makes the velocity axis explicit: on SSM-actual the speed-scaling filter is the specialist (→ 0.048), the policy also helps via avoidance (→ 0.112) because keeping further from the human lowers the required margin, and the SVF veto does not help at all (0.148 ≈ baseline). Figure 6.3b then plots both axes together: only the policy-plus-speed-scaling composition reaches the both-safe corner (low proximity *and* low SSM-actual), and it is strictly better than the SVF-veto composition, which *degraded* proximity with no velocity gain. A complementary-axis, model-based filter composes additively where a redundant-axis, out-of-distribution *learned* filter destroyed.

The composition's success cost (0.85 → 0.44, analysed in Section 6.2.4) should be read as *the*

persistent-co-location regime's ceiling, not as a defect of the composition: when the coworker is co-located most of the time, any separation-triggered speed response is active most of the time (measured directly in Section 6.4), so the reactive cost is paid almost everywhere and stacks with the proactive cost. Under this regime, both-axis ISO 15066 safety is purchasable only at stacked cost; whether a smarter trigger (a learned critic gating the scaler) can avoid that cost is exactly the question the intermittent-co-location chapter answers in the affirmative, and Section 6.4 answers in the negative *for this regime*.



(a) The velocity (SSM-actual) axis: the graded speed-scaling filter is the specialist; the policy also reduces it via avoidance; the veto does not maintain the graded margin and leaves it at baseline.

(b) Joint coverage: proximity (x, policy's axis) vs. SSM-actual (y, filter's axis). Only the composition reaches the both-safe corner, at success 0.44; it beats neither specialist on that specialist's own axis.

Figure 6.3: The two-axis division of labour on `saucepan_to_hob` (60-episode cells, `noisy` perception, COWORKER distribution) and what composing the specialists buys. The policy owns the proximity axis, the graded speed-scaling filter owns the velocity axis, and only their composition reaches the corner where both ISO 15066 axes improve at once, at a success cost ($0.85 \rightarrow 0.44$) that Section 6.4 shows to be the persistent regime's ceiling rather than a tuning defect.

6.2.7 The Policy Does Not Internalise the Filter

If the policy and the SVF veto were complementary in the originally hypothesised sense, the filter's required intervention rate would *fall* as the Lagrangian policy is trained, the policy progressively taking over the filter's job. E4.3 evaluates the SVF veto's intervention rate on intermediate snapshots of the fixed-λ policy across training. It does *not* fall: it stays at ∼40–60 % throughout (mean ∼49 %, Figure 6.4), *higher* than the ∼15 % intervention on the unconstrained baseline, not lower. The reason is the distribution shift of Section 6.2.5: the constrained policy's state–action distribution is precisely where the baseline-collected critic is least calibrated, so as the policy avoids *more*, the critic vetoes *more*. This is the corroborating mechanism behind the counterproductive veto composition: there is no internalisation curve to report because the baseline-calibrated filter never becomes redundant in the benign direction; it becomes *more* active and counterproductive.

The deployment implication inverts the originally intended one: a filter intended to compose with a learned policy must either be collected on that policy's rollouts (still insufficient here, Section 6.2.5) or operate on an axis the policy does not move along, which is exactly why the graded speed-scaling filter, on the velocity axis, composes where the veto does not.

6.2.8 The Avoidance Survives Realistic Perception

To test whether the learned avoidance depends on perfect state estimation, we re-evaluated the *same* constrained policy (fixed-λ, Section 6.2.4) under two perception regimes: `oracle` (ground-truth human pose) and `noisy` (the calibrated mock-BodySLAM++ estimate with measurement noise and a 3-step latency). The avoidance is *not* perception-bottlenecked: the proximity-violation rate is statistically indistinguishable across regimes (0.236 oracle vs. 0.198 noisy), both far below

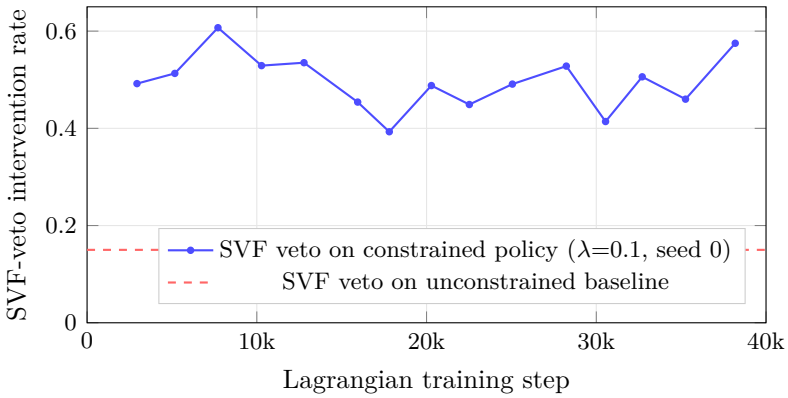


Figure 6.4: Experiment E4.3. The SVF veto's intervention rate on the constrained policy does *not* fall during training; it stays at ∼40–60 % (blue), above the ∼15 % it requires on the unconstrained baseline (red dashed). A complementary filter would show a falling curve; this flat, elevated curve is direct evidence that the baseline-collected critic is out of distribution on the avoiding policy and over-vetoes, the mechanism behind the counterproductive SVF-veto composition (Section 6.2.5).

the unconstrained baseline's ≈0.30, and task success is comparable (0.82 vs. 0.75). The baseline, by contrast, shows *no* oracle–noisy gap (0.302 vs. 0.296): it does not use the human channel to avoid, so perception quality cannot help it. That the policy retains its full reduction under noisy, lagged perception, and is if anything marginally better there, indicates a robust avoidance behaviour rather than one that exploits perfect state (Table 6.3).

Table 6.3: Experiment E3.6. The constrained policy (seed-0 trained) and the unconstrained baseline re-evaluated under `oracle` and `noisy` perception (`saucepan_to_hob`, `coworker_train`; each cell is 60 episodes, 3 evaluation seeds × 20). The policy's proximity reduction is preserved under realistic noisy, lagged perception (bootstrap CIs across modes overlap: 0.236 [0.17, 0.30] oracle vs. 0.198 [0.14, 0.26] noisy), and the baseline shows no oracle–noisy gap because it does not use the channel to avoid. The headline E4.1 table reports the pooled three-trained-seed `noisy` numbers.

	Unconstrained baseline		Constrained policy (λ=0.1)	
Perception	`succ.`	`prox.viol`	`succ.`	`prox.viol`
`oracle`	0.77	0.302	0.82	0.236
`noisy`	0.85	0.296	0.75	0.198

A third, "perception-off" mode is architecturally inapplicable (removing the channel changes the input size, 288 → 264, so the trained weights cannot run); a genuine no-human-observation condition is the separately trained behaviour-cloning ablation (Section 6.2.1). The perception sweep is thus a two-mode comparison by construction.

6.2.9 The Worst-Case Tail Is Exogenous and Uncontrollable

ISO 15066 compliance is a worst-case rather than an average property, so the tail of the per-episode separation distribution is the safety-relevant quantity. The central tail finding is that the policy lowers the *mean* and *frequency* of proximity but *not* the worst-case tail. Mean separation improves modestly (baseline 0.132 m → policy 0.140 m), but the worst-5 % closest approach is essentially unchanged across the unconstrained baseline, the no-shaping baseline, and the constrained policy alike: $\mathrm{CVaR}_{0.95}$ of `ep_min_separation` ≈ 0.005 m and the single worst approach ≈ 0.002 m in every configuration (Table 6.4).

This is the exogenous decomposition of Section 5.5.1 seen from its tail end: ≈58 % of proximity is robot-driven (suppressible by avoidance) and ≈42 % is human-driven, and the worst single approach lives entirely in the human-driven remainder. The policy captures roughly 40 % of the reducible part; no robot-side mechanism, and no configuration evaluated anywhere in this report, policy, veto, dodge, speed-scaling, or composition, moves $\mathrm{CVaR}_{0.95}$ of the closest approach. Honest reporting of this floor is what makes the −22.8 % headline a real, bounded number rather

Table 6.4: Experiment E5.1. Separation statistics across the headline configurations (COWORKER, `noisy`). The policy improves the *mean* separation but the worst-case tail ($\text{CVaR}_{0.95}$ and the single worst approach) is unchanged, because the closest approach is the coworker's lunge toward a robot it is co-located with, an *exogenous* event no robot-action policy or filter can remove. The controllable quantity is the dwell and frequency of proximity, not the single worst-case approach.

Configuration	mean sep. (m)	$\text{CVaR}_{0.95}$ min-sep (m)	worst min-sep (m)
Unconstrained baseline	0.132	≈ 0.005	≈ 0.002
Proactive policy (λ=0.1)	0.140	≈ 0.005	≈ 0.002

than an over-claim, and it motivates the Power-and-Force-Limiting collision-brake future work (Section 8.1.2), the only mechanism class that could address the exogenous near-contact tail.

6.2.10 Generalisation to a Held-Out Coworker Distribution

Re-evaluating on a held-out coworker distribution (E5.2; `coworker_eval`: broader and gentler, closest 0.6–1.8 m, reach 3–9 s, walk 0.6–2.2 m/s) tests whether the policy's reduction is specific to the training coworker. It is not: the reduction *generalises*, and is proportionally larger on the gentler distribution. The absolute proximity is much lower there because the gentler coworker approaches less, but the policy still cuts it by 31 % (0.084 → 0.058, against the in-distribution 0.296 → 0.228). The success cost is *larger* (0.92 → 0.70) because $\lambda = 0.1$ was tuned for the harder training coworker and over-constrains the gentler one: the operating point is task- and distribution-specific, but the qualitative result holds out of distribution. The SVF-veto composition is again worse than the policy alone here (54 % intervention), consistent with the in-distribution finding.

A third generalisation axis, *cross-task* transfer of the full pipeline (base curriculum → Lagrangian → eval) to `dishwasher_close` and `drawers_open_all` (E5.3), resolved *negative*: the saucepan recipe does not replicate there, no budget binds usefully, and the best constrained operating point is a marginal, within-noise improvement (Section 6.3.1). The word "reproducibly" in this chapter's headline therefore means *across the three saucepan training seeds*, not across tasks. The cross-task story is the regime map of Section 6.4, and the failure of this transfer is one of its two arms rather than an unexplained null.

6.3 Intermittent Co-location: Gate the Runtime Backstop

This section is the other arm of the regime map. The coworker distribution is unchanged (COWORKER, `noisy`, 3 evaluation seeds × 20 episodes per cell), but here the close encounters arrive in bursts: the co-location measurement of Section 6.4 puts a separation-triggered gate's activity at 19–27 % of steps, versus 51–63 % on saucepan. The claim this section establishes is the regime map's second half: *where close encounters arrive in bursts separated by genuinely-safe windows, constrained RL has nothing to bind on, and a learned critic gating a graded speed-scaling backstop halves SSM violations at a fraction of the unconditional filter's success cost.* Transplanting the persistent-regime recipe fails in both directions: the constrained policy that carried the proximity axis on saucepan finds no feasible budget here (E3.3), while the runtime-filter family that was bounded to a backstop role there produces, once gated by the learned critic, the best safety–throughput operating point in the entire evaluation (E2.6). This section works through that reversal mechanism by mechanism.

6.3.1 Constrained RL Finds No Feasible Budget (WCSAC Corroborates)

The first budget sweep on these tasks bracketed the policies' natural rolling cost (≈0.25 on the SSM-margin cost signal) from above, with budgets $d \in \{0.1, 0.3, 0.5\}$: budgets above the natural cost left $\lambda = 0$ (constraint inert, baseline behaviour), and $d = 0.1$ drove $\lambda \to 21.5$ and 0 % success (task-fatal). E3.3 re-targets the budget *into* the unsampled window just below the natural cost, $d \in \{0.16, 0.19, 0.22\}$ on both tasks, to test whether a binding-but-feasible operating point exists

at all (Table 6.5).

Table 6.5: Experiment E3.3. Cost-budget retarget sweep on the intermittent-co-location tasks (40k frames, single training seed per cell, warm-started from the stage-2 curriculum snapshot; cost is the graded SSM-margin signal). Re-targeting makes the constraint *bind* ($\lambda > 0$ everywhere), but the rolling cost floors at ≈0.21–0.25, at or above every budget, so λ integrates upward and the task collapses. There is no budget that both binds and preserves the task; even the floor budget has no *stable* feasible point (`dish` d=0.22 was at 1.00 success at 37k frames, $\lambda = 0.69$, and collapsed to 0.30 by 40k as λ reached 1.5).

Task	Budget d	Final λ	Rolling cost	In-train success	Outcome
dishwasher	0.16	13.8	0.225	0.30	binds hard, collapsed
dishwasher	0.19	9.3	0.235	0.10	binds hard, collapsed
dishwasher	0.22	1.5	0.252	0.30	binds late, collapsed
drawers	0.16	10.4	0.209	0.00	binds hard, collapsed
drawers	0.19	5.2	0.231	0.30	binds hard, collapsed
drawers	0.22	1.3	0.236	0.60	binds late, partial

Deployment of the best cells confirms the cliff. Benchmarking each $d = 0.22$ cell's best λ-active basin checkpoint and its final checkpoint under the deployment-faithful harness (`noisy`, 60 episodes): on dishwasher the basin checkpoint is statistically the baseline (success $0.82\,[0.72, 0.92]$, proximity $0.243\,[0.15, 0.34]$ vs. baseline 0.246, a $-1\,\%$ change), and the final checkpoint trades $-7\,\%$ proximity for success 0.48. On drawers the single best feasible point in the whole sweep, the $d = 0.22$ basin, reaches $-13\,\%$ proximity ($0.211 \to 0.184\,[0.14, 0.24]$) at -5 pt success, with a CI that overlaps the baseline: a marginal, within-noise improvement, not a robust win. The relationship is a monotonic trade, proximity falls only as success falls, with no graceful elbow anywhere in the swept range.

When λ binds, the robot gets *faster*, not more careful. The mechanism of the collapse matters for what it rules out. Under a binding multiplier the dishwasher policy's mean velocity rises from 0.44 to 0.79 m/s and its max from 2.46 to 4.14 m/s: the cost pressure degrades action selection into fast, erratic motion rather than producing careful slowing. A scalar penalty on *expected* cost evidently cannot, on these tasks, separate task-necessary proximity from gratuitous risk; it can only destroy the behaviour that incurs cost, which here is the task itself.

The trainability confound, stated honestly, and the external check. These are sparse-reward tasks on which the base policy is fragile without the demonstration warm-start and curriculum, so "constrained RL fails here" could in principle mean "RL barely trains here." The WCSAC external baseline (E3.7, Section 4.5) speaks to both readings at once (Table 6.6). Trained from scratch, WCSAC *learns dishwasher* (0.47 success at the tightest CVaR budget, proximity-violation 3.6 %), so the external reference is non-degenerate and the pipeline sound; but it reaches 0 % on drawers at *every* budget, corroborating base-task fragility (its low violation numbers there are trivial: a policy that never engages the task stays far from the human). The verdict is therefore *no feasible budget exists on these tasks*, compounded by base-task fragility, not *constrained RL is universally ineffective*: the saucepan policy of Section 6.2 is the standing counterexample, and the regime-level account of the flip is taken up in Section 6.4.

6.3.2 A Binary Veto Breaks a Chunked Policy

The learned safety critic itself is sound here. The shared dish/drawers SVF (Section 4.2) separates safe from unsafe transitions in-sample (mean Q of 3.14 on safe vs. 2.26 on near-contact transitions; at $R = 2.75$ it catches 92 % of near-contact transitions while vetoing 22 % of safe ones). What fails is the *intervention*: a binary veto that zeroes velocity mid-chunk interacts destructively with the temporal-ensemble action execution of the CQN-AS policy (Section 4.10). The robot stops

Table 6.6: Experiment E3.7. WCSAC [1] external reference, reimplemented faithfully and trained from scratch (150k frames, no demonstrations, single seed per cell; 30 evaluation episodes, `oracle` human-state, so these are upper bounds with respect to perception). WCSAC is non-degenerate on dishwasher but fails to learn drawers at any budget; the apparent budget ordering on dishwasher (d=30 at 0.03) is within single-seed from-scratch SAC variance and is not interpreted further. From-scratch WCSAC also sits well below the demo-warm-started value-based Lagrangian (0.47 vs. 0.82 peak on dishwasher), quantifying how much of task competence on these tasks comes from the demonstration prior.

Task	Budget	Final λ	Success	Prox. viol.	Worst min-sep (m)
dishwasher	5	100 (max)	0.47	0.036	0.025
dishwasher	15	4.5	0.43	0.102	0.009
dishwasher	30	0	0.03	0.021	0.031
drawers	5	92	0.00	0.000	1.31
drawers	15	0	0.00	0.016	0.035
drawers	30	0	0.00	0.000	0.015

inside a blended action chunk, the ensemble's open-loop state runs ahead, and when the veto lifts the policy fires a large catch-up action. Max velocity *rises*, 2.46 → 3.77 m/s at $R = 2.25$ and to 6.03 m/s at $R = 3.0$ (where the veto is active on 84 % of steps and success has collapsed to 0.12); the minimum SSM margin goes sharply negative (0.16 → −1.69, −6.84); and the SSM-violation rate it was meant to reduce does *not* improve (0.176 → 0.174 on dishwasher; 0.100 → 0.130, worse, on drawers, where success falls 0.82 → 0.43). The override *form* is the design axis that matters (Section 2.3.3): a hard switch is the wrong intervention for chunked visuomotor policies, independent of how good the critic that triggers it is.

6.3.3 Unconditional Speed-Scaling Owns the Velocity Axis, at Heavy Cost

Replacing the veto with the graded ISO 15066 speed-scaler (Equation 4.4) removes the overshoot by construction: it scales motion *magnitude* while preserving the policy's action direction, and max velocity is identical to the baseline's (2.46 → 2.46 m/s). Applied unconditionally it is the first mechanism in this entire research programme to substantially improve the robot-controllable axis: SSM-violation falls 0.176 → 0.087 (−50 %) on dishwasher and 0.100 → 0.059 (−41 %) on drawers, with the minimum stopping margin improved on both. The cost is indiscriminate: it slows the robot at *every* close approach, safe or not, and success falls 0.77 → 0.52 and 0.82 → 0.38. That cost profile is exactly what a gate should fix, where the task structure leaves the gate something to find.

6.3.4 Critic-Gated Speed-Scaling: The When/How Split Recovers the Task

The critic-gated filter (Equation 4.5) assigns each half of the decision to the component that is good at it: the SVF decides *when* (intervene only when $Q_{\text{safe}} < R$), the graded scaler decides *how* (slow smoothly, never veto). On these tasks, where genuinely-safe windows exist for the critic to find, the split works (Figure 6.5, Table 6.7): on dishwasher the gated filter at $R = 2.75$ retains the unconditional scaler's *full* −50 % SSM-violation reduction at less than half its success cost (0.67 vs. 0.52, against a 0.77 baseline); on drawers the recommended aggressive point ($R = 3.0$, $d_{\text{slow}} = 0.8$) reaches −22 % at −0.09 success, and a near-free point ($R = 2.75$, $d_{\text{slow}} = 0.8$) gives −10 % at −0.02 success. The gate-active fractions confirm the mechanism: the gate fires on 27 % (dishwasher) and 19–27 % (drawers, threshold-dependent) of steps, so most steps pass at full speed.

The gated result should be read with its scope attached: the gate recovers throughput *where genuinely-safe windows exist for the critic to find*. That scope is not a rhetorical hedge; it is the regime variable, and Section 6.4 measures it and tests it on a third task.

6.3.5 Gate Ablations: Four Refinements Confirm the Recipe

Four follow-ups probed whether a better configuration was being missed; all four converge back on the recommended recipe, and each yields a portable lesson.

Table 6.7: The intermittent-co-location method comparison (60 episodes per cell, `noisy`, COWORKER; both safety axes reported per the two-axis protocol). Δ columns are relative to each task's unconstrained baseline. The Lagrangian rows are the best basin checkpoints from the seed-unstable budget sweep and carry that caveat (Section 6.3.1); WCSAC rows are the external from-scratch reference at its best budget (single seed, `oracle` perception). Only critic-gated speed-scaling reduces the SSM-velocity axis substantially while keeping success within 0.10 of baseline.

Task	Method	Success	Δ succ.	SSM-actual	Δ SSM	Prox. viol.
dishwasher	Baseline (unconstrained)	0.77	–	0.176	–	0.246
dishwasher	Lagrangian (best basin, E3.3)	0.82	+0.05	0.184	+5 %	0.243
dishwasher	SVF veto (R=2.25)	0.63	−0.14	0.174	−1 %	–
dishwasher	Speed-scale (unconditional)	0.52	−0.25	0.087	−50 %	–
dishwasher	**Critic-gated (R=2.75)**	**0.67**	**−0.10**	**0.088**	**−50 %**	–
dishwasher	*WCSAC (external, d=5)*	*0.47*	−0.30	–	–	*0.036*
drawers	Baseline (unconstrained)	0.82	–	0.100	–	0.211
drawers	Lagrangian (best basin, E3.3)	0.77	−0.05	0.082	−18 %	0.184
drawers	SVF veto (R=2.25)	0.43	−0.39	0.130	+30 %	–
drawers	Speed-scale (unconditional)	0.38	−0.44	0.059	−41 %	–
drawers	**Critic-gated (R=3.0, d_{slow}=0.8)**	**0.73**	**−0.09**	**0.078**	**−22 %**	–
drawers	Critic-gated, near-free (R=2.75, d_{slow}=0.8)	0.80	−0.02	0.090	−10 %	–
drawers	*WCSAC (external, d=5)*	*0.00*	−0.82	–	–	*0.000*

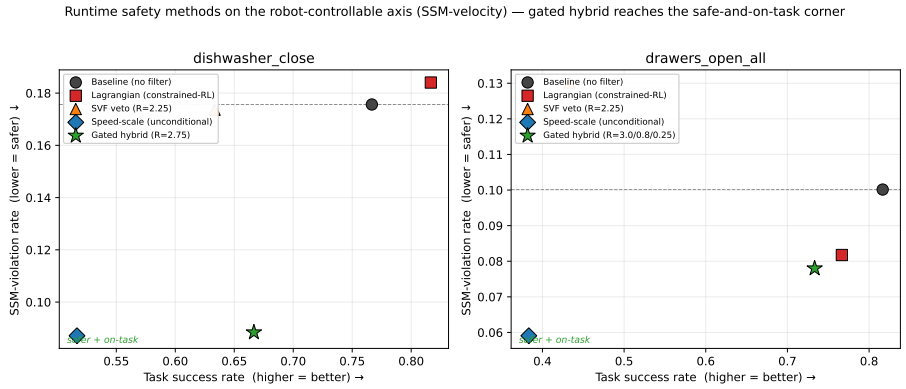


Figure 6.5: The intermittent-co-location headline: each method on the success (x) / SSM-violation (y) plane; the desirable corner is bottom-right (high success, low violation), and only the critic-gated filter reaches it on both tasks. Two points need their context. The *drawers Lagrangian* point sits near the gated one, but it is a basin-selected checkpoint from seed-unstable, single-seed training on which no feasible budget exists (PID-λ is inert or task-fatal across the budget sweep); gating-on-Lagrangian ≈ gating-on-baseline (ablation 3, Section 6.3.5) shows it learned no genuinely proactive avoidance; and it offers no deployable dial, whereas the gate threshold R traces a tunable frontier (Figure 6.6). The *unconditional speed-scale* points reach the lowest violation rates but at 2.5–5× the gated filter's success cost.

1. *Pareto surface (d_{slow} × R, dishwasher).* Across $d_{\text{slow}} \in \{0.4, 0.5, 0.6, 0.8\}$ m, the gate threshold R is the dominant dial and d_{slow} is secondary ($R = 2.75$ gives −49 to −51 % SSM at 0.63–0.67 success regardless). The R-sweep *is* the frontier. One interaction matters: $R = 3.0$ with $d_{\text{slow}} = 0.4$ re-introduces veto-like overshoot (max velocity 6.10), so $d_{\text{slow}} \geq 0.5$ is kept, an echo of the veto's coherence break (Section 6.3.2) inside the gated filter itself.

2. *Anticipatory critic labels.* Re-labelling the critic at looser thresholds (0.20 / 0.30 m, so it warns earlier) does not improve the operating point; the earlier-warning critics slide along the same success/safety trade (dishwasher $R = 2.75$: 0.63/−35 % and 0.63/−44 % vs. the 0.10 m critic's 0.67/−50 %). The most *selective* near-contact label is the best gate; anticipation belongs to the policy, not the gate.

3. *Gating on the Lagrangian policy.* Running the gated backstop on the best Lagrangian basin checkpoint instead of the curriculum baseline is a wash (dishwasher 0.70/−36 % vs.

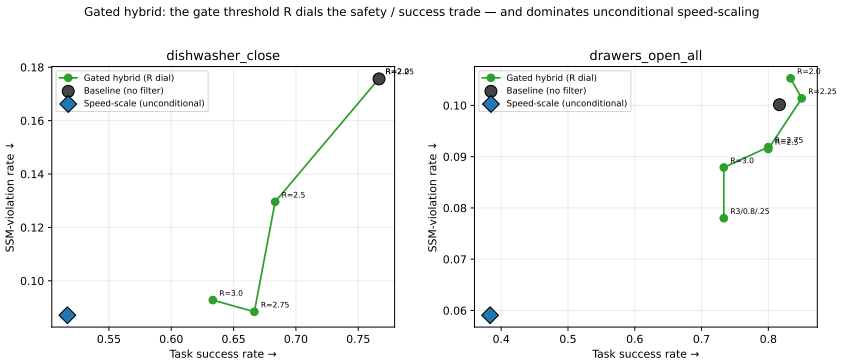


Figure 6.6: The gate threshold R is a clean operating-point dial: low R is selective (near-baseline success, small SSM gain), high R approaches unconditional scaling. The dominance claim is task-scoped, per the two-axis protocol: on *dishwasher* the gated frontier matches unconditional scaling's SSM-violation at strictly higher success; on *drawers* the gated frontier never reaches unconditional's absolute floor (0.059 vs. best gated 0.078) but pays far less success at every *shared* SSM-violation level.

0.67/−50 %; drawers 0.77/−25 % vs. 0.80/−8 % at matched settings; neither dominates). This is the direct evidence that the budget-sweep Lagrangian produced no genuinely proactive-avoiding policy on these tasks, and it is what licenses the drawers-Lagrangian caveat on Figure 6.5.

4. *DAgger-style re-collection on filtered rollouts.* Retraining the critic on the *filtered* policy's own rollouts, the natural "fix the distribution shift" move, is null to harmful: the re-collected critic *under*-gates (dishwasher −2 % SSM vs. −50 %), because the filtered distribution already looks safe (the filter slowed the robot during collection), so the critic rarely fires. A runtime gate must be trained on the *unfiltered* policy's behaviour: it has to recognise what the raw policy *would* have done.

6.4 The Boundary Condition: One Law, Measured Across Three Tasks

The two preceding sections carry opposite headlines: on saucepan the constrained policy is the working arm and every runtime filter pays an unacceptable cost, while on dishwasher/drawers the critic-gated filter is the working arm and constrained RL fails. The claim this section establishes is that *the two headlines are one finding, and the variable that decides between them is measurable.* The experiment that closes the loop is the cross-task gating test (E4.4): run the exact intermittent-regime winner, critic-gated speed-scaling, on `saucepan_to_hob`, under a decision rule fixed in advance: *the gate counts as recovering throughput iff some threshold reaches success* ≥ 0.60 *and SSM-actual* ≤ 0.08.

6.4.1 The Gate Does Not Recover Throughput on Saucepan

Table 6.8 reports the full sweep (the saucepan SVF critic, $R \in \{0 \ldots 3.0\}$, plus an on-policy re-collected critic and the unconditional control; 180 episodes per row). **No row passes the pre-registered rule.** As R rises, the operating point moves monotonically from policy-alone (0.722 success, 0.138 SSM-actual) to the unconditional point (0.433, 0.072); the closest rows are $R = 3.0$ (0.472/0.084, below the success floor) and $R = 2.5$ (0.567/0.105, below both bars). The gated curve *tracks* the unconditional scaler rather than dominating it; no high-success, low-violation corner opens. The decision rule failed on saucepan, and that outcome was the *predicted* failure mode under the regime map, not an anomaly to be explained away.

6.4.2 The Mechanism, Measured

The explanation is not asserted from task descriptions; it is read off the gate itself (Figure 6.8). At the matched threshold $R = 2.75$, the gate is active on 61.5 % [57.2, 65.4] of steps on saucepan,

Table 6.8: Experiment E4.4. Critic-gated speed-scaling on `saucepan_to_hob` (180 episodes per row, 3 trained seeds; `noisy`; `snapshot_best` checkpoints, see the caveat box). The pre-registered rule, success ≥ 0.60 *and* SSM-actual ≤ 0.08, is met by no row: gating slides from policy-alone toward the unconditional point without opening a corner. The on-policy re-collected critic (v3op) does not help either, at $R = 2.0$ its SSM-actual (0.149) is *worse* than policy-alone (0.138), confirming the limit is task structure, not critic calibration.

Critic	R	Success	Prox. viol.	SSM-actual	Reads as
v3	0.0	0.722	0.271	0.138	policy alone (gate never fires)
v3	1.5	0.717	0.299	0.127	gate barely active
v3	2.0	0.706	0.295	0.125	
v3	2.5	0.567	0.282	0.105	success falling
v3	2.75	0.472	0.275	0.102	
v3	3.0	0.472	0.283	0.084	approaching unconditional
uncond.	–	0.433	0.276	0.072	the gate's $R \to \infty$ limit
v3op	2.0	0.733	0.297	0.149	on-policy critic: no help
v3op	2.5	0.644	0.279	0.130	on-policy critic: no help

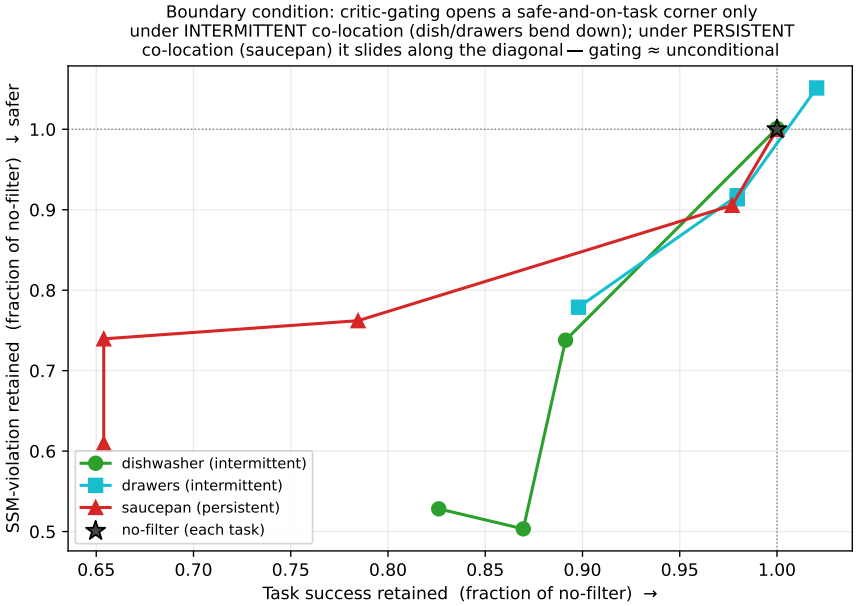


Figure 6.7: The boundary condition in one picture: each task's gated R-dial trajectory on the success/SSM-violation plane, normalised to its own no-filter point. Dishwasher and drawers *bend down*, SSM-violation falls steeply while success is retained, because the gate passes the genuinely-safe majority of steps at full speed. Saucepan *slides along the diagonal* toward its unconditional endpoint, success and SSM-violation falling together, because the gate is active on most steps and gating degenerates to unconditional scaling.

versus 26.5 % [16.3, 36.7] on dishwasher and 19.2 % [12.7, 25.3] on drawers; at each task's recommended operating point the contrast is 51–63 % versus ≈27 %. On saucepan the gate fires *more often* than the unconditional scaler's own proximity trigger (44.2 % of steps within $d_{\text{slow}} = 0.40$ m): the critic has no selectivity left to offer, so the "gated" filter is the unconditional filter with extra steps. This is the same mechanism the persistent-co-location section identified from the other side, where the *unconditional* scaler's success floor was traced to a separation-triggered response that "fires almost constantly" under persistent co-location (Section 6.2.4). The two arcs of this report discovered one law from opposite directions:

> **The boundary condition.** Critic-gated filtering recovers task throughput if and only if human–robot co-location is intermittent, because the gate's value *is* the existence of genuinely-safe windows to pass through at full speed. Under persistent co-location no such windows exist, every separation-triggered mechanism (gated or

not) is active almost continuously, and the only remaining lever on the exposure axis is anticipatory avoidance, which must be trained into the policy.

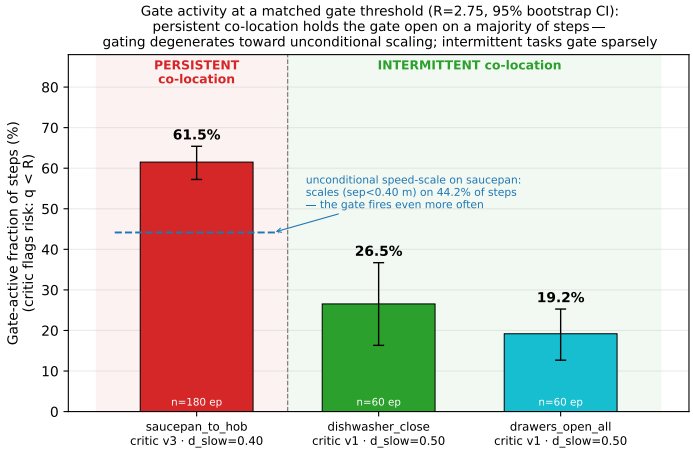


Figure 6.8: The regime variable, measured: per-step gate-active fraction of the critic-gated filter at the matched threshold $R = 2.75$ (bars; 95 % bootstrap CIs; 180/60/60 episodes). The saucepan gate fires on 61.5 % of steps, exceeding even the unconditional scaler's proximity trigger rate (44.2 %, dashed), so gating degenerates to unconditional scaling; on dishwasher/drawers the gate passes 73–81 % of steps at full speed, which is exactly the recovered throughput of Table 6.7.

6.4.3 The Regime Map

Figure 6.9 assembles the full cross-task picture. Two sentences in it are deliberately blunt. First, **critic-gating does not rescue the saucepan composition**: the policy–filter composition's 0.85 → 0.44 success cost (Section 6.2.6) stands, an on-policy critic does not change it, and the contribution of E4.4 is the *explanation* of that ceiling and its validation across tasks, not a workaround for it. Second, the regime map is the deployment rule: measure (or simulate) the co-location pattern first, then pick the arm, a constrained policy where co-location is persistent, a critic-gated backstop where it is intermittent, and never a binary veto on a chunked policy.

Snapshot caveat (scope of the cross-task numbers). The exact persistent-task basin checkpoints were deleted in a storage cleanup before E4.4 ran; the sweep used each seed's `snapshot_best`, which peak-success selection makes *less* avoiding (its $R = 0$ row deploys at proximity 0.271 vs. the basin's 0.228). The verdict transfers for a measurable reason: the unconditional control on `snapshot_best` reproduces the composition's success/velocity operating point almost exactly (0.433/0.072 vs. 0.44/0.065), and the gating question is a statement about the *filter* on exactly those two axes; only proximity, the policy's axis, differs. A more-avoiding basin policy would also leave the gate *less* velocity to cut, so any residual bias runs against gating helping more, not less. A basin-checkpoint rerun requires retraining the three policies and is recorded as a verification item, not a threat to the conclusion.

6.5 Summary of Results

RQ1, the persistent regime: safety must be trained in. Where co-location is persistent, no runtime mechanism reduces exposure gracefully: vetoes freeze, dodges flee, and every separation-triggered filter is active on most steps. The fixed-multiplier constrained policy cuts proximity violations by 22.8 % (0.296 → 0.228) at 0.76 success across three seeds. The reduction survives noisy, lagged perception and a held-out coworker distribution (−31 % there), while the worst-case separation tail is exogenous, the coworker's own approach, and is moved by no mechanism evaluated: proximity *frequency* is controllable, the single worst approach is not. Both axes

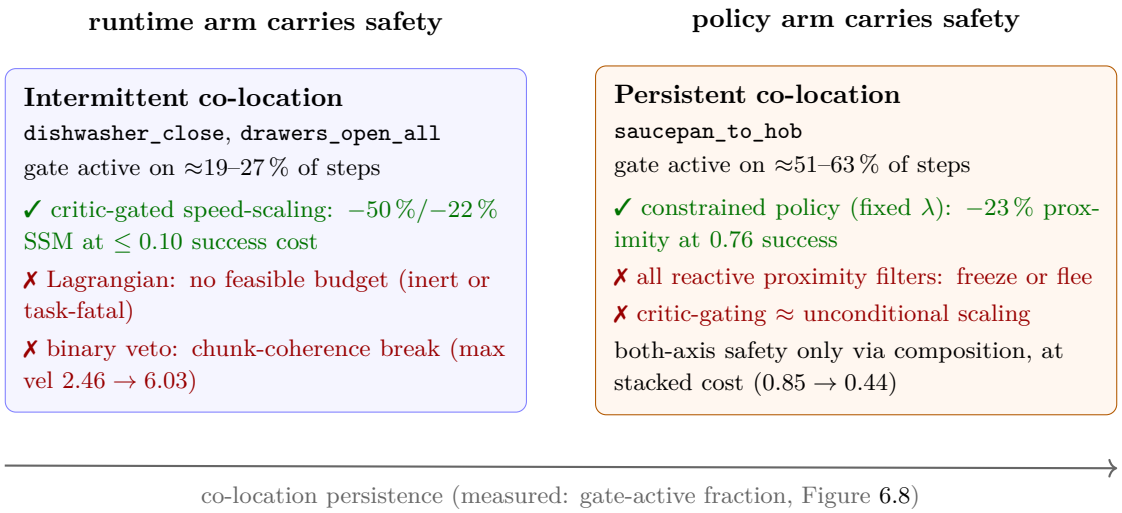


Figure 6.9: The regime map (the thesis in one image). The observable regime variable, how persistently the coworker is co-located, measured as the fraction of steps a separation/risk-triggered filter would be active, determines which arm of the Hybrid Safety Critic carries the safety burden. The boundary was validated by the pre-registered E4.4 rule, which failed on saucepan exactly as the regime map predicted. Evidence per cell: gated wins (§6.3.4), Lagrangian infeasibility (§6.3.1), veto coherence break (§6.3.2), policy win and composition ceiling (§6.2.4, §6.2.6), freeze/flee taxonomy (§6.2.5), gating ≈ unconditional (Table 6.8). On the intermittent tasks the exposure axis itself is left without a graceful mechanism, bounded by the exogenous floor (§6.3.1).

at once are available only by composing policy and graded scaler at a stacked success cost (0.85 → 0.44), the regime's ceiling.

RQ2, the intermittent regime: gate the runtime backstop. Where encounters arrive in bursts, constrained RL finds no feasible budget: the constraint is inert above the natural cost and task-fatal below it, a binding multiplier makes motion *faster* (mean velocity 0.44 → 0.79 m/s), and from-scratch WCSAC corroborates the trainability difficulty externally. A binary veto breaks chunked-policy coherence (max velocity 2.46 → 6.03 m/s). The working mechanism is the learned critic gating the graded speed-scaler: −50 % SSM-violation at −0.10 success on dishwasher, −22 % at −0.09 (or −10 % at −0.02) on drawers, with the gate threshold as a deployable dial (Figure 6.10).

RQ3, the boundary: co-location persistence decides. The gate-active fraction separates the regimes cleanly (61.5 % of steps on saucepan versus 19–27 % on dishwasher/drawers at the matched threshold), it predicts which arm carries the safety burden, and the pre-registered cross-task rule failed exactly where the map predicts: under persistent co-location, gating degenerates to unconditional scaling, and constrained-RL feasibility flips across the same boundary.

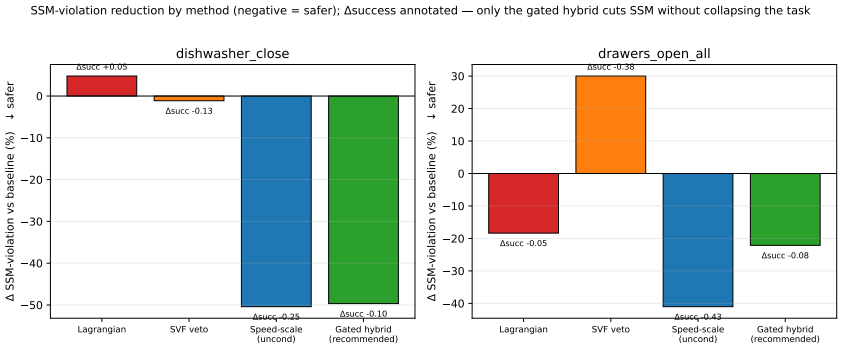


Figure 6.10: Summary of the intermittent-co-location arm: SSM-violation reduction by method with the success cost annotated. Only critic-gated speed-scaling combines a substantial velocity-axis reduction with a small success cost; the unconditional scaler buys slightly more reduction at several times the cost, the veto and the Lagrangian buy little or none. The persistent-regime counterpart of this figure is Table 6.1.

7 Discussion

7.1 What the Answers Mean

The factual answers to the three research questions are summarised in Section 6.5; this section adds the interpretation that does not belong in a results table.

On RQ1, the persistent regime. Two points sit behind the headline numbers. First, the observation pilot and the perception sweep together say something sharper than either alone: *adding human-state observation to an imitation pipeline without a safety-reward gradient is not a safety mechanism* (the channel was inert or counterproductive under behaviour cloning), and yet once a constraint gradient exists, the learned avoidance does not depend on perfect state estimation (statistically indistinguishable under `oracle` and `noisy` perception). Imitation methods marginalise observations uncorrelated with the demonstrated action; constrained RL has a use for the channel. Second, the training lessons travel beyond this benchmark: fix or schedule the multiplier when the only feasible budget sits on a feasibility boundary, and select constrained checkpoints by a safety-aware rule, because peak-success selection picks *against* the constraint.

On RQ2, the intermittent regime. *Why* the same constrained method flips from working to inert-or-fatal across regimes admits a mechanism-level hypothesis, stated as discussion rather than demonstrated result: persistent co-location makes the cost signal *dense* (nearly every step carries graded cost, so the rolling cost moves smoothly and a feasible region exists just below the inherent cost), while intermittent co-location delivers cost in sparse bursts, leaving a cliff-shaped budget landscape (inert on one side, windup on the other) with nothing graded for the multiplier to push against. Base-task fragility compounds this, as from-scratch WCSAC corroborates. Both halves are testable, and the persistence dial (Section 8.1.1) is the controlled test.

On RQ3, the boundary. The override form decides feasibility, the gate decides cost, and the regime decides whether the gate has anything to find. The honest claim is therefore not "the hybrid dominates" but "each arm dominates its own regime, and both-axis safety under persistent co-location has an explicit, tunable price". What no arm moves is the exogenous tail: proximity *frequency* is controllable, the single closest approach is not, and only a Power-and-Force-Limiting collision brake (Section 8.1.2) could address it. The regime classification itself rests on three tasks, a deliberate limitation examined below (Section 7.2.7).

7.2 Threats to Validity

This section is deliberately frank: the work has well-defined limits, and naming them precisely is part of the result, because each one bounds where the regime map and the deployable recipes can be trusted.

7.2.1 The PFL Contact-Detection Limitation

The ISO 15066 compliance claim is qualified to "geometric / SSM-only." The PFL side of the standard requires biomechanical force monitoring; the project plumbs the PFL force limits, the cost signal, the labeller, and the runtime filter end to end, but the underlying contact detection mechanism produces `data.ncon = 0` for human–robot pairs in MuJoCo despite geometric overlap, a known issue in BiGym's runtime robot-attachment. Therefore $c_t^{PFL} \equiv 0$ throughout this work. *Resolving this is a 1–2 week BiGym-internals project and is the highest-priority follow-up (Section 8.1.2)*; without it, no PFL-side compliance claim in this report is valid, and the contact-force tail metrics (originally planned for Section 6.2.9) are omitted rather than reported as 0.

7.2.2 Curriculum Dependence

The headline results presuppose the three-stage curriculum of Section 4.6. Direct training on the full COWORKER disruption collapses to evacuation; every headline result generalises only to settings where a comparable staged curriculum is feasible. This is not a fatal limit (curriculum learning is a standard solution to long-horizon sparse-reward RL [41–43]), but it should temper any reading of the architecture as "trainable from scratch on arbitrary human-co-located tasks."

7.2.3 Sim-to-Real: Three Orthogonal Gaps

Perception: the mock-BodySLAM++ model is calibrated against published characteristics [25] but is not real perception; a rendered-image run feeding a real estimator is the natural closure (Section 8.1.4). **Dynamics:** MuJoCo's H1 differs from the real platform in unmeasured ways. Domain randomisation [70, 71] is the standard follow-up and remains the deployed recipe for real-world humanoid RL [81], and the gap must be acknowledged before any real-world claim. **Contact fidelity:** we found no validation of any rigid-body engine's contact forces against measured human–robot collision forces. Simulator validation on real impacts is trajectory-level only and finds MuJoCo [82] nearly insensitive to its contact-stiffness parameter [83], force-level MuJoCo validation exists only outside HRC [84], and industrial PFL validation relies on physical biofidelic measurement precisely because simulating constrained contacts is considered unreliable [35, 85]. The benchmark's contact forces are therefore safety-shaped rather than safety-validated, even once the PFL detection limitation is resolved.

7.2.4 The Runtime Filter's Role Is Axis- and Regime-Conditional

The results force a precise statement of what a runtime filter contributes, narrower than the field's default framing on one side and broader on the other. *Narrower*: on the proximity (exposure) axis under persistent co-location, no reactive mechanism helps gracefully, the fallback sub-study (Section 6.2.3) shows the only proximity-cutting fallback is a task-abandoning retreat that worsens the SSM-actual rate, and this freeze-versus-flee limit is one the runtime-filter literature reaches independently (Section 2.4.1): a worst-case reactive filter over-restricts to stay safe [64], relaxing conservatism is provably benign only for a least-restrictive filter [63], and trusting a human-motion model to relax it fails exactly when the human is unpredictable [64], as a co-located coworker is. The internalisation analysis (Section 6.2.7) adds the composition half: a learned veto's intervention rate does not fall as the policy learns to avoid, it rises, because the avoiding policy exits the critic's calibration distribution. *Broader*: on the velocity axis with the right override form, the filter is not a backstop but the *primary* working mechanism wherever co-location is intermittent, the critic-gated scaler is the best operating point in the entire evaluation there, and even under persistent co-location the unconditional scaler is the only route to velocity-axis compliance, at a price the regime sets. Our contribution is to establish both halves empirically on a high-DOF manipulator under ISO 15066, and to name the variable that separates them. The reading is calibrated on the COWORKER distribution and these base policies; a distribution with more robot-initiated approaches would shift the exogenous share and require re-calibrating filter operating points (R, d_{slow}) against the deployed policy.

7.2.5 Generalisation Scope and the Resolved Cross-Task Transfer

The constrained-RL evidence supports the claim that, on the persistent-co-location task under this backbone, a fixed-λ Lagrangian at a feasible budget achieves a reproducible proximity reduction where both auto-tuned budgets and reactive filters do not (RQ1). The cost-form ablation bounds the scope: the cost-signal *form* (continuous vs. binary) is not the operative variable, the budget relative to the task's inherent cost is, so we do not claim a general "continuous beats binary" result. *Cross-task transfer* of the full saucepan pipeline (E5.3) resolved *negative*: on `dishwasher_close` and `drawers_open_all` no budget binds usefully and the best constrained point is within noise (Section 6.3.1). Under the regime map this negative is informative rather than anomalous, but its converse limits should be stated equally plainly: "reproducibly" in the

persistent-regime headline means across three saucepan training seeds, not across tasks; and whether the fixed-λ recipe holds under a different backbone, a one-shot-intrusion disruption pattern, or a different embodiment remains untested.

7.2.6 External Baseline (WCSAC): Implemented, with Stated Limits

The WCSAC [1] comparison (Section 6.3.1) is reported with four explicit limits. (i) *Single seed per cell*: from-scratch SAC is high-variance, and the anomalous dishwasher d=30 cell (0.03 success at $\lambda = 0$) is best read as that variance, not as a budget effect; a clean WCSAC trade-off curve needs ≥ 3 seeds. (ii) *From-scratch by necessity*: WCSAC is actor-critic and cannot consume the critic-only warm-start snapshots, so the comparison conflates "safe-RL formulation" with "demonstration prior", which is exactly why it is presented as an external reference for trainability and not as a like-for-like ablation. (iii) *Oracle perception at evaluation*, an upper bound with respect to the noisy condition the project's own methods are evaluated under. (iv) *No Safety-Gym revalidation*: the reimplementation was cross-validated against the training-time evaluator on these tasks (and a de-normalisation confound was found and fixed before any number was recorded, Section 4.5), but it was not re-run on Safety-Gym to reproduce published numbers, so residual reimplementation error cannot be fully excluded. Within those limits the result is robust: WCSAC's dishwasher success demonstrates the external method is viable on the easier task, and its uniform drawers failure corroborates the base-task-fragility reading of the budget-retarget result (Section 6.3.1).

7.2.7 The Regime Classification Is Observational, on Three Tasks

The regime map rests on three tasks, and the regime variable is *measured* but not *manipulated*: persistence co-varies with task identity, geometry, episode length, and policy speed, so a confound-free causal reading is not yet available. Three things mitigate, without eliminating, the concern: the boundary condition was pre-registered before the cross-task gating test ran (the saucepan failure case is a prediction satisfied, not a post-hoc classification); the mechanism is measured at the step level (Figure 6.8), and the degeneration it implies is visible *within*-task as R rises (Table 6.8); and the same boundary predicts an unrelated phenomenon, the constrained-RL feasibility flip, which a task-identity explanation would have to explain separately. The clean falsification test is the *persistence dial* (Section 8.1.1): hold the task fixed, manipulate only the coworker's dwell schedule, and watch whether the dishwasher's bent frontier flattens onto the diagonal. Until it runs, the regime map should be read as a strongly-evidenced empirical regularity with a measured mechanism, not a proven causal law.

7.2.8 Headline-Architecture Caveat (B-value-mean vs. B-CVaR)

The headline architecture is B-value-mean (mean-cost dual-critic). The standard critique of mean-cost Lagrangian methods is that they bound only $\mathbb{E}[Z_c]$, not the tail [1]. We report $\text{CVaR}_{0.95}$ as a diagnostic and find that the mean-trained policy's worst-case separation tail is no worse than the baseline's, but the reason is specific to this setting and does *not* vindicate mean-cost training in general: the tail here is *exogenous* (Section 5.5.1), the human walking into a co-located robot, so it lies outside what *any* robot-side training objective, mean or CVaR, can move. A CVaR-trained policy would not reduce this tail either; the limitation is the reactive control surface, not the risk measure. On a heavier-tailed cost distribution whose tail *is* robot-driven (explicit adversarial coworker behaviours, contact-rich tasks where the robot's own motion creates the worst case), the mean-trained policy would likely under-perform a CVaR-trained one, and the future-work direction (Section 8.1.3) is motivated by that scenario rather than by this one.

7.2.9 Computational Feasibility

At 76 DOF, a constraint-projection CBF quadratic program would have to be constructed and solved over many human-link pairs every control step, a substantially heavier per-step cost than a single MLP forward pass and one that additionally requires hand-specified barrier functions; this

project instead uses a learned safety value function (for the veto) and a closed-form speed-scaling rule as the runtime filters, both of which are negligible alongside the policy's own forward pass. The control loop runs at 20 Hz (a 50 ms per-step budget); the policy forward pass, the filter evaluation, and the fallbacks are simple tensor operations that sit comfortably within that budget, so no component was separately micro-profiled. The substantive computational argument is comparative: the learned filter avoids the per-step QP that a 76-DOF CBF would demand.

7.3 Broader Impact

7.3.1 Responsible Deployment and Misuse Risk

The strongest practical implication of the regime map is also the largest ethical warning: a runtime safety wrapper can make a learned humanoid *look* safer without making it certifiably safe. The project improves measurable SSM-velocity behaviour in simulation, but it does not validate real contact forces, real perception failures, or real human reactions. A company deploying a humanoid in a home, warehouse, or care setting could misuse these results by presenting a wrapper as evidence of ISO compliance. That would be a category error. The report's claim is narrower: the mechanisms improve specific measured axes in a simulated co-working benchmark, and the PFL force loop remains unevaluated.

The regime map adds one concrete pre-deployment check: *measure the co-location pattern first*. The gate-active fraction of a candidate filter on logged or simulated traffic is cheap to compute and, on this evidence, predicts whether critic-gated scaling can recover throughput or whether persistent co-location will force a heavy safety–throughput trade. This should be treated as a screening test, not a certificate. If the pattern is persistent, a deployer should expect either slowed operation, proactive task replanning, or a redesign of the workspace rather than a transparent safety layer.

A second lesson, from E4.3, is that validation must be joint. A runtime filter cannot be assumed to "hand over" to a policy trained to be safe, because the policy may move out of the critic's calibration distribution. Real-world validation must therefore test the exact policy–filter composition on the exact deployment distribution, with the filter re-calibrated whenever the policy, perception stack, task, or coworker pattern changes.

7.3.2 Vulnerable Users and Domestic Environments

ISO 15066 targets industrial collaboration, where users are trained, adult, aware of the robot, and operating under a safety process. Domestic or care environments violate those assumptions. Children may approach unpredictably, elderly users or disabled users may have slower reaction times, and bystanders may not understand the robot's intended workspace. The exogenous-proximity finding is therefore especially important: if a vulnerable person walks into a stationary robot, a robot-side policy cannot remove the exposure, and only a last-resort PFL brake, physical workspace design, supervision, or external sensing can reduce the residual risk.

For this reason the report should not be read as a case for deploying learned humanoid manipulators in unconstrained homes. It is a step towards evaluating such systems. Extending it to vulnerable-user deployment would require stricter cost budgets, real BodySLAM++ or multi-camera perception tests, contact-force validation, emergency-stop hardware, human-subjects ethics review, and a regulatory answer to which standard applies outside the industrial-collaboration setting. The safety–throughput trade should also be disclosed to users: in the persistent regime, the safer configuration may be substantially slower or less successful, and hiding that cost would encourage unsafe expectations about what the robot can do.

7.3.3 Reproducibility as Honesty

The full reproducibility checklist (Appendix B) documents commit hashes, hyperparameters, dataset versions, and GPU hours per phase, and the codebase is open-source under the MIT licence at https://github.com/ayushpatel4/safety-bigym-2, together with the BiGym fork

carrying the collision-channel and PD-actuator fixes (patches submitted upstream). This is a small but non-zero contribution to the safe-RL community's reproducibility crisis.

8 Conclusions and Future Work

ISO 15066-referenced safety for high-DOF humanoid manipulation under human co-working has not been systematically addressed in the safe-RL literature. The field has strong ingredients, including constrained RL and runtime filtering, but it did not have a rule for when each should be used. This work contributes five linked pieces, in the order of the introduction: the regime map, an empirical rule for matching the safety mechanism to the human–robot co-location pattern; a two-axis measurement protocol built on the finding that ≈42 % of close proximity is human-driven and therefore outside robot control; `safety_bigym`, a safety-aware extension of the BiGym [6] benchmark on which the question can be asked at all; a value-based re-derivation of the constrained objective for critic-only agents, with a proved C51 support-bound condition for dense shaping; and a set of mechanism-level design lessons with deployable operating points. Which mechanism helps is governed by the human–robot co-location regime, not by mechanism sophistication. Under *persistent* co-location (`saucepan_to_hob`, gate-active ≈62 % of steps), only the anticipatory constrained policy reduces exposure gracefully: a fixed $\lambda = 0.1$ value-based Lagrangian cuts the proximity-violation rate by 22.8 % (0.296 → 0.228) at 0.76 success across three seeds, where PID auto-tuning is seed-unstable at the feasibility boundary; every reactive filter pays a freeze-or-flee cost on that axis, and both measured safety axes are improved only by composing policy and graded speed-scaling at a multiplicative success cost (0.85 → 0.44). Under *intermittent* co-location (`dishwasher_close`, `drawers_open_all`, gate-active 19–27 %), constrained RL finds no feasible budget, and when its multiplier binds the robot gets *faster*, not safer, with the from-scratch WCSAC [1] reference corroborating the trainability difficulty. A binary veto breaks chunked-policy coherence (max velocity 2.46 → 6.03 m/s), and the working mechanism is a learned critic *gating* an ISO 15066 speed-scaling backstop: −50 % SSM-violation at −0.10 success on dishwasher, −22 % at −0.09 on drawers, with the gate threshold as a deployable dial. The boundary was validated by a pre-registered rule whose failure case occurred on saucepan, where gating measurably degenerates to unconditional scaling. The worst-case separation tail is exogenous and unmoved by every mechanism: exposure frequency is controllable; the single closest approach is not.

As humanoid platforms move from research labs into commercial deployment (and the Unitree G1's deployment without collaborative-operation validation [9, 13] is a representative data point), principled methods for safety under imperfect perception and sustained human co-presence are urgently needed. The most useful output of this work is a decision rule where the field had only a default. Runtime filtering is not a universal safety layer, and constrained RL is not a universal alternative. The co-location pattern, cheap to measure on logged or simulated traffic, predicts which arm is likely to work and what the safety–throughput trade will cost. The positioning table (Table 2.2) situates the work honestly within the emerging literature rather than claiming primacy. The practitioner lessons are concrete: gate the scaler, do not veto a chunked policy; train the gate critic on unfiltered rollouts; fix the multiplier when the feasible budget lies on a boundary; select constrained checkpoints by a safety-aware rule; and check the support-bounding invariant before adding dense shaping to a distributional critic.

8.1 Future Work

8.1.1 The Persistence Dial: From Regime Map to Causal Test

The single highest-value next experiment, and the named next step of this project. The regime map currently rests on three tasks in which co-location persistence co-varies with task identity (Section 7.2.7). The persistence dial removes the confound: hold the task fixed (`dishwasher_close`, the cheapest), and run the *same* gated R-sweep under two scripted-coworker schedules, the

existing intermittent patrol and a persistent variant in which the coworker's near-zone dwell is extended to span the episode. The map makes a falsifiable prediction: the dishwasher's bent frontier (Figure 6.7) flattens onto the diagonal as the gate-active fraction rises toward the saucepan range, with the transition tracking the measured gate activity rather than any task property. A confirming result upgrades the boundary condition to a controlled intervention on the regime variable; a disconfirming one localises what task identity contributes. The infrastructure exists (the dwell knobs are `ParameterSpace` axes, Table 3.2); the cost is one gated sweep per schedule.

A companion question on the same tasks: the intermittent regime's *exposure* axis is left without a working mechanism (the Lagrangian is inert or task-fatal there, and the gate by design does not reduce exposure). The open lever is a potential-based separation reward, which sidesteps the budget-cliff mechanism entirely. The equally concrete risk is that the task geometry (the robot must occupy the appliance the coworker walks through) imposes its own exposure ceiling, in which case the honest outcome is a measured ceiling rather than a win. Either result sharpens the regime map's exposure column.

8.1.2 Closing the Force Loop: Fixing PFL

The highest-priority infrastructure follow-up. Fix BiGym's runtime robot-attachment to expose human–robot contacts via `data.ncon`, then re-run the pipeline with `use_pfl=true`. Estimated effort: 1–2 weeks of focused BiGym-internals work. Until this is done, the project's ISO 15066 compliance claim is SSM-only; with it, the full standard is in scope, and a *last-resort collision-imminent brake*, firing only at near-contact, becomes evaluable. That brake is the only mechanism class that could address the exogenous near-contact tail that no current mechanism moves (Section 6.2.9), completing the division of labour: policy → exposure, filter → velocity, PFL brake → contact. The schema is forward-compatible: the labeller, cost signal, and runtime filter all read c_t^{PFL} unconditionally.

8.1.3 Stronger Guarantees: CVaR Training and Formal Filters

On the training side, the mean-cost headline formulation lifts to an explicit distributional cost critic (Section 4.4.4): a Gaussian parametrisation (WCSAC-style [1]) or quantile head (QR-DQN-style [48]), with ready instruments in off-policy trust-region CVaR constraints [86] and distributional multi-constraint methods [87]. The cost critic is already C51-shaped, so a parallel head slots in cleanly, subject to the support invariant of Proposition 4.1 holding for the cost distribution. The motivating scenario is a cost tail that is *robot-driven* (contact-rich tasks, adversarial coworkers); on the current benchmark the tail is exogenous and no risk measure can move it (Section 7.2.8).

On the runtime side, our filters demonstrate velocity-axis behaviour empirically but certify nothing. A CBF-QP with a certified barrier or an HJ-reachability filter [50, 53] would add provable separation/velocity guarantees; the concrete obstacles are a control-affine dynamics model under *exogenous* human motion and the curse of dimensionality at 76 DOF, and the well-posed target is the velocity axis, where the filter's role is unambiguous. The sharpest unmeasured design, surfaced by the literature survey behind Section 2.4.1, combines confidence-aware fallback [64] with this report's override finding: an *anticipatory filter whose conservative fallback is graded slow-down rather than freeze*, trusting a human-motion model only under online misspecification monitoring. We found no published work that measures that combination. Two smaller items complete the thread: a proportional-damping fallback where a veto is unavoidable, and the implemented-but-unrun filter-during-training ablation (Section 4.4.8), expected to trade fewer catastrophic exploration trajectories for some robustness cost [24, 54].

8.1.4 Real-Robot Transfer and the Coworker Realism Spectrum

The large-scale extension: a real Unitree H1, real BodySLAM++ [25] from on-robot RGB + IMU, and real human collaborators under ethics review with a safety supervisor in the loop; the

project contributes the calibrated noise model, the regime map, and the curriculum, and the open question is whether the map survives deployment. Two cheaper studies sit on the way: a *coworker-aggressiveness spectrum* (the `coworker_eval` axes swept continuously), charting where reactive filtering becomes sufficient as the human grows less adversarial, and an SMPL-H [78] coworker replication of the headline cells via the benchmark's body-agnostic selector (Section 3.4), a ∼15 GPU-hour check on embodiment-specificity.

Declarations

Originality. I declare that this report and the work it describes are my own, carried out under the supervision of Dr Stephen James, and that all material from other sources, published or unpublished, is acknowledged and cited in the references.

Use of generative AI. Generative AI tools were used as assistive tools for literature search, code scaffolding, experiment-log summarisation, plotting-script drafts, LaTeX formatting, and prose drafting/proofreading. They were not used as an autonomous source of experimental results or as a substitute for authorial judgement: all citations were checked against primary sources, all correctness-bearing code was reviewed and tested by the author, all numerical claims were traced to run logs or CSV artifacts, and all report prose was edited and endorsed by the author. The full declaration, including observed failure modes and verification procedures, is in Appendix D.

Ethics. The project involves no human-subjects research and no personal data; all human-motion data is from the publicly released AMASS dataset under its research licence. The full ethical considerations, including the principal risk of over-claiming safety compliance, are in Appendix E.

Sustainability. Total compute was roughly 450–500 GPU-hours on A100s, corresponding to approximately 40 kg CO_2e; the accounting and the practices that bounded it are in Appendix F.

Code and data availability. The code is open-source under the MIT licence at https://github.com/ayushpatel4/safety-bigym-2; model-weight and dataset availability are detailed in Appendix G.

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A Hyperparameter Tables

A.1 Phase 2: Offline CQL Safety Critic

Table A.1: Phase 2 offline CQL safety-value-function hyperparameters. The dataset is collected under `noisy` perception because a runtime filter must learn the conditions it faces at deployment (Section 5.3).

Hyperparameter	Value	Note
Architecture	MLP $[256, 256, 256]$ + scaled sigmoid	bounded value head
Optimiser	Adam, lr 3×10^{-4}	
CQL regulariser	$\alpha_{\text{CQL}} = 5.0$	ablated in E2.1
Target violation rate	0.30	matches τ_{prox} calibration
Discount	$\gamma = 0.99$	
Target-net Polyak	$\tau = 0.005$	
Dataset	v3: CQN-AS rollouts, `coworker_train`, `noisy`	labelled at τ_{prox}=0.3 m ($\sim$16.8 % unsafe)
Sampler	violation-balanced, target 1:1	
Training / batch	200,000 steps / 512	

The four collection-versus-deployment faithfulness fixes that make this critic's deployment decisions valid (action de-normalisation, execution-mode matching, the 500 $\rightarrow$ 20 Hz control-frequency root cause, and coworker-scenario alignment) are described in Section 4.10; the $R = 0$ sweep agreement ($0.286 \approx 0.296$) is the validation that the fixes succeeded.

A.2 Phase 3: Constrained Lagrangian Agent (Headline: Fixed λ)

Table A.2 gives the headline configuration. The headline multiplier is *fixed* at $\lambda = 0.1$; the PID gains below are used *only* for the E3.2 auto-tuning runs that establish the seed-instability finding (Section 6.2.2), and are set to zero for the headline. There is no active cost budget d in the headline, because pinning λ removes the dual-update loop entirely; $d = 0.3$ is the budget used in the PID-ablation runs.

Table A.2: Phase 3 headline hyperparameters (fixed-λ constrained CQN-AS). The cost critic is fully decoupled (its own optimiser and its own features, the RoboBase shared-encoder constraint) and cold-started.

Group	Value	Note
CQN-AS backbone	C51, 101 atoms, $\gamma = 0.99$	[4]
Value support	$v_{\min} = -6$, $v_{\max} = +2$	satisfies Prop. 4.1
Cost critic	mirrored C51, same support	decoupled optimiser & features; cold-started
Replay / batch / stack	1M / 256 / 4	
Demonstrations	num_demos = 36	
Multiplier (headline)	**fixed** $\lambda = 0.1$	dual loop removed
PID gains (E3.2 ablation only)	K_I=10^{-3}, K_P=10^{-2}, K_D=0	$\lambda_{\max}$=100; zero in headline
Cost budget (E3.2 ablation only)	$d = 0.3$	not active in headline
Workspace shaping	β=0.05, r_{ws}=0.4 m, c_{ws}=1.0 m	confound-controlled (E4.1)
Lagrangian fine-tune	up to 40,000 frames	on the stage-2 base
Checkpoint selection	**safety-aware**	lowest deploy proximity at success floor 0.75
Eval interval	2,000 frames, 20 eval ep	basin at $\sim$20k–35k frames

A.3 Phase 4: Per-Regime Deployment Configurations

The deployed runtime configuration is regime-dependent (Section 6.4). On the persistent-co-location task, the both-axis configuration is the fixed-λ policy composed with *unconditional* graded speed-scaling ($d_{\text{slow}} = 0.40$ m); gating does not help there. On the intermittent-co-location

tasks the deployment filter is *critic-gated* speed-scaling at the recommended per-task operating points. The learned SVF *veto* ($R = 2.25$, $\alpha_{\text{CQL}} = 5.0$) is a negative result everywhere (velocity backstop at best, OOD on the avoiding policy, coherence break on chunked policies) and is never the deployment filter; Table A.3 lists the configurations.

Table A.3: Recommended runtime deployment configuration per co-location regime, as established by the results chapter: the constrained policy carries safety under persistent co-location, the critic-gated speed-scaler under intermittent co-location. The veto and the model-based dodges are taxonomy points, not deployment configurations.

Task (regime)	Filter	Operating point	Critic
saucepan (persistent)	uncond. speed-scaling on fixed-λ policy	d_{slow}=0.40, d_{stop}=0.15 m	none (ungated)
dishwasher (intermittent)	**critic-gated speed-scaling**	R=2.75, d_{slow}=0.50, d_{stop}=0.15 m	`svf_dish_drawers_v1`
drawers (intermittent)	**critic-gated speed-scaling**	R=3.0, d_{slow}=0.80, d_{stop}=0.25 m	`svf_dish_drawers_v1`
drawers (near-free option)	critic-gated speed-scaling	R=2.75, d_{slow}=0.80 m	`svf_dish_drawers_v1`
(all tasks) SVF veto	negative result; not deployed	R=2.25	per-task SVF
(saucepan) model-based dodge	taxonomy point (flee); not deployed	d_{target}=0.35 m	none

Shared dish/drawers SVF critic. Same architecture and CQL recipe as Table A.1, trained once for both tasks: 52,794 transitions (random + on-policy curriculum rollouts under COWORKER), near-contact label at min-separation < 0.10 m, 200k training steps. Anticipatory-label variants (0.20/0.30 m) and a DAgger-style re-collection on filtered rollouts were trained and rejected by the ablations of Section 6.3.5.

WCSAC external baseline. Faithful reimplementation of [1] (the Gaussian variant; the journal extension adds a quantile-regression critic [47]): Gaussian distributional cost critic, CVaR_α constraint with $\alpha = 0.9$, budgets $d \in \{5, 15, 30\}$ on the discounted cost return, adaptive multiplier ($\lambda_{\max} = 100$); trained from scratch (num_demos = 0), 150k frames, single seed per cell; evaluated with identity action-normalisation statistics (Section 4.5) over 30 episodes per cell under `oracle` perception.

B Reproducibility Checklist

Code

https://github.com/ayushpatel4/safety-bigym-2 (MIT licence). Headline experiments were run from the `phase3` branch at commit `b3b2ce5`; in addition, the snapshot-evaluation harness stamps the exact git SHA into every benchmark CSV row, so each reported cell is traceable to the code that produced it.

Upstream assets

BiGym [6] is consumed via a public fork at https://github.com/ayushpatel4/bigym, which carries the collision-channel and PD-actuator fixes of Section 3.4. AMASS [80] (CMU subset, downloaded 18 February 2026) drives the demonstration-replay human-state channel. CQN-AS [4] is vendored at upstream commit `8cf806e` (Section 4.9).

Hardware and software

A single NVIDIA A100 (40 GB), CUDA 12.8, PyTorch 2.10.0; MuJoCo via BiGym. Every script has a CPU-only smoke gate (Section 5.6).

Seeds

Trained seeds $\{0, 1, 2\}$ for all three-seed cells; evaluation uses seeds $\{0, 1, 2\}$ with 20 episodes each (Section 5.6).

Headline recipe (R1)

(i) Train the three-stage curriculum (Table 4.6; ≈19 h). (ii) Fine-tune with fixed $\lambda = 0.1$ for up to 40k frames per seed (≈9 h each; Table A.2). (iii) Benchmark every saved checkpoint and select per seed by the safety-aware rule (lowest deployment proximity at success ≥ 0.75), which yields the headline checkpoints `lam0p1_seed{0,1,2}` at steps 30,546 / 8,225 / 32,696. (iv) Reproduce Table 6.1 by running `scripts/benchmark_policy.py` on those checkpoints (minutes per cell; no retraining).

Headline recipe (R2/R3)

All intermittent-co-location and boundary numbers are snapshot evaluations through the same harness: the budget sweep via `scripts/dispatch_budget_sweep.py` (artifacts under `results/budget_sweep_eval/`); the veto, unconditional, gated, and ablation sweeps under `results/{svf_sweep, speedscale, gated_sweep, improve, anticipatory_sweep, dagger_sweep}/`; the boundary sweep via `scripts/dispatch_gated_saucepan.py` (under `results/e4_1/gated_saucepan/`); WCSAC via `scripts/dispatch_wcsac.py` and `scripts/eval_wcsac.sh` (under `results/wcsac_eval/`). Figures regenerate via `scripts/make_report_figures.py` and `scripts/make_f7_gate_activity.py`.

Hyperparameters

Appendix A; the Hydra configs in the repository pin every value not listed there.

C Experiment Code Index

Experiments are referred to in the body of the report by what they show; the repository, run logs, and figure-data notes identify them by phase-prefixed codes (E$n.m$), stated once in parentheses where each experiment is reported. Table C.1 maps the codes to the report sections carrying their evidence.

Table C.1: Experiment index. Each is identified by a phase-prefixed code (E$n.m$) and reported in Chapter 6; R1 = persistent co-location (`saucepan_to_hob`), R2 = intermittent co-location (`dishwasher_close`, `drawers_open_all`), R3 = the cross-task boundary condition.

ID	Description	Reported in
E1.1	BC observation ablation (off / oracle / noisy), no penalty	§6.2.1
E2.1–E2.3	SVF CQL-α / filter-on-baseline / threshold-R Pareto (R1)	§6.2.3
E2.4	SVF binary veto on the chunked policy (R2)	§6.3.2
E2.5	Unconditional speed-scaling (R2)	§6.3.3
E2.6	Critic-gated speed-scaling, R-dial sweep + ablations (R2)	§6.3.4, §6.3.5
E3.1	Cost-signal form vs. budget feasibility (R1)	§6.2.2
E3.2	Cost budget sweep and feasibility boundary (R1)	§6.2.2
E3.3	Cost-budget retarget sweep, $d \in \{0.16, 0.19, 0.22\}$ (R2)	§6.3.1
E3.6	Perception robustness (oracle vs. noisy) of the constrained policy	§6.2.8
E3.7	External baseline: WCSAC, from scratch, 2 tasks × 3 budgets	§6.3.1
E4.1	**R1 headline: proactive / reactive / composed comparison**	§6.2.4
E4.3	Filter intervention rate during Lagrangian training (R1)	§6.2.7
E4.4	**R3: critic-gated scaling on saucepan (pre-registered rule)**	§6.4
E5.1	Tail-risk metrics across the headline configurations (R1)	§6.2.9
E5.2	OOD generalisation on the wider `coworker_eval` band (R1)	§6.2.10
E5.3	Cross-task transfer of the saucepan pipeline (resolved negative)	§6.3.1

D Use of Generative AI

Generative AI tools (Anthropic Claude, OpenAI ChatGPT) were used during this project, and their use is declared here in full. Their role was assistive and reviewable:

- **Literature search and triage.** AI tools helped shortlist candidate work in safe RL, CBF filtering, shielding, and human-aware robotics. Every retained citation was then checked manually against a primary source (arXiv, IEEE/ACM/PMLR/AAAI/NeurIPS/JMLR, ISO catalogue, or publisher page). This verification step changed several entries; the audit is recorded in `REFERENCES_AUDIT.md`.
- **Code scaffolding.** AI tools drafted repetitive infrastructure such as data-loading helpers, wrapper boilerplate, and plotting scripts. Correctness-bearing code, including training loops, loss functions, cost signals, evaluation metrics, and aggregation logic, was reviewed, modified where necessary, and tested by the author before use.
- **Experiment-log summarisation.** AI tools helped summarise large collections of CSVs, JSONL logs, and draft result notes into candidate tables or prose. Every numerical claim in the report was checked back to an artifact in `results/`, a documented script, or a saved figure-data note before inclusion.
- **LaTeX and prose.** AI tools assisted with LaTeX formatting, restructuring, drafting, and proofreading. No AI-generated passage is included verbatim without author editing; the author chose the final framing, claims, section ordering, and wording.

The tools' limitations shaped the workflow. They sometimes overstated ISO compliance or treated simulated SSM improvements as real-world certification, so all safety claims were edited back to the qualified SSM/proximity language of Section 7.2.1. They also surfaced inaccurate bibliographic metadata, which is why the reference audit was performed. They did not diagnose the C51 support-saturation failure (Section 4.4.5) or the CUDA `linspace` integer-overflow bug (Section 4.9); both required direct code, simulator, and log inspection. All experimental design decisions, all choices of what to run and measure, all quantitative results, and all scientific claims are the author's responsibility.

E Ethical Considerations

The project involves no human-subjects research and no personal data. All human-motion data is from the publicly released AMASS dataset [80], used under its research-only licence. No real-robot deployment was conducted; the simulated humanoid–human contact forces are not safety-validated for real-world use. The work targets a safety standard (ISO 15066) and aims to improve safety outcomes; the principal ethical risk is over-claiming compliance and thereby prompting under-validated deployment. This is mitigated by the qualified, SSM-only compliance claim of Section 7.2.1, by the honest reporting of the safety–throughput trade rather than presenting the safe configuration as free, and by the explicit recommendation in Section 7.3 that any real-world deployment require independent validation and, for vulnerable users, stronger guarantees than this work provides.

F Sustainability

Total compute was roughly 450–500 GPU-hours on NVIDIA A100 (40 GB) GPUs (Section 5.7): ≈220 h for the persistent-co-location arc, reconstructed from run logs, plus ≈230–280 h for the intermittent-co-location arc and the WCSAC external baseline, of which the six 150k-frame from-scratch WCSAC runs dominate and are estimated rather than log-reconstructed. At an assumed mean board power of 0.3 kW and a data-centre power-usage effectiveness of 1.4, the midpoint (≈475 h) is approximately 200 kWh; at the 2024 UK grid carbon intensity of 0.21 kg CO_2e/kWh, approximately 40 kg CO_2e. Several practices kept this figure low: resumable training with checkpoint reuse; CPU smoke gates before every multi-hour training launch; deliberate seed-count selection (3–5) balancing statistical confidence against compute; warm-starting the constrained policy from the Phase 1 base-policy snapshot rather than retraining the task from scratch (Section 4.4.9); and snapshot-only Phase 2 dataset collection, which avoids a separate, slow random-rollout pass. The dominant cost is policy training, not evaluation; the snapshot harness makes re-evaluation effectively free relative to a retrain.

G Code and Data Availability

The code, including the `safety_bigym` benchmark, all runtime-filter implementations, the training entry points, and the snapshot-evaluation harness (`scripts/benchmark_policy.py`, Section 3.7), is open-source under the MIT licence at https://github.com/ayushpatel4/safety-bigym-2. The BiGym simulator fork carrying the collision-channel and PD-actuator fixes of Section 3.4 is public at https://github.com/ayushpatel4/bigym. Pre-trained weights for the headline fixed-λ policy (the three selected checkpoints listed in Appendix B), the saucepan (v3) and shared dish/drawers (v1) SVF critics together with their dataset shards (37,883 and 52,794 transitions respectively), the WCSAC checkpoints, and the per-cell benchmark CSVs behind every results table (including the gated, ablation, and boundary sweeps) are archived on the project's experiment workstation and are available from the author on request while a hosted release is prepared; every quantitative table in this report is regenerable from those snapshots with the harness, without any retraining. One exception is recorded for completeness: the three R1 basin checkpoints were lost to a storage cleanup after their benchmark CSVs were written (Section 6.4); their numbers remain reproducible from the archived CSVs, and regenerating the checkpoints themselves requires re-running the (deterministic-recipe but stochastic-outcome) fixed-λ fine-tune.